

SpectrumKV 全量实验数据提取

- 数据生成时间: 2026-06-05 14:43:40
- 总记录数: 10248

文件清单

| #  | 文件名                              | 大小    | 内容                            | 记录数                        |
|----|----------------------------------|-------|-------------------------------|----------------------------|
| 1  | exp1_pp1_qwen7b.json             | 63KB  | Qwen PPL精细预算扫描                | 240                        |
| 2  | exp1_pp1_mistral7b.json          | 63KB  | Mistral PPL精细预算扫描             | 240                        |
| 3  | exp1_pp1_gemma9b.json            | 42KB  | Gemma PPL精细预算扫描               | 160                        |
| 4  | exp2_niah_qwen7b.json            | 15KB  | Qwen NIAH检索评测                 | 680                        |
| 5  | exp2_niah_mistral7b.json         | 701KB | Mistral NIAH检索评测              | 580 rates + 2900 trials    |
| 6  | exp2_niah_gemma9b.json           | 972KB | Gemma NIAH检索评测                | 684 rates + 3420 trials    |
| 7  | exp2_spectrumkv_mistral7b.json   | 143KB | Mistral SpectrumKV Adaptive验证 | 475 trials                 |
| 8  | exp2_spectrumkv_gemma9b.json     | 163KB | Gemma SpectrumKV Adaptive验证   | 475 trials                 |
| 9  | exp3_quant_error_qwen7b.json     | 63KB  | Qwen逐层量化误差                    | per_layer:28 + per_head:28 |
| 10 | exp3_quant_error_mistral7b.json  | 80KB  | Mistral逐层量化误差                 | per_layer:36 + per_head:36 |
| 11 | exp3_quant_error_gemma9b.json    | 93KB  | Gemma逐层量化误差                   | per_layer:42 + per_head:42 |
| 12 | exp4_layer_budget_all.json       | 69KB  | 全量层级预算汇总                      | 18                         |
| 13 | exp4_layer_budget_mistral7b.json | 69KB  | Mistral层级预算分配                 | 18                         |
| 14 | exp4_layer_budget_qwen7b.json    | 55KB  | Qwen层级预算分配                    | 18                         |
| 15 | experiment_log_v2.md             | 30KB  | 实验记录 (补充数据源)                  | RQ1-16                     |

RQ1: KV Access Skewness (KV访问偏度)

| 数据来源: experiment_log_v2.md (Exp0 Locality Characterization, 仿真+初步GPU实验) |               |
|-------------------------------------------------------------------------|---------------|
| 指标                                                                      | 数值            |
| Gini系数范围                                                                | 0.866 ~ 0.952 |

RQ2: Sink Token Importance (Sink token重要性)

| 数据来源: experiment_log_v2.md (Exp0, attention hook提取) |        |                                      |
|-----------------------------------------------------|--------|--------------------------------------|
| 模型                                                  | Sink模式 | 特征描述                                 |
| Mistral-7B                                          | Sink   | sink token集中了极高权重, 是模型attention的核心锚点 |
| Qwen-7B                                             | Local  | sink token权重相对较低, attention更均匀分布     |
| Gemma-9B                                            | Hybrid | sink token+local token混合模式           |

关键数据: Mistral全budget NIAH=89.5% (=FP16基线), 删除sink的方法 (PDTrim/SWS) 显著退化

RQ3: Attention Pattern Classification (注意力模式分类)

| 数据来源: experiment_log_v2.md (Exp0, transmission_aware_attention.py提取) |            |                           |                 |
|----------------------------------------------------------------------|------------|---------------------------|-----------------|
| 模式                                                                   | 代表模型       | 特征                        | SpectrumKV策略    |
| Local                                                                | Qwen-7B    | attention集中在局部窗口, sink权重低 | 2-tier (禁INT4)  |
| Sink                                                                 | Mistral-7B | 首层sink token占极高权重, 中间层均匀  | 3-tier (全开)     |
| Hybrid                                                               | Gemma-9B   | sink+local混合, 不同层表现不同     | 3-tier (可用INT4) |

RQ4: Hot Set Capacity Multiplier (热集容量倍数)

| 数据来源: experiment_log_v2.md (仿真数据) |             |
|-----------------------------------|-------------|
| 序列长度                              | Hot Set容量倍数 |
| 32K                               | 130x        |

RQ5: Tier-Aware Adapter Overhead (分层适配器开销)

| 数据来源: experiment_log_v2.md (仿真/GPU实测) |        |
|---------------------------------------|--------|
| 指标                                    | 数值     |
| Tier-aware adapter overhead           | <0.04% |

RQ6: Cross-Model Generality (跨模型通用性)

| 数据来源: experiment_log_v2.md + Exp1-4三模型同步实验 |        |                         |        |
|--------------------------------------------|--------|-------------------------|--------|
| 模型                                         | 注意力模式  | 最优SpectrumKV配置          | INT4耐受 |
| Qwen-7B                                    | Local  | 2-tier (FP16+INT8)      | 不耐受    |
| Mistral-7B                                 | Sink   | 3-tier (FP16+INT8+INT4) | 耐受     |
| Gemma-9B                                   | Hybrid | 3-tier (可用INT4但需控制比例)   | 部分耐受   |

RQ7: Strategy Comparison at b=0.5 (策略对比: 50%预算PPL)

|                                                                            |  |
|----------------------------------------------------------------------------|--|
| 数据来源: exp1_pp1_qwen7b.json, exp1_pp1_mistral7b.json, exp1_pp1_gemma9b.json |  |
|----------------------------------------------------------------------------|--|

RQ7 完整表格: b=0.5, seq\_len=2048

| 方法                     | Qwen PPL | Qwen Δ% | Qwen FP16% | Qwen INT8% | Qwen INT4% | Mistral PPL | Mistral Δ% | Mistral FP16% | Mistral INT8% | Mistral INT4% | Gemma PPL | Gemma Δ% | Gemma FP16% | Gemma INT8% | Gemma INT4% |
|------------------------|----------|---------|------------|------------|------------|-------------|------------|---------------|---------------|---------------|-----------|----------|-------------|-------------|-------------|
| Full_FP16              | 7.0415   | 0.00    | 1.0000     | 0.0000     | 0.0000     | 7.9568      | 0.00       | 1.0000        | 0.0000        | 0.0000        | 11.2042   | 0.00     | 1.0000      | 0.0000      | 0.0000      |
| Uniform_INT8           | 7.1805   | 1.97    | 0.0000     | 1.0000     | 0.0000     | 7.9517      | -0.06      | 0.0000        | 1.0000        | 0.0000        | 11.1548   | -0.44    | 0.0000      | 1.0000      | 0.0000      |
| Uniform_INT4           | 未测试      | 未测试     | 未测试        | 未测试        | 未测试        | 未测试         | 未测试        | 未测试           | 未测试           | 未测试           | 未测试       | 未测试      | 未测试         | 未测试         | 未测试         |
| PDTrim                 | 8.8614   | 25.85   | 0.5000     | 0.0000     | 0.5000     | 9.7132      | 22.07      | 0.5000        | 0.0000        | 0.5000        | 15.1960   | 35.63    | 0.5000      | 0.0000      | 0.5000      |
| SWS_Original           | 9.2159   | 30.88   | 0.5000     | 0.0000     | 0.5000     | 10.6025     | 33.25      | 0.5000        | 0.0000        | 0.5000        | 16.1092   | 43.78    | 0.5000      | 0.0000      | 0.5000      |
| SWS_ValueAware         | 9.2159   | 30.88   | 0.5000     | 0.0000     | 0.5000     | 10.6025     | 33.25      | 0.5000        | 0.0000        | 0.5000        | 16.1092   | 43.78    | 0.5000      | 0.0000      | 0.5000      |
| SpectrumKV_Greedy      | 7.1805   | 1.97    | 0.0000     | 1.0000     | 0.0000     | 7.9517      | -0.06      | 0.0000        | 1.0000        | 0.0000        | 11.1548   | -0.44    | 0.0000      | 1.0000      | 0.0000      |
| SpectrumKV_Balanced    | 535.6371 | 7506.84 | 0.1997     | 0.3999     | 0.4004     | 8.1721      | 2.71       | 0.1997        | 0.3999        | 0.4004        | 11.4623   | 2.30     | 0.1997      | 0.3999      | 0.4004      |
| SpectrumKV_SinkProtect | 7.1397   | 1.39    | 0.0020     | 0.9980     | 0.0000     | 7.9492      | -0.10      | 0.0020        | 0.9980        | 0.0000        | 11.1514   | -0.47    | 0.0020      | 0.9980      | 0.0000      |
| Random_Tier            | 7.1805   | 1.97    | 0.0000     | 1.0000     | 0.0000     | 7.9517      | -0.06      | 0.0000        | 1.0000        | 0.0000        | 11.1548   | -0.44    | 0.0000      | 1.0000      | 0.0000      |

RQ8: Budget Sweep (预算扫描)

|                                                                            |  |
|----------------------------------------------------------------------------|--|
| 数据来源: exp1_pp1_qwen7b.json, exp1_pp1_mistral7b.json, exp1_pp1_gemma9b.json |  |
|----------------------------------------------------------------------------|--|

Qwen-7B seq\_len=2048

| 方法           | Budget | PPL    | Δ%   | FP16%  | INT8%  | INT4%  |
|--------------|--------|--------|------|--------|--------|--------|
| Full_FP16    | 1.0    | 7.0415 | 0.00 | 1.0000 | 0.0000 | 0.0000 |
| Uniform_INT8 | 0.25   | 未测试    | 未测试  | 未测试    | 未测试    | 未测试    |
| Uniform_INT8 | 0.3    | 未测试    | 未测试  | 未测试    | 未测试    | 未测试    |
| Uniform_INT8 | 0.35   | 未测试    | 未测试  | 未测试    | 未测试    | 未测试    |

|                        |      |             |            |        |        |
|------------------------|------|-------------|------------|--------|--------|
| Uniform_INT8           | 0.4  | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT8           | 0.45 | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT8           | 0.5  | 7.1805      | 1.97       | 0.0000 | 1.0000 |
| Uniform_INT8           | 0.55 | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT8           | 0.6  | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT8           | 0.65 | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT8           | 0.7  | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT8           | 0.75 | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT4           | 0.25 | 45000.2004  | 638969.64  | 0.0000 | 0.0000 |
| Uniform_INT4           | 0.3  | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT4           | 0.35 | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT4           | 0.4  | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT4           | 0.45 | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT4           | 0.5  | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT4           | 0.55 | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT4           | 0.6  | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT4           | 0.65 | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT4           | 0.7  | 未测试         | 未测试        | 未测试    | 未测试    |
| Uniform_INT4           | 0.75 | 未测试         | 未测试        | 未测试    | 未测试    |
| PDTrim                 | 0.25 | 10.6661     | 51.47      | 0.2500 | 0.0000 |
| PDTrim                 | 0.3  | 10.0565     | 42.82      | 0.2998 | 0.0000 |
| PDTrim                 | 0.35 | 9.3540      | 32.84      | 0.3496 | 0.0000 |
| PDTrim                 | 0.4  | 8.9481      | 27.08      | 0.3999 | 0.0000 |
| PDTrim                 | 0.45 | 8.9223      | 26.71      | 0.4497 | 0.0000 |
| PDTrim                 | 0.5  | 8.8614      | 25.85      | 0.5000 | 0.0000 |
| PDTrim                 | 0.55 | 8.7026      | 23.59      | 0.5498 | 0.0000 |
| PDTrim                 | 0.6  | 8.3235      | 18.21      | 0.5996 | 0.0000 |
| PDTrim                 | 0.65 | 8.2650      | 17.38      | 0.6499 | 0.0000 |
| PDTrim                 | 0.7  | 8.3190      | 18.14      | 0.6997 | 0.0000 |
| PDTrim                 | 0.75 | 8.2279      | 16.85      | 0.7500 | 0.0000 |
| SWS_Original           | 0.25 | 11.3096     | 60.61      | 0.2500 | 0.0000 |
| SWS_Original           | 0.3  | 11.0382     | 56.76      | 0.2998 | 0.0000 |
| SWS_Original           | 0.35 | 10.4365     | 48.21      | 0.3496 | 0.0000 |
| SWS_Original           | 0.4  | 9.8041      | 39.23      | 0.3999 | 0.0000 |
| SWS_Original           | 0.45 | 9.4686      | 34.47      | 0.4497 | 0.0000 |
| SWS_Original           | 0.5  | 9.2159      | 30.88      | 0.5000 | 0.0000 |
| SWS_Original           | 0.55 | 8.8986      | 26.37      | 0.5498 | 0.0000 |
| SWS_Original           | 0.6  | 8.6040      | 22.19      | 0.5996 | 0.0000 |
| SWS_Original           | 0.65 | 8.4135      | 19.48      | 0.6499 | 0.0000 |
| SWS_Original           | 0.7  | 8.1491      | 15.73      | 0.6997 | 0.0000 |
| SWS_Original           | 0.75 | 8.0093      | 13.74      | 0.7500 | 0.0000 |
| SWS_ValueAware         | 0.25 | 11.6068     | 64.83      | 0.2500 | 0.0000 |
| SWS_ValueAware         | 0.3  | 11.0382     | 56.76      | 0.2998 | 0.0000 |
| SWS_ValueAware         | 0.35 | 10.4365     | 48.21      | 0.3496 | 0.0000 |
| SWS_ValueAware         | 0.4  | 9.8041      | 39.23      | 0.3999 | 0.0000 |
| SWS_ValueAware         | 0.45 | 9.4686      | 34.47      | 0.4497 | 0.0000 |
| SWS_ValueAware         | 0.5  | 9.2159      | 30.88      | 0.5000 | 0.0000 |
| SWS_ValueAware         | 0.55 | 8.8986      | 26.37      | 0.5498 | 0.0000 |
| SWS_ValueAware         | 0.6  | 8.6040      | 22.19      | 0.5996 | 0.0000 |
| SWS_ValueAware         | 0.65 | 8.4135      | 19.48      | 0.6499 | 0.0000 |
| SWS_ValueAware         | 0.7  | 8.1491      | 15.73      | 0.6997 | 0.0000 |
| SWS_ValueAware         | 0.75 | 8.0093      | 13.74      | 0.7500 | 0.0000 |
| SpectrumKV_Greedy      | 0.25 | 45000.2004  | 638969.64  | 0.0000 | 0.0000 |
| SpectrumKV_Greedy      | 0.3  | 16063.8680  | 228030.77  | 0.0000 | 0.1997 |
| SpectrumKV_Greedy      | 0.35 | 4675.4921   | 66298.93   | 0.0000 | 0.3999 |
| SpectrumKV_Greedy      | 0.4  | 568.6381    | 7975.51    | 0.0000 | 0.5996 |
| SpectrumKV_Greedy      | 0.45 | 88.1980     | 1152.54    | 0.0000 | 0.7998 |
| SpectrumKV_Greedy      | 0.5  | 7.1805      | 1.97       | 0.0000 | 1.0000 |
| SpectrumKV_Greedy      | 0.55 | 7.1096      | 0.97       | 0.1001 | 0.8999 |
| SpectrumKV_Greedy      | 0.6  | 7.1033      | 0.88       | 0.2002 | 0.7998 |
| SpectrumKV_Greedy      | 0.65 | 7.0871      | 0.65       | 0.2998 | 0.7002 |
| SpectrumKV_Greedy      | 0.7  | 7.0745      | 0.47       | 0.3999 | 0.6001 |
| SpectrumKV_Greedy      | 0.75 | 7.0624      | 0.30       | 0.5000 | 0.5000 |
| SpectrumKV_Balanced    | 0.25 | 45000.2004  | 638969.64  | 0.0000 | 0.0000 |
| SpectrumKV_Balanced    | 0.3  | 25582.2877  | 363206.46  | 0.0396 | 0.0796 |
| SpectrumKV_Balanced    | 0.35 | 15260.5693  | 216622.74  | 0.0796 | 0.1597 |
| SpectrumKV_Balanced    | 0.4  | 6166.9184   | 87479.39   | 0.1196 | 0.2397 |
| SpectrumKV_Balanced    | 0.45 | 2139.0505   | 30277.69   | 0.1597 | 0.3198 |
| SpectrumKV_Balanced    | 0.5  | 535.6371    | 7506.84    | 0.1997 | 0.3999 |
| SpectrumKV_Balanced    | 0.55 | 188.8520    | 2581.98    | 0.2397 | 0.4800 |
| SpectrumKV_Balanced    | 0.6  | 53.2028     | 655.56     | 0.2798 | 0.5596 |
| SpectrumKV_Balanced    | 0.65 | 14.9510     | 112.33     | 0.3198 | 0.6396 |
| SpectrumKV_Balanced    | 0.7  | 7.1364      | 1.35       | 0.3999 | 0.5996 |
| SpectrumKV_Balanced    | 0.75 | 7.0624      | 0.30       | 0.5000 | 0.5000 |
| SpectrumKV_SinkProtect | 0.25 | 122908.0602 | 1745376.90 | 0.0020 | 0.0000 |
| SpectrumKV_SinkProtect | 0.3  | 15804.9121  | 224353.21  | 0.0020 | 0.1978 |
| SpectrumKV_SinkProtect | 0.35 | 4648.2637   | 65912.24   | 0.0020 | 0.3979 |
| SpectrumKV_SinkProtect | 0.4  | 564.9075    | 7922.53    | 0.0020 | 0.5977 |
| SpectrumKV_SinkProtect | 0.45 | 84.0683     | 1093.89    | 0.0020 | 0.7979 |
| SpectrumKV_SinkProtect | 0.5  | 7.1397      | 1.39       | 0.0020 | 0.9980 |
| SpectrumKV_SinkProtect | 0.55 | 7.1096      | 0.97       | 0.1001 | 0.8999 |
| SpectrumKV_SinkProtect | 0.6  | 7.1033      | 0.88       | 0.2002 | 0.7998 |
| SpectrumKV_SinkProtect | 0.65 | 7.0871      | 0.65       | 0.2998 | 0.7002 |
| SpectrumKV_SinkProtect | 0.7  | 7.0745      | 0.47       | 0.3999 | 0.6001 |
| SpectrumKV_SinkProtect | 0.75 | 7.0624      | 0.30       | 0.5000 | 0.5000 |
| Random_Tier            | 0.25 | 45000.2004  | 638969.64  | 0.0000 | 0.0000 |
| Random_Tier            | 0.3  | 10177.9676  | 144442.25  | 0.0000 | 0.1997 |
| Random_Tier            | 0.35 | 4640.1027   | 65796.35   | 0.0000 | 0.3999 |
| Random_Tier            | 0.4  | 1406.9671   | 19881.02   | 0.0000 | 0.5996 |
| Random_Tier            | 0.45 | 269.2142    | 3723.24    | 0.0000 | 0.7998 |
| Random_Tier            | 0.5  | 7.1805      | 1.97       | 0.0000 | 1.0000 |
| Random_Tier            | 0.55 | 7.1429      | 1.44       | 0.1001 | 0.8999 |
| Random_Tier            | 0.6  | 7.1580      | 1.65       | 0.2002 | 0.7998 |
| Random_Tier            | 0.65 | 7.1188      | 1.10       | 0.2998 | 0.7002 |
| Random_Tier            | 0.7  | 7.1152      | 1.05       | 0.3999 | 0.6001 |
| Random_Tier            | 0.75 | 7.1028      | 0.87       | 0.5000 | 0.5000 |

| 方法                     | Budget | PPL        | Δ%         | FP16%  | INT8%  | INT4%  |
|------------------------|--------|------------|------------|--------|--------|--------|
| Full_FP16              | 1.0    | 6.0084     | 0.00       | 1.0000 | 0.0000 | 0.0000 |
| Uniform_INT8           | 0.25   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.3    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.35   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.4    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.45   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.5    | 6.1554     | 2.45       | 0.0000 | 1.0000 | 0.0000 |
| Uniform_INT8           | 0.55   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.6    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.65   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.7    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.75   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.25   | 25205.8339 | 419411.44  | 0.0000 | 0.0000 | 1.0000 |
| Uniform_INT4           | 0.3    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.35   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.4    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.45   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.5    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.55   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.6    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.65   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.7    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.75   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| PDTrim                 | 0.25   | 7.7744     | 29.39      | 0.2500 | 0.0000 | 0.7500 |
| PDTrim                 | 0.3    | 7.2328     | 20.38      | 0.2998 | 0.0000 | 0.7002 |
| PDTrim                 | 0.35   | 6.9756     | 16.10      | 0.3499 | 0.0000 | 0.6501 |
| PDTrim                 | 0.4    | 7.0719     | 17.70      | 0.3999 | 0.0000 | 0.6001 |
| PDTrim                 | 0.45   | 6.9901     | 16.34      | 0.4500 | 0.0000 | 0.5500 |
| PDTrim                 | 0.5    | 6.9987     | 16.48      | 0.5000 | 0.0000 | 0.5000 |
| PDTrim                 | 0.55   | 6.9944     | 16.41      | 0.5498 | 0.0000 | 0.4502 |
| PDTrim                 | 0.6    | 6.7967     | 13.12      | 0.5999 | 0.0000 | 0.4001 |
| PDTrim                 | 0.65   | 6.6855     | 11.27      | 0.6499 | 0.0000 | 0.3501 |
| PDTrim                 | 0.7    | 6.5466     | 8.96       | 0.7000 | 0.0000 | 0.3000 |
| PDTrim                 | 0.75   | 6.5244     | 8.59       | 0.7500 | 0.0000 | 0.2500 |
| SWS_Original           | 0.25   | 8.0790     | 34.46      | 0.2500 | 0.0000 | 0.7500 |
| SWS_Original           | 0.3    | 7.3638     | 22.56      | 0.2998 | 0.0000 | 0.7002 |
| SWS_Original           | 0.35   | 7.1548     | 19.08      | 0.3499 | 0.0000 | 0.6501 |
| SWS_Original           | 0.4    | 6.7994     | 13.17      | 0.3999 | 0.0000 | 0.6001 |
| SWS_Original           | 0.45   | 6.6976     | 11.47      | 0.4500 | 0.0000 | 0.5500 |
| SWS_Original           | 0.5    | 6.5845     | 9.59       | 0.5000 | 0.0000 | 0.5000 |
| SWS_Original           | 0.55   | 6.3201     | 5.19       | 0.5498 | 0.0000 | 0.4502 |
| SWS_Original           | 0.6    | 6.2359     | 3.79       | 0.5999 | 0.0000 | 0.4001 |
| SWS_Original           | 0.65   | 6.1882     | 2.99       | 0.6499 | 0.0000 | 0.3501 |
| SWS_Original           | 0.7    | 6.2092     | 3.34       | 0.7000 | 0.0000 | 0.3000 |
| SWS_Original           | 0.75   | 6.2580     | 4.15       | 0.7500 | 0.0000 | 0.2500 |
| SWS_ValueAware         | 0.25   | 8.0790     | 34.46      | 0.2500 | 0.0000 | 0.7500 |
| SWS_ValueAware         | 0.3    | 7.3638     | 22.56      | 0.2998 | 0.0000 | 0.7002 |
| SWS_ValueAware         | 0.35   | 7.1548     | 19.08      | 0.3499 | 0.0000 | 0.6501 |
| SWS_ValueAware         | 0.4    | 6.7994     | 13.17      | 0.3999 | 0.0000 | 0.6001 |
| SWS_ValueAware         | 0.45   | 6.6976     | 11.47      | 0.4500 | 0.0000 | 0.5500 |
| SWS_ValueAware         | 0.5    | 6.5845     | 9.59       | 0.5000 | 0.0000 | 0.5000 |
| SWS_ValueAware         | 0.55   | 6.3201     | 5.19       | 0.5498 | 0.0000 | 0.4502 |
| SWS_ValueAware         | 0.6    | 6.2359     | 3.79       | 0.5999 | 0.0000 | 0.4001 |
| SWS_ValueAware         | 0.65   | 6.1882     | 2.99       | 0.6499 | 0.0000 | 0.3501 |
| SWS_ValueAware         | 0.7    | 6.2092     | 3.34       | 0.7000 | 0.0000 | 0.3000 |
| SWS_ValueAware         | 0.75   | 6.2580     | 4.15       | 0.7500 | 0.0000 | 0.2500 |
| SpectrumKV_Greedy      | 0.25   | 25205.8339 | 419411.44  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Greedy      | 0.3    | 14765.8672 | 245654.62  | 0.0000 | 0.2000 | 0.8000 |
| SpectrumKV_Greedy      | 0.35   | 4155.8759  | 69068.01   | 0.0000 | 0.3999 | 0.6001 |
| SpectrumKV_Greedy      | 0.4    | 637.2627   | 10506.23   | 0.0000 | 0.5999 | 0.4001 |
| SpectrumKV_Greedy      | 0.45   | 95.3731    | 1487.33    | 0.0000 | 0.7998 | 0.2002 |
| SpectrumKV_Greedy      | 0.5    | 6.1554     | 2.45       | 0.0000 | 1.0000 | 0.0000 |
| SpectrumKV_Greedy      | 0.55   | 6.1021     | 1.56       | 0.1001 | 0.8999 | 0.0000 |
| SpectrumKV_Greedy      | 0.6    | 6.0886     | 1.34       | 0.2000 | 0.8000 | 0.0000 |
| SpectrumKV_Greedy      | 0.65   | 6.0779     | 1.16       | 0.3000 | 0.7000 | 0.0000 |
| SpectrumKV_Greedy      | 0.7    | 6.0584     | 0.83       | 0.3999 | 0.6001 | 0.0000 |
| SpectrumKV_Greedy      | 0.75   | 6.0542     | 0.76       | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_Balanced    | 0.25   | 25205.8339 | 419411.44  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Balanced    | 0.3    | 16476.5664 | 274126.52  | 0.0398 | 0.0798 | 0.8804 |
| SpectrumKV_Balanced    | 0.35   | 13690.6433 | 227759.21  | 0.0798 | 0.1599 | 0.7603 |
| SpectrumKV_Balanced    | 0.4    | 5217.4910  | 86736.93   | 0.1199 | 0.2400 | 0.6401 |
| SpectrumKV_Balanced    | 0.45   | 1732.2128  | 28729.96   | 0.1599 | 0.3198 | 0.5203 |
| SpectrumKV_Balanced    | 0.5    | 603.0251   | 9936.40    | 0.2000 | 0.3999 | 0.4001 |
| SpectrumKV_Balanced    | 0.55   | 180.1888   | 2898.96    | 0.2400 | 0.4800 | 0.2800 |
| SpectrumKV_Balanced    | 0.6    | 65.0693    | 982.98     | 0.2798 | 0.5598 | 0.1604 |
| SpectrumKV_Balanced    | 0.65   | 14.9261    | 148.42     | 0.3198 | 0.6399 | 0.0403 |
| SpectrumKV_Balanced    | 0.7    | 6.1848     | 2.94       | 0.3999 | 0.5999 | 0.0002 |
| SpectrumKV_Balanced    | 0.75   | 6.0542     | 0.76       | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.25   | 67852.3852 | 1129196.17 | 0.0010 | 0.0000 | 0.9990 |
| SpectrumKV_SinkProtect | 0.3    | 14186.5531 | 236012.85  | 0.0010 | 0.1990 | 0.8000 |
| SpectrumKV_SinkProtect | 0.35   | 4052.1307  | 67341.34   | 0.0010 | 0.3989 | 0.6001 |
| SpectrumKV_SinkProtect | 0.4    | 639.0712   | 10536.33   | 0.0010 | 0.5989 | 0.4001 |
| SpectrumKV_SinkProtect | 0.45   | 93.3853    | 1454.25    | 0.0010 | 0.7988 | 0.2002 |
| SpectrumKV_SinkProtect | 0.5    | 6.1144     | 1.76       | 0.0010 | 0.9990 | 0.0000 |
| SpectrumKV_SinkProtect | 0.55   | 6.1021     | 1.56       | 0.1001 | 0.8999 | 0.0000 |
| SpectrumKV_SinkProtect | 0.6    | 6.0886     | 1.34       | 0.2000 | 0.8000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.65   | 6.0779     | 1.16       | 0.3000 | 0.7000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.7    | 6.0584     | 0.83       | 0.3999 | 0.6001 | 0.0000 |
| SpectrumKV_SinkProtect | 0.75   | 6.0542     | 0.76       | 0.5000 | 0.5000 | 0.0000 |
| Random_Tier            | 0.25   | 25205.8339 | 419411.44  | 0.0000 | 0.0000 | 1.0000 |
| Random_Tier            | 0.3    | 7702.6674  | 128098.78  | 0.0000 | 0.2000 | 0.8000 |
| Random_Tier            | 0.35   | 4835.9110  | 80386.13   | 0.0000 | 0.3999 | 0.6001 |
| Random_Tier            | 0.4    | 2051.5531  | 34044.87   | 0.0000 | 0.5999 | 0.4001 |
| Random_Tier            | 0.45   | 292.2698   | 4764.37    | 0.0000 | 0.7998 | 0.2002 |
| Random_Tier            | 0.5    | 6.1554     | 2.45       | 0.0000 | 1.0000 | 0.0000 |
| Random_Tier            | 0.55   | 6.1314     | 2.05       | 0.1001 | 0.8999 | 0.0000 |

|             |      |        |      |        |        |        |
|-------------|------|--------|------|--------|--------|--------|
| Random_Tier | 0.6  | 6.1160 | 1.79 | 0.2000 | 0.8000 | 0.0000 |
| Random_Tier | 0.65 | 6.1017 | 1.55 | 0.3000 | 0.7000 | 0.0000 |
| Random_Tier | 0.7  | 6.0846 | 1.27 | 0.3999 | 0.6001 | 0.0000 |
| Random_Tier | 0.75 | 6.0767 | 1.14 | 0.5000 | 0.5000 | 0.0000 |

Qwen-7B seq\_len=8192

| 方法                     | Budget | PPL        | Δ%         | FP16%  | INT8%  | INT4%  |
|------------------------|--------|------------|------------|--------|--------|--------|
| Full_FP16              | 1.0    | 6.2145     | 0.00       | 1.0000 | 0.0000 | 0.0000 |
| Uniform_INT8           | 0.25   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.3    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.35   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.4    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.45   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.5    | 6.3949     | 2.90       | 0.0000 | 1.0000 | 0.0000 |
| Uniform_INT8           | 0.55   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.6    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.65   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.7    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.75   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.25   | 24512.2691 | 394338.36  | 0.0000 | 0.0000 | 1.0000 |
| Uniform_INT4           | 0.3    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.35   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.4    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.45   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.5    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.55   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.6    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.65   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.7    | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.75   | 未测试        | 未测试        | 未测试    | 未测试    | 未测试    |
| PDTrim                 | 0.25   | 7.6597     | 23.26      | 0.2500 | 0.0000 | 0.7500 |
| PDTrim                 | 0.3    | 7.3588     | 18.41      | 0.2999 | 0.0000 | 0.7001 |
| PDTrim                 | 0.35   | 6.9888     | 12.46      | 0.3500 | 0.0000 | 0.6500 |
| PDTrim                 | 0.4    | 6.8349     | 9.98       | 0.3999 | 0.0000 | 0.6001 |
| PDTrim                 | 0.45   | 6.7264     | 8.24       | 0.4500 | 0.0000 | 0.5500 |
| PDTrim                 | 0.5    | 6.6644     | 7.24       | 0.5000 | 0.0000 | 0.5000 |
| PDTrim                 | 0.55   | 6.5959     | 6.14       | 0.5499 | 0.0000 | 0.4501 |
| PDTrim                 | 0.6    | 6.5139     | 4.82       | 0.6000 | 0.0000 | 0.4000 |
| PDTrim                 | 0.65   | 6.3716     | 2.53       | 0.6499 | 0.0000 | 0.3501 |
| PDTrim                 | 0.7    | 6.4601     | 3.95       | 0.7000 | 0.0000 | 0.3000 |
| PDTrim                 | 0.75   | 6.3827     | 2.71       | 0.7500 | 0.0000 | 0.2500 |
| SWS_Original           | 0.25   | 7.8118     | 25.70      | 0.2500 | 0.0000 | 0.7500 |
| SWS_Original           | 0.3    | 7.4766     | 20.31      | 0.2999 | 0.0000 | 0.7001 |
| SWS_Original           | 0.35   | 7.4004     | 19.08      | 0.3500 | 0.0000 | 0.6500 |
| SWS_Original           | 0.4    | 7.2176     | 16.14      | 0.3999 | 0.0000 | 0.6001 |
| SWS_Original           | 0.45   | 7.0886     | 14.07      | 0.4500 | 0.0000 | 0.5500 |
| SWS_Original           | 0.5    | 7.0160     | 12.90      | 0.5000 | 0.0000 | 0.5000 |
| SWS_Original           | 0.55   | 6.9736     | 12.22      | 0.5499 | 0.0000 | 0.4501 |
| SWS_Original           | 0.6    | 6.8162     | 9.68       | 0.6000 | 0.0000 | 0.4000 |
| SWS_Original           | 0.65   | 6.6891     | 7.64       | 0.6499 | 0.0000 | 0.3501 |
| SWS_Original           | 0.7    | 6.6598     | 7.17       | 0.7000 | 0.0000 | 0.3000 |
| SWS_Original           | 0.75   | 6.5833     | 5.94       | 0.7500 | 0.0000 | 0.2500 |
| SWS_ValueAware         | 0.25   | 7.8118     | 25.70      | 0.2500 | 0.0000 | 0.7500 |
| SWS_ValueAware         | 0.3    | 7.4766     | 20.31      | 0.2999 | 0.0000 | 0.7001 |
| SWS_ValueAware         | 0.35   | 7.4004     | 19.08      | 0.3500 | 0.0000 | 0.6500 |
| SWS_ValueAware         | 0.4    | 7.2176     | 16.14      | 0.3999 | 0.0000 | 0.6001 |
| SWS_ValueAware         | 0.45   | 7.0886     | 14.07      | 0.4500 | 0.0000 | 0.5500 |
| SWS_ValueAware         | 0.5    | 7.0160     | 12.90      | 0.5000 | 0.0000 | 0.5000 |
| SWS_ValueAware         | 0.55   | 6.9736     | 12.22      | 0.5499 | 0.0000 | 0.4501 |
| SWS_ValueAware         | 0.6    | 6.8162     | 9.68       | 0.6000 | 0.0000 | 0.4000 |
| SWS_ValueAware         | 0.65   | 6.6891     | 7.64       | 0.6499 | 0.0000 | 0.3501 |
| SWS_ValueAware         | 0.7    | 6.6598     | 7.17       | 0.7000 | 0.0000 | 0.3000 |
| SWS_ValueAware         | 0.75   | 6.5833     | 5.94       | 0.7500 | 0.0000 | 0.2500 |
| SpectrumKV_Greedy      | 0.25   | 24512.2691 | 394338.36  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Greedy      | 0.3    | 14683.0871 | 236172.41  | 0.0000 | 0.2000 | 0.8000 |
| SpectrumKV_Greedy      | 0.35   | 4688.5412  | 75345.50   | 0.0000 | 0.3999 | 0.6001 |
| SpectrumKV_Greedy      | 0.4    | 736.1136   | 11745.15   | 0.0000 | 0.6000 | 0.4000 |
| SpectrumKV_Greedy      | 0.45   | 137.0993   | 2106.13    | 0.0000 | 0.7999 | 0.2001 |
| SpectrumKV_Greedy      | 0.5    | 6.3949     | 2.90       | 0.0000 | 1.0000 | 0.0000 |
| SpectrumKV_Greedy      | 0.55   | 6.3589     | 2.32       | 0.1000 | 0.9000 | 0.0000 |
| SpectrumKV_Greedy      | 0.6    | 6.3465     | 2.12       | 0.2000 | 0.8000 | 0.0000 |
| SpectrumKV_Greedy      | 0.65   | 6.3213     | 1.72       | 0.3000 | 0.7000 | 0.0000 |
| SpectrumKV_Greedy      | 0.7    | 6.3011     | 1.39       | 0.4000 | 0.6000 | 0.0000 |
| SpectrumKV_Greedy      | 0.75   | 6.2880     | 1.18       | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_Balanced    | 0.25   | 24512.2691 | 394338.36  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Balanced    | 0.3    | 16206.8953 | 260692.72  | 0.0399 | 0.0800 | 0.8801 |
| SpectrumKV_Balanced    | 0.35   | 13353.2549 | 214773.46  | 0.0800 | 0.1599 | 0.7601 |
| SpectrumKV_Balanced    | 0.4    | 5275.3282  | 84787.77   | 0.1200 | 0.2400 | 0.6400 |
| SpectrumKV_Balanced    | 0.45   | 1836.9158  | 29458.67   | 0.1599 | 0.3199 | 0.5201 |
| SpectrumKV_Balanced    | 0.5    | 711.7830   | 11353.63   | 0.2000 | 0.3999 | 0.4001 |
| SpectrumKV_Balanced    | 0.55   | 218.8949   | 3422.34    | 0.2400 | 0.4800 | 0.2800 |
| SpectrumKV_Balanced    | 0.6    | 85.9605    | 1283.23    | 0.2799 | 0.5599 | 0.1602 |
| SpectrumKV_Balanced    | 0.65   | 18.9343    | 204.68     | 0.3199 | 0.6399 | 0.0402 |
| SpectrumKV_Balanced    | 0.7    | 6.3412     | 2.04       | 0.3999 | 0.6000 | 0.0001 |
| SpectrumKV_Balanced    | 0.75   | 6.2880     | 1.18       | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.25   | 70535.4578 | 1134918.97 | 0.0005 | 0.0000 | 0.9995 |
| SpectrumKV_SinkProtect | 0.3    | 13567.6217 | 218222.93  | 0.0005 | 0.1995 | 0.8000 |
| SpectrumKV_SinkProtect | 0.35   | 4796.7152  | 77086.18   | 0.0005 | 0.3994 | 0.6001 |
| SpectrumKV_SinkProtect | 0.4    | 712.8986   | 11371.58   | 0.0005 | 0.5995 | 0.4000 |
| SpectrumKV_SinkProtect | 0.45   | 129.7738   | 1988.25    | 0.0005 | 0.7994 | 0.2001 |
| SpectrumKV_SinkProtect | 0.5    | 6.3722     | 2.54       | 0.0005 | 0.9995 | 0.0000 |
| SpectrumKV_SinkProtect | 0.55   | 6.3589     | 2.32       | 0.1000 | 0.9000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.6    | 6.3465     | 2.12       | 0.2000 | 0.8000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.65   | 6.3213     | 1.72       | 0.3000 | 0.7000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.7    | 6.3011     | 1.39       | 0.4000 | 0.6000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.75   | 6.2880     | 1.18       | 0.5000 | 0.5000 | 0.0000 |
| Random_Tier            | 0.25   | 24512.2691 | 394338.36  | 0.0000 | 0.0000 | 1.0000 |

|             |      |           |           |        |        |        |
|-------------|------|-----------|-----------|--------|--------|--------|
| Random_Tier | 0.3  | 9771.4426 | 157136.84 | 0.0000 | 0.2000 | 0.8000 |
| Random_Tier | 0.35 | 5496.6131 | 88348.57  | 0.0000 | 0.3999 | 0.6001 |
| Random_Tier | 0.4  | 2474.5538 | 39719.20  | 0.0000 | 0.6000 | 0.4000 |
| Random_Tier | 0.45 | 922.4395  | 14743.40  | 0.0000 | 0.7999 | 0.2001 |
| Random_Tier | 0.5  | 6.3949    | 2.90      | 0.0000 | 1.0000 | 0.0000 |
| Random_Tier | 0.55 | 6.3729    | 2.55      | 0.1000 | 0.9000 | 0.0000 |
| Random_Tier | 0.6  | 6.3464    | 2.12      | 0.2000 | 0.8000 | 0.0000 |
| Random_Tier | 0.65 | 6.3288    | 1.84      | 0.3000 | 0.7000 | 0.0000 |
| Random_Tier | 0.7  | 6.3141    | 1.60      | 0.4000 | 0.6000 | 0.0000 |
| Random_Tier | 0.75 | 6.3003    | 1.38      | 0.5000 | 0.5000 | 0.0000 |

Mistral-7B seq\_len=2048

| 方法                     | Budget | PPL     | Δ%    | FP16%  | INT8%  | INT4%  |
|------------------------|--------|---------|-------|--------|--------|--------|
| Full_FP16              | 1.0    | 7.9568  | 0.00  | 1.0000 | 0.0000 | 0.0000 |
| Uniform_INT8           | 0.25   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.3    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.35   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.4    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.45   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.5    | 7.9517  | -0.06 | 0.0000 | 1.0000 | 0.0000 |
| Uniform_INT8           | 0.55   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.6    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.65   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.7    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8           | 0.75   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.25   | 8.4840  | 6.63  | 0.0000 | 0.0000 | 1.0000 |
| Uniform_INT4           | 0.3    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.35   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.4    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.45   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.5    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.55   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.6    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.65   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.7    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4           | 0.75   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| PDTrim                 | 0.25   | 11.4444 | 43.83 | 0.2500 | 0.0000 | 0.7500 |
| PDTrim                 | 0.3    | 10.4155 | 30.90 | 0.2998 | 0.0000 | 0.7002 |
| PDTrim                 | 0.35   | 10.1289 | 27.30 | 0.3496 | 0.0000 | 0.6504 |
| PDTrim                 | 0.4    | 9.8788  | 24.15 | 0.3999 | 0.0000 | 0.6001 |
| PDTrim                 | 0.45   | 9.5993  | 20.64 | 0.4497 | 0.0000 | 0.5503 |
| PDTrim                 | 0.5    | 9.7132  | 22.07 | 0.5000 | 0.0000 | 0.5000 |
| PDTrim                 | 0.55   | 9.5299  | 19.77 | 0.5498 | 0.0000 | 0.4502 |
| PDTrim                 | 0.6    | 9.3901  | 18.01 | 0.5996 | 0.0000 | 0.4004 |
| PDTrim                 | 0.65   | 9.1521  | 15.02 | 0.6499 | 0.0000 | 0.3501 |
| PDTrim                 | 0.7    | 9.0991  | 14.36 | 0.6997 | 0.0000 | 0.3003 |
| PDTrim                 | 0.75   | 9.1788  | 15.36 | 0.7500 | 0.0000 | 0.2500 |
| SWS_Original           | 0.25   | 12.2075 | 53.42 | 0.2500 | 0.0000 | 0.7500 |
| SWS_Original           | 0.3    | 11.7988 | 48.29 | 0.2998 | 0.0000 | 0.7002 |
| SWS_Original           | 0.35   | 11.2358 | 41.21 | 0.3496 | 0.0000 | 0.6504 |
| SWS_Original           | 0.4    | 10.8922 | 36.89 | 0.3999 | 0.0000 | 0.6001 |
| SWS_Original           | 0.45   | 10.7690 | 35.34 | 0.4497 | 0.0000 | 0.5503 |
| SWS_Original           | 0.5    | 10.6025 | 33.25 | 0.5000 | 0.0000 | 0.5000 |
| SWS_Original           | 0.55   | 10.4638 | 31.51 | 0.5498 | 0.0000 | 0.4502 |
| SWS_Original           | 0.6    | 10.0165 | 25.89 | 0.5996 | 0.0000 | 0.4004 |
| SWS_Original           | 0.65   | 9.7982  | 23.14 | 0.6499 | 0.0000 | 0.3501 |
| SWS_Original           | 0.7    | 9.5044  | 19.45 | 0.6997 | 0.0000 | 0.3003 |
| SWS_Original           | 0.75   | 9.4619  | 18.92 | 0.7500 | 0.0000 | 0.2500 |
| SWS_ValueAware         | 0.25   | 12.3359 | 55.04 | 0.2500 | 0.0000 | 0.7500 |
| SWS_ValueAware         | 0.3    | 11.7988 | 48.29 | 0.2998 | 0.0000 | 0.7002 |
| SWS_ValueAware         | 0.35   | 11.2358 | 41.21 | 0.3496 | 0.0000 | 0.6504 |
| SWS_ValueAware         | 0.4    | 10.8922 | 36.89 | 0.3999 | 0.0000 | 0.6001 |
| SWS_ValueAware         | 0.45   | 10.7690 | 35.34 | 0.4497 | 0.0000 | 0.5503 |
| SWS_ValueAware         | 0.5    | 10.6025 | 33.25 | 0.5000 | 0.0000 | 0.5000 |
| SWS_ValueAware         | 0.55   | 10.4638 | 31.51 | 0.5498 | 0.0000 | 0.4502 |
| SWS_ValueAware         | 0.6    | 10.0165 | 25.89 | 0.5996 | 0.0000 | 0.4004 |
| SWS_ValueAware         | 0.65   | 9.7982  | 23.14 | 0.6499 | 0.0000 | 0.3501 |
| SWS_ValueAware         | 0.7    | 9.5044  | 19.45 | 0.6997 | 0.0000 | 0.3003 |
| SWS_ValueAware         | 0.75   | 9.4619  | 18.92 | 0.7500 | 0.0000 | 0.2500 |
| SpectrumKV_Greedy      | 0.25   | 8.4840  | 6.63  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Greedy      | 0.3    | 8.3874  | 5.41  | 0.0000 | 0.1997 | 0.8003 |
| SpectrumKV_Greedy      | 0.35   | 8.2790  | 4.05  | 0.0000 | 0.3999 | 0.6001 |
| SpectrumKV_Greedy      | 0.4    | 8.1613  | 2.57  | 0.0000 | 0.5996 | 0.4004 |
| SpectrumKV_Greedy      | 0.45   | 8.0722  | 1.45  | 0.0000 | 0.7998 | 0.2002 |
| SpectrumKV_Greedy      | 0.5    | 7.9517  | -0.06 | 0.0000 | 1.0000 | 0.0000 |
| SpectrumKV_Greedy      | 0.55   | 7.9552  | -0.02 | 0.1001 | 0.8999 | 0.0000 |
| SpectrumKV_Greedy      | 0.6    | 7.9628  | 0.08  | 0.2002 | 0.7998 | 0.0000 |
| SpectrumKV_Greedy      | 0.65   | 7.9569  | 0.00  | 0.2998 | 0.7002 | 0.0000 |
| SpectrumKV_Greedy      | 0.7    | 7.9591  | 0.03  | 0.3999 | 0.6001 | 0.0000 |
| SpectrumKV_Greedy      | 0.75   | 7.9583  | 0.02  | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_Balanced    | 0.25   | 8.4840  | 6.63  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Balanced    | 0.3    | 8.3968  | 5.53  | 0.0396 | 0.0796 | 0.8809 |
| SpectrumKV_Balanced    | 0.35   | 8.4197  | 5.82  | 0.0796 | 0.1597 | 0.7607 |
| SpectrumKV_Balanced    | 0.4    | 8.2825  | 4.09  | 0.1196 | 0.2397 | 0.6406 |
| SpectrumKV_Balanced    | 0.45   | 8.2306  | 3.44  | 0.1597 | 0.3198 | 0.5205 |
| SpectrumKV_Balanced    | 0.5    | 8.1721  | 2.71  | 0.1997 | 0.3999 | 0.4004 |
| SpectrumKV_Balanced    | 0.55   | 8.1110  | 1.94  | 0.2397 | 0.4800 | 0.2803 |
| SpectrumKV_Balanced    | 0.6    | 8.0839  | 1.60  | 0.2798 | 0.5596 | 0.1606 |
| SpectrumKV_Balanced    | 0.65   | 7.9811  | 0.31  | 0.3198 | 0.6396 | 0.0405 |
| SpectrumKV_Balanced    | 0.7    | 7.9590  | 0.03  | 0.3999 | 0.5996 | 0.0005 |
| SpectrumKV_Balanced    | 0.75   | 7.9583  | 0.02  | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.25   | 8.4243  | 5.88  | 0.0020 | 0.0000 | 0.9980 |
| SpectrumKV_SinkProtect | 0.3    | 8.3770  | 5.28  | 0.0020 | 0.1978 | 0.8003 |
| SpectrumKV_SinkProtect | 0.35   | 8.2973  | 4.28  | 0.0020 | 0.3979 | 0.6001 |
| SpectrumKV_SinkProtect | 0.4    | 8.1586  | 2.54  | 0.0020 | 0.5977 | 0.4004 |
| SpectrumKV_SinkProtect | 0.45   | 8.0679  | 1.40  | 0.0020 | 0.7979 | 0.2002 |
| SpectrumKV_SinkProtect | 0.5    | 7.9492  | -0.10 | 0.0020 | 0.9980 | 0.0000 |

|                        |      |        |       |        |        |        |
|------------------------|------|--------|-------|--------|--------|--------|
| SpectrumKV_SinkProtect | 0.55 | 7.9552 | -0.02 | 0.1001 | 0.8999 | 0.0000 |
| SpectrumKV_SinkProtect | 0.6  | 7.9628 | 0.08  | 0.2002 | 0.7998 | 0.0000 |
| SpectrumKV_SinkProtect | 0.65 | 7.9569 | 0.00  | 0.2998 | 0.7002 | 0.0000 |
| SpectrumKV_SinkProtect | 0.7  | 7.9591 | 0.03  | 0.3999 | 0.6001 | 0.0000 |
| SpectrumKV_SinkProtect | 0.75 | 7.9583 | 0.02  | 0.5000 | 0.5000 | 0.0000 |
| Random_Tier            | 0.25 | 8.4840 | 6.63  | 0.0000 | 0.0000 | 1.0000 |
| Random_Tier            | 0.3  | 8.3475 | 4.91  | 0.0000 | 0.1997 | 0.8003 |
| Random_Tier            | 0.35 | 8.2009 | 3.07  | 0.0000 | 0.3999 | 0.6001 |
| Random_Tier            | 0.4  | 8.1544 | 2.48  | 0.0000 | 0.5996 | 0.4004 |
| Random_Tier            | 0.45 | 8.0618 | 1.32  | 0.0000 | 0.7998 | 0.2002 |
| Random_Tier            | 0.5  | 7.9517 | -0.06 | 0.0000 | 1.0000 | 0.0000 |
| Random_Tier            | 0.55 | 7.9490 | -0.10 | 0.1001 | 0.8999 | 0.0000 |
| Random_Tier            | 0.6  | 7.9475 | -0.12 | 0.2002 | 0.7998 | 0.0000 |
| Random_Tier            | 0.65 | 7.9545 | -0.03 | 0.2998 | 0.7002 | 0.0000 |
| Random_Tier            | 0.7  | 7.9533 | -0.04 | 0.3999 | 0.6001 | 0.0000 |
| Random_Tier            | 0.75 | 7.9522 | -0.06 | 0.5000 | 0.5000 | 0.0000 |

Mistral-7B seq\_len=4096

| 方法                  | Budget | PPL    | Δ%    | FP16%  | INT8%  | INT4%  |
|---------------------|--------|--------|-------|--------|--------|--------|
| Full_FP16           | 1.0    | 6.2379 | 0.00  | 1.0000 | 0.0000 | 0.0000 |
| Uniform_INT8        | 0.25   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.3    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.35   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.4    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.45   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.5    | 6.2301 | -0.13 | 0.0000 | 1.0000 | 0.0000 |
| Uniform_INT8        | 0.55   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.6    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.65   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.7    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.75   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.25   | 6.7814 | 8.71  | 0.0000 | 0.0000 | 1.0000 |
| Uniform_INT4        | 0.3    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.35   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.4    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.45   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.5    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.55   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.6    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.65   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.7    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.75   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| PDTrim              | 0.25   | 8.5290 | 36.73 | 0.2500 | 0.0000 | 0.7500 |
| PDTrim              | 0.3    | 8.2656 | 32.50 | 0.2998 | 0.0000 | 0.7002 |
| PDTrim              | 0.35   | 7.7398 | 24.08 | 0.3499 | 0.0000 | 0.6501 |
| PDTrim              | 0.4    | 7.4996 | 20.23 | 0.3999 | 0.0000 | 0.6001 |
| PDTrim              | 0.45   | 7.3080 | 17.15 | 0.4500 | 0.0000 | 0.5500 |
| PDTrim              | 0.5    | 7.0919 | 13.69 | 0.5000 | 0.0000 | 0.5000 |
| PDTrim              | 0.55   | 7.1581 | 14.75 | 0.5498 | 0.0000 | 0.4502 |
| PDTrim              | 0.6    | 7.1313 | 14.32 | 0.5999 | 0.0000 | 0.4001 |
| PDTrim              | 0.65   | 6.9969 | 12.17 | 0.6499 | 0.0000 | 0.3501 |
| PDTrim              | 0.7    | 6.8358 | 9.58  | 0.7000 | 0.0000 | 0.3000 |
| PDTrim              | 0.75   | 6.7689 | 8.51  | 0.7500 | 0.0000 | 0.2500 |
| SWS_Original        | 0.25   | 8.8034 | 41.13 | 0.2500 | 0.0000 | 0.7500 |
| SWS_Original        | 0.3    | 8.5382 | 36.87 | 0.2998 | 0.0000 | 0.7002 |
| SWS_Original        | 0.35   | 8.2464 | 32.20 | 0.3499 | 0.0000 | 0.6501 |
| SWS_Original        | 0.4    | 7.9447 | 27.36 | 0.3999 | 0.0000 | 0.6001 |
| SWS_Original        | 0.45   | 7.6231 | 22.21 | 0.4500 | 0.0000 | 0.5500 |
| SWS_Original        | 0.5    | 7.2948 | 16.94 | 0.5000 | 0.0000 | 0.5000 |
| SWS_Original        | 0.55   | 7.0011 | 12.23 | 0.5498 | 0.0000 | 0.4502 |
| SWS_Original        | 0.6    | 6.8130 | 9.22  | 0.5999 | 0.0000 | 0.4001 |
| SWS_Original        | 0.65   | 6.7948 | 8.93  | 0.6499 | 0.0000 | 0.3501 |
| SWS_Original        | 0.7    | 6.8209 | 9.34  | 0.7000 | 0.0000 | 0.3000 |
| SWS_Original        | 0.75   | 6.6927 | 7.29  | 0.7500 | 0.0000 | 0.2500 |
| SWS_ValueAware      | 0.25   | 8.8034 | 41.13 | 0.2500 | 0.0000 | 0.7500 |
| SWS_ValueAware      | 0.3    | 8.5382 | 36.87 | 0.2998 | 0.0000 | 0.7002 |
| SWS_ValueAware      | 0.35   | 8.2464 | 32.20 | 0.3499 | 0.0000 | 0.6501 |
| SWS_ValueAware      | 0.4    | 7.9447 | 27.36 | 0.3999 | 0.0000 | 0.6001 |
| SWS_ValueAware      | 0.45   | 7.6231 | 22.21 | 0.4500 | 0.0000 | 0.5500 |
| SWS_ValueAware      | 0.5    | 7.2948 | 16.94 | 0.5000 | 0.0000 | 0.5000 |
| SWS_ValueAware      | 0.55   | 7.0011 | 12.23 | 0.5498 | 0.0000 | 0.4502 |
| SWS_ValueAware      | 0.6    | 6.8130 | 9.22  | 0.5999 | 0.0000 | 0.4001 |
| SWS_ValueAware      | 0.65   | 6.7948 | 8.93  | 0.6499 | 0.0000 | 0.3501 |
| SWS_ValueAware      | 0.7    | 6.8209 | 9.34  | 0.7000 | 0.0000 | 0.3000 |
| SWS_ValueAware      | 0.75   | 6.6927 | 7.29  | 0.7500 | 0.0000 | 0.2500 |
| SpectrumKV_Greedy   | 0.25   | 6.7814 | 8.71  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Greedy   | 0.3    | 6.6707 | 6.94  | 0.0000 | 0.2000 | 0.8000 |
| SpectrumKV_Greedy   | 0.35   | 6.5146 | 4.44  | 0.0000 | 0.3999 | 0.6001 |
| SpectrumKV_Greedy   | 0.4    | 6.4100 | 2.76  | 0.0000 | 0.5999 | 0.4001 |
| SpectrumKV_Greedy   | 0.45   | 6.3147 | 1.23  | 0.0000 | 0.7998 | 0.2002 |
| SpectrumKV_Greedy   | 0.5    | 6.2301 | -0.13 | 0.0000 | 1.0000 | 0.0000 |
| SpectrumKV_Greedy   | 0.55   | 6.2401 | 0.03  | 0.1001 | 0.8999 | 0.0000 |
| SpectrumKV_Greedy   | 0.6    | 6.2419 | 0.06  | 0.2000 | 0.8000 | 0.0000 |
| SpectrumKV_Greedy   | 0.65   | 6.2412 | 0.05  | 0.3000 | 0.7000 | 0.0000 |
| SpectrumKV_Greedy   | 0.7    | 6.2396 | 0.03  | 0.3999 | 0.6001 | 0.0000 |
| SpectrumKV_Greedy   | 0.75   | 6.2421 | 0.07  | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_Balanced | 0.25   | 6.7814 | 8.71  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Balanced | 0.3    | 6.7072 | 7.52  | 0.0398 | 0.0798 | 0.8804 |
| SpectrumKV_Balanced | 0.35   | 6.6796 | 7.08  | 0.0798 | 0.1599 | 0.7603 |
| SpectrumKV_Balanced | 0.4    | 6.5247 | 4.60  | 0.1199 | 0.2400 | 0.6401 |
| SpectrumKV_Balanced | 0.45   | 6.4734 | 3.77  | 0.1599 | 0.3198 | 0.5203 |
| SpectrumKV_Balanced | 0.5    | 6.4118 | 2.79  | 0.2000 | 0.3999 | 0.4001 |
| SpectrumKV_Balanced | 0.55   | 6.3605 | 1.97  | 0.2400 | 0.4800 | 0.2800 |
| SpectrumKV_Balanced | 0.6    | 6.3115 | 1.18  | 0.2798 | 0.5598 | 0.1604 |
| SpectrumKV_Balanced | 0.65   | 6.2646 | 0.43  | 0.3198 | 0.6399 | 0.0403 |
| SpectrumKV_Balanced | 0.7    | 6.2398 | 0.03  | 0.3999 | 0.5999 | 0.0002 |
| SpectrumKV_Balanced | 0.75   | 6.2421 | 0.07  | 0.5000 | 0.5000 | 0.0000 |

|                        |      |        |       |        |        |        |
|------------------------|------|--------|-------|--------|--------|--------|
| SpectrumKV_SinkProtect | 0.25 | 6.8073 | 9.13  | 0.0010 | 0.0000 | 0.9990 |
| SpectrumKV_SinkProtect | 0.3  | 6.6879 | 7.21  | 0.0010 | 0.1990 | 0.8000 |
| SpectrumKV_SinkProtect | 0.35 | 6.5125 | 4.40  | 0.0010 | 0.3989 | 0.6001 |
| SpectrumKV_SinkProtect | 0.4  | 6.4007 | 2.61  | 0.0010 | 0.5989 | 0.4001 |
| SpectrumKV_SinkProtect | 0.45 | 6.3203 | 1.32  | 0.0010 | 0.7988 | 0.2002 |
| SpectrumKV_SinkProtect | 0.5  | 6.2397 | 0.03  | 0.0010 | 0.9990 | 0.0000 |
| SpectrumKV_SinkProtect | 0.55 | 6.2401 | 0.03  | 0.1001 | 0.8999 | 0.0000 |
| SpectrumKV_SinkProtect | 0.6  | 6.2419 | 0.06  | 0.2000 | 0.8000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.65 | 6.2412 | 0.05  | 0.3000 | 0.7000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.7  | 6.2396 | 0.03  | 0.3999 | 0.6001 | 0.0000 |
| SpectrumKV_SinkProtect | 0.75 | 6.2421 | 0.07  | 0.5000 | 0.5000 | 0.0000 |
| Random_Tier            | 0.25 | 6.7814 | 8.71  | 0.0000 | 0.0000 | 1.0000 |
| Random_Tier            | 0.3  | 6.6692 | 6.91  | 0.0000 | 0.2000 | 0.8000 |
| Random_Tier            | 0.35 | 6.5457 | 4.93  | 0.0000 | 0.3999 | 0.6001 |
| Random_Tier            | 0.4  | 6.4553 | 3.48  | 0.0000 | 0.5999 | 0.4001 |
| Random_Tier            | 0.45 | 6.3180 | 1.28  | 0.0000 | 0.7998 | 0.2002 |
| Random_Tier            | 0.5  | 6.2301 | -0.13 | 0.0000 | 1.0000 | 0.0000 |
| Random_Tier            | 0.55 | 6.2315 | -0.10 | 0.1001 | 0.8999 | 0.0000 |
| Random_Tier            | 0.6  | 6.2338 | -0.07 | 0.2000 | 0.8000 | 0.0000 |
| Random_Tier            | 0.65 | 6.2414 | 0.06  | 0.3000 | 0.7000 | 0.0000 |
| Random_Tier            | 0.7  | 6.2376 | -0.00 | 0.3999 | 0.6001 | 0.0000 |
| Random_Tier            | 0.75 | 6.2386 | 0.01  | 0.5000 | 0.5000 | 0.0000 |

Mistral-7B seq\_len=8192

| 方法                  | Budget | PPL    | Δ%    | FP16%  | INT8%  | INT4%  |
|---------------------|--------|--------|-------|--------|--------|--------|
| Full_FP16           | 1.0    | 6.4185 | 0.00  | 1.0000 | 0.0000 | 0.0000 |
| Uniform_INT8        | 0.25   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.3    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.35   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.4    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.45   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.5    | 6.3998 | -0.29 | 0.0000 | 1.0000 | 0.0000 |
| Uniform_INT8        | 0.55   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.6    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.65   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.7    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8        | 0.75   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.25   | 6.9655 | 8.52  | 0.0000 | 0.0000 | 1.0000 |
| Uniform_INT4        | 0.3    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.35   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.4    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.45   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.5    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.55   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.6    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.65   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.7    | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4        | 0.75   | 未测试    | 未测试   | 未测试    | 未测试    | 未测试    |
| PDTrim              | 0.25   | 7.5371 | 17.43 | 0.2500 | 0.0000 | 0.7500 |
| PDTrim              | 0.3    | 7.3525 | 14.55 | 0.2999 | 0.0000 | 0.7001 |
| PDTrim              | 0.35   | 7.0620 | 10.03 | 0.3500 | 0.0000 | 0.6500 |
| PDTrim              | 0.4    | 6.9723 | 8.63  | 0.3999 | 0.0000 | 0.6001 |
| PDTrim              | 0.45   | 7.0515 | 9.86  | 0.4500 | 0.0000 | 0.5500 |
| PDTrim              | 0.5    | 6.9286 | 7.95  | 0.5000 | 0.0000 | 0.5000 |
| PDTrim              | 0.55   | 6.9824 | 8.79  | 0.5499 | 0.0000 | 0.4501 |
| PDTrim              | 0.6    | 6.9687 | 8.57  | 0.6000 | 0.0000 | 0.4000 |
| PDTrim              | 0.65   | 6.9864 | 8.85  | 0.6499 | 0.0000 | 0.3501 |
| PDTrim              | 0.7    | 6.8336 | 6.47  | 0.7000 | 0.0000 | 0.3000 |
| PDTrim              | 0.75   | 6.8321 | 6.44  | 0.7500 | 0.0000 | 0.2500 |
| SWS_Original        | 0.25   | 7.9990 | 24.63 | 0.2500 | 0.0000 | 0.7500 |
| SWS_Original        | 0.3    | 7.6880 | 19.78 | 0.2999 | 0.0000 | 0.7001 |
| SWS_Original        | 0.35   | 7.2063 | 12.28 | 0.3500 | 0.0000 | 0.6500 |
| SWS_Original        | 0.4    | 6.8717 | 7.06  | 0.3999 | 0.0000 | 0.6001 |
| SWS_Original        | 0.45   | 6.9728 | 8.64  | 0.4500 | 0.0000 | 0.5500 |
| SWS_Original        | 0.5    | 6.9818 | 8.78  | 0.5000 | 0.0000 | 0.5000 |
| SWS_Original        | 0.55   | 6.8441 | 6.63  | 0.5499 | 0.0000 | 0.4501 |
| SWS_Original        | 0.6    | 6.9220 | 7.84  | 0.6000 | 0.0000 | 0.4000 |
| SWS_Original        | 0.65   | 6.9134 | 7.71  | 0.6499 | 0.0000 | 0.3501 |
| SWS_Original        | 0.7    | 6.9359 | 8.06  | 0.7000 | 0.0000 | 0.3000 |
| SWS_Original        | 0.75   | 6.9169 | 7.77  | 0.7500 | 0.0000 | 0.2500 |
| SWS_ValueAware      | 0.25   | 7.9990 | 24.63 | 0.2500 | 0.0000 | 0.7500 |
| SWS_ValueAware      | 0.3    | 7.6880 | 19.78 | 0.2999 | 0.0000 | 0.7001 |
| SWS_ValueAware      | 0.35   | 7.2063 | 12.28 | 0.3500 | 0.0000 | 0.6500 |
| SWS_ValueAware      | 0.4    | 6.8717 | 7.06  | 0.3999 | 0.0000 | 0.6001 |
| SWS_ValueAware      | 0.45   | 6.9728 | 8.64  | 0.4500 | 0.0000 | 0.5500 |
| SWS_ValueAware      | 0.5    | 6.9818 | 8.78  | 0.5000 | 0.0000 | 0.5000 |
| SWS_ValueAware      | 0.55   | 6.8441 | 6.63  | 0.5499 | 0.0000 | 0.4501 |
| SWS_ValueAware      | 0.6    | 6.9220 | 7.84  | 0.6000 | 0.0000 | 0.4000 |
| SWS_ValueAware      | 0.65   | 6.9134 | 7.71  | 0.6499 | 0.0000 | 0.3501 |
| SWS_ValueAware      | 0.7    | 6.9359 | 8.06  | 0.7000 | 0.0000 | 0.3000 |
| SWS_ValueAware      | 0.75   | 6.9169 | 7.77  | 0.7500 | 0.0000 | 0.2500 |
| SpectrumKV_Greedy   | 0.25   | 6.9655 | 8.52  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Greedy   | 0.3    | 6.8406 | 6.58  | 0.0000 | 0.2000 | 0.8000 |
| SpectrumKV_Greedy   | 0.35   | 6.6840 | 4.14  | 0.0000 | 0.3999 | 0.6001 |
| SpectrumKV_Greedy   | 0.4    | 6.5973 | 2.79  | 0.0000 | 0.6000 | 0.4000 |
| SpectrumKV_Greedy   | 0.45   | 6.4835 | 1.01  | 0.0000 | 0.7999 | 0.2001 |
| SpectrumKV_Greedy   | 0.5    | 6.3998 | -0.29 | 0.0000 | 1.0000 | 0.0000 |
| SpectrumKV_Greedy   | 0.55   | 6.4174 | -0.02 | 0.1000 | 0.9000 | 0.0000 |
| SpectrumKV_Greedy   | 0.6    | 6.4182 | -0.00 | 0.2000 | 0.8000 | 0.0000 |
| SpectrumKV_Greedy   | 0.65   | 6.4182 | -0.00 | 0.3000 | 0.7000 | 0.0000 |
| SpectrumKV_Greedy   | 0.7    | 6.4177 | -0.01 | 0.4000 | 0.6000 | 0.0000 |
| SpectrumKV_Greedy   | 0.75   | 6.4184 | -0.00 | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_Balanced | 0.25   | 6.9655 | 8.52  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Balanced | 0.3    | 6.8794 | 7.18  | 0.0399 | 0.0800 | 0.8801 |
| SpectrumKV_Balanced | 0.35   | 6.8217 | 6.28  | 0.0800 | 0.1599 | 0.7601 |
| SpectrumKV_Balanced | 0.4    | 6.7361 | 4.95  | 0.1200 | 0.2400 | 0.6400 |
| SpectrumKV_Balanced | 0.45   | 6.6654 | 3.85  | 0.1599 | 0.3199 | 0.5201 |

|                        |      |        |       |        |        |        |
|------------------------|------|--------|-------|--------|--------|--------|
| SpectrumKV_Balanced    | 0.5  | 6.6014 | 2.85  | 0.2000 | 0.3999 | 0.4001 |
| SpectrumKV_Balanced    | 0.55 | 6.5439 | 1.95  | 0.2400 | 0.4800 | 0.2800 |
| SpectrumKV_Balanced    | 0.6  | 6.4732 | 0.85  | 0.2799 | 0.5599 | 0.1602 |
| SpectrumKV_Balanced    | 0.65 | 6.4355 | 0.26  | 0.3199 | 0.6399 | 0.0402 |
| SpectrumKV_Balanced    | 0.7  | 6.4176 | -0.01 | 0.3999 | 0.6000 | 0.0001 |
| SpectrumKV_Balanced    | 0.75 | 6.4184 | -0.00 | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.25 | 7.0045 | 9.13  | 0.0005 | 0.0000 | 0.9995 |
| SpectrumKV_SinkProtect | 0.3  | 6.8573 | 6.84  | 0.0005 | 0.1995 | 0.8000 |
| SpectrumKV_SinkProtect | 0.35 | 6.7038 | 4.45  | 0.0005 | 0.3994 | 0.6001 |
| SpectrumKV_SinkProtect | 0.4  | 6.6014 | 2.85  | 0.0005 | 0.5995 | 0.4000 |
| SpectrumKV_SinkProtect | 0.45 | 6.4902 | 1.12  | 0.0005 | 0.7994 | 0.2001 |
| SpectrumKV_SinkProtect | 0.5  | 6.4136 | -0.08 | 0.0005 | 0.9995 | 0.0000 |
| SpectrumKV_SinkProtect | 0.55 | 6.4174 | -0.02 | 0.1000 | 0.9000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.6  | 6.4182 | -0.00 | 0.2000 | 0.8000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.65 | 6.4182 | -0.00 | 0.3000 | 0.7000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.7  | 6.4177 | -0.01 | 0.4000 | 0.6000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.75 | 6.4184 | -0.00 | 0.5000 | 0.5000 | 0.0000 |
| Random_Tier            | 0.25 | 6.9655 | 8.52  | 0.0000 | 0.0000 | 1.0000 |
| Random_Tier            | 0.3  | 6.8377 | 6.53  | 0.0000 | 0.2000 | 0.8000 |
| Random_Tier            | 0.35 | 6.7337 | 4.91  | 0.0000 | 0.3999 | 0.6001 |
| Random_Tier            | 0.4  | 6.6330 | 3.34  | 0.0000 | 0.6000 | 0.4000 |
| Random_Tier            | 0.45 | 6.5206 | 1.59  | 0.0000 | 0.7999 | 0.2001 |
| Random_Tier            | 0.5  | 6.3998 | -0.29 | 0.0000 | 1.0000 | 0.0000 |
| Random_Tier            | 0.55 | 6.3972 | -0.33 | 0.1000 | 0.9000 | 0.0000 |
| Random_Tier            | 0.6  | 6.3953 | -0.36 | 0.2000 | 0.8000 | 0.0000 |
| Random_Tier            | 0.65 | 6.4200 | 0.02  | 0.3000 | 0.7000 | 0.0000 |
| Random_Tier            | 0.7  | 6.4212 | 0.04  | 0.4000 | 0.6000 | 0.0000 |
| Random_Tier            | 0.75 | 6.4196 | 0.02  | 0.5000 | 0.5000 | 0.0000 |

Gemma-9B seq\_len=2048

| 方法                | Budget | PPL     | Δ%     | FP16%  | INT8%  | INT4%  |
|-------------------|--------|---------|--------|--------|--------|--------|
| Full_FP16         | 1.0    | 11.2042 | 0.00   | 1.0000 | 0.0000 | 0.0000 |
| Uniform_INT8      | 0.25   | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.3    | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.35   | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.4    | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.45   | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.5    | 11.1548 | -0.44  | 0.0000 | 1.0000 | 0.0000 |
| Uniform_INT8      | 0.55   | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.6    | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.65   | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.7    | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.75   | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.25   | 11.7357 | 4.74   | 0.0000 | 0.0000 | 1.0000 |
| Uniform_INT4      | 0.3    | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.35   | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.4    | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.45   | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.5    | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.55   | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.6    | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.65   | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.7    | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.75   | 未测试     | 未测试    | 未测试    | 未测试    | 未测试    |
| PDTrim            | 0.25   | 22.2431 | 98.52  | 0.2500 | 0.0000 | 0.7500 |
| PDTrim            | 0.3    | 20.3624 | 81.74  | 0.2998 | 0.0000 | 0.7002 |
| PDTrim            | 0.35   | 17.2144 | 53.64  | 0.3496 | 0.0000 | 0.6504 |
| PDTrim            | 0.4    | 16.7967 | 49.91  | 0.3999 | 0.0000 | 0.6001 |
| PDTrim            | 0.45   | 15.1745 | 35.44  | 0.4497 | 0.0000 | 0.5503 |
| PDTrim            | 0.5    | 15.1960 | 35.63  | 0.5000 | 0.0000 | 0.5000 |
| PDTrim            | 0.55   | 14.3719 | 28.27  | 0.5498 | 0.0000 | 0.4502 |
| PDTrim            | 0.6    | 14.3937 | 28.47  | 0.5996 | 0.0000 | 0.4004 |
| PDTrim            | 0.65   | 13.6415 | 21.75  | 0.6499 | 0.0000 | 0.3501 |
| PDTrim            | 0.7    | 12.9193 | 15.31  | 0.6997 | 0.0000 | 0.3003 |
| PDTrim            | 0.75   | 13.1871 | 17.70  | 0.7500 | 0.0000 | 0.2500 |
| SWS_Original      | 0.25   | 26.2737 | 134.50 | 0.2500 | 0.0000 | 0.7500 |
| SWS_Original      | 0.3    | 22.5515 | 101.28 | 0.2998 | 0.0000 | 0.7002 |
| SWS_Original      | 0.35   | 21.0771 | 88.12  | 0.3496 | 0.0000 | 0.6504 |
| SWS_Original      | 0.4    | 19.0884 | 70.37  | 0.3999 | 0.0000 | 0.6001 |
| SWS_Original      | 0.45   | 17.4821 | 56.03  | 0.4497 | 0.0000 | 0.5503 |
| SWS_Original      | 0.5    | 16.1092 | 43.78  | 0.5000 | 0.0000 | 0.5000 |
| SWS_Original      | 0.55   | 15.3281 | 36.81  | 0.5498 | 0.0000 | 0.4502 |
| SWS_Original      | 0.6    | 14.8839 | 32.84  | 0.5996 | 0.0000 | 0.4004 |
| SWS_Original      | 0.65   | 14.5101 | 29.51  | 0.6499 | 0.0000 | 0.3501 |
| SWS_Original      | 0.7    | 13.7234 | 22.48  | 0.6997 | 0.0000 | 0.3003 |
| SWS_Original      | 0.75   | 13.1751 | 17.59  | 0.7500 | 0.0000 | 0.2500 |
| SWS_ValueAware    | 0.25   | 26.2613 | 134.39 | 0.2500 | 0.0000 | 0.7500 |
| SWS_ValueAware    | 0.3    | 22.5515 | 101.28 | 0.2998 | 0.0000 | 0.7002 |
| SWS_ValueAware    | 0.35   | 21.0771 | 88.12  | 0.3496 | 0.0000 | 0.6504 |
| SWS_ValueAware    | 0.4    | 19.0884 | 70.37  | 0.3999 | 0.0000 | 0.6001 |
| SWS_ValueAware    | 0.45   | 17.4821 | 56.03  | 0.4497 | 0.0000 | 0.5503 |
| SWS_ValueAware    | 0.5    | 16.1092 | 43.78  | 0.5000 | 0.0000 | 0.5000 |
| SWS_ValueAware    | 0.55   | 15.3281 | 36.81  | 0.5498 | 0.0000 | 0.4502 |
| SWS_ValueAware    | 0.6    | 14.8839 | 32.84  | 0.5996 | 0.0000 | 0.4004 |
| SWS_ValueAware    | 0.65   | 14.5101 | 29.51  | 0.6499 | 0.0000 | 0.3501 |
| SWS_ValueAware    | 0.7    | 13.7234 | 22.48  | 0.6997 | 0.0000 | 0.3003 |
| SWS_ValueAware    | 0.75   | 13.1751 | 17.59  | 0.7500 | 0.0000 | 0.2500 |
| SpectrumKV_Greedy | 0.25   | 11.7357 | 4.74   | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Greedy | 0.3    | 11.4788 | 2.45   | 0.0000 | 0.1997 | 0.8003 |
| SpectrumKV_Greedy | 0.35   | 11.3758 | 1.53   | 0.0000 | 0.3999 | 0.6001 |
| SpectrumKV_Greedy | 0.4    | 11.2622 | 0.52   | 0.0000 | 0.5996 | 0.4004 |
| SpectrumKV_Greedy | 0.45   | 11.2121 | 0.07   | 0.0000 | 0.7998 | 0.2002 |
| SpectrumKV_Greedy | 0.5    | 11.1548 | -0.44  | 0.0000 | 1.0000 | 0.0000 |
| SpectrumKV_Greedy | 0.55   | 11.1913 | -0.12  | 0.1001 | 0.8999 | 0.0000 |
| SpectrumKV_Greedy | 0.6    | 11.2208 | 0.15   | 0.2002 | 0.7998 | 0.0000 |
| SpectrumKV_Greedy | 0.65   | 11.2002 | -0.04  | 0.2998 | 0.7002 | 0.0000 |
| SpectrumKV_Greedy | 0.7    | 11.2011 | -0.03  | 0.3999 | 0.6001 | 0.0000 |



|                        |      |         |       |        |        |        |
|------------------------|------|---------|-------|--------|--------|--------|
| SpectrumKV_Greedy      | 0.75 | 11.2088 | 0.04  | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_Balanced    | 0.25 | 11.7357 | 4.74  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Balanced    | 0.3  | 12.0381 | 7.44  | 0.0396 | 0.0796 | 0.8809 |
| SpectrumKV_Balanced    | 0.35 | 11.4151 | 1.88  | 0.0796 | 0.1597 | 0.7607 |
| SpectrumKV_Balanced    | 0.4  | 11.2765 | 0.65  | 0.1196 | 0.2397 | 0.6406 |
| SpectrumKV_Balanced    | 0.45 | 11.3556 | 1.35  | 0.1597 | 0.3198 | 0.5205 |
| SpectrumKV_Balanced    | 0.5  | 11.4623 | 2.30  | 0.1997 | 0.3999 | 0.4004 |
| SpectrumKV_Balanced    | 0.55 | 11.3010 | 0.86  | 0.2397 | 0.4800 | 0.2803 |
| SpectrumKV_Balanced    | 0.6  | 11.2503 | 0.41  | 0.2798 | 0.5596 | 0.1606 |
| SpectrumKV_Balanced    | 0.65 | 11.2134 | 0.08  | 0.3198 | 0.6396 | 0.0405 |
| SpectrumKV_Balanced    | 0.7  | 11.2041 | -0.00 | 0.3999 | 0.5996 | 0.0005 |
| SpectrumKV_Balanced    | 0.75 | 11.2088 | 0.04  | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.25 | 11.6218 | 3.73  | 0.0020 | 0.0000 | 0.9980 |
| SpectrumKV_SinkProtect | 0.3  | 11.4917 | 2.57  | 0.0020 | 0.1978 | 0.8003 |
| SpectrumKV_SinkProtect | 0.35 | 11.2199 | 0.14  | 0.0020 | 0.3979 | 0.6001 |
| SpectrumKV_SinkProtect | 0.4  | 11.3846 | 1.61  | 0.0020 | 0.5977 | 0.4004 |
| SpectrumKV_SinkProtect | 0.45 | 11.2122 | 0.07  | 0.0020 | 0.7979 | 0.2002 |
| SpectrumKV_SinkProtect | 0.5  | 11.1514 | -0.47 | 0.0020 | 0.9980 | 0.0000 |
| SpectrumKV_SinkProtect | 0.55 | 11.1913 | -0.12 | 0.1001 | 0.8999 | 0.0000 |
| SpectrumKV_SinkProtect | 0.6  | 11.2208 | 0.15  | 0.2002 | 0.7998 | 0.0000 |
| SpectrumKV_SinkProtect | 0.65 | 11.2002 | -0.04 | 0.2998 | 0.7002 | 0.0000 |
| SpectrumKV_SinkProtect | 0.7  | 11.2011 | -0.03 | 0.3999 | 0.6001 | 0.0000 |
| SpectrumKV_SinkProtect | 0.75 | 11.2088 | 0.04  | 0.5000 | 0.5000 | 0.0000 |
| Random_Tier            | 0.25 | 11.7357 | 4.74  | 0.0000 | 0.0000 | 1.0000 |
| Random_Tier            | 0.3  | 11.4871 | 2.52  | 0.0000 | 0.1997 | 0.8003 |
| Random_Tier            | 0.35 | 11.4138 | 1.87  | 0.0000 | 0.3999 | 0.6001 |
| Random_Tier            | 0.4  | 11.3077 | 0.92  | 0.0000 | 0.5996 | 0.4004 |
| Random_Tier            | 0.45 | 11.2657 | 0.55  | 0.0000 | 0.7998 | 0.2002 |
| Random_Tier            | 0.5  | 11.1548 | -0.44 | 0.0000 | 1.0000 | 0.0000 |
| Random_Tier            | 0.55 | 11.1201 | -0.75 | 0.1001 | 0.8999 | 0.0000 |
| Random_Tier            | 0.6  | 11.2236 | 0.17  | 0.2002 | 0.7998 | 0.0000 |
| Random_Tier            | 0.65 | 11.1951 | -0.08 | 0.2998 | 0.7002 | 0.0000 |
| Random_Tier            | 0.7  | 11.2033 | -0.01 | 0.3999 | 0.6001 | 0.0000 |
| Random_Tier            | 0.75 | 11.2039 | -0.00 | 0.5000 | 0.5000 | 0.0000 |

Gemma-9B seq\_len=4096

| 方法                | Budget | PPL     | Δ%    | FP16%  | INT8%  | INT4%  |
|-------------------|--------|---------|-------|--------|--------|--------|
| Full_FP16         | 1.0    | 7.9672  | 0.00  | 1.0000 | 0.0000 | 0.0000 |
| Uniform_INT8      | 0.25   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.3    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.35   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.4    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.45   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.5    | 7.9781  | 0.14  | 0.0000 | 1.0000 | 0.0000 |
| Uniform_INT8      | 0.55   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.6    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.65   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.7    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT8      | 0.75   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.25   | 8.2133  | 3.09  | 0.0000 | 0.0000 | 1.0000 |
| Uniform_INT4      | 0.3    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.35   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.4    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.45   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.5    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.55   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.6    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.65   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.7    | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| Uniform_INT4      | 0.75   | 未测试     | 未测试   | 未测试    | 未测试    | 未测试    |
| PDTrim            | 0.25   | 11.2550 | 41.27 | 0.2500 | 0.0000 | 0.7500 |
| PDTrim            | 0.3    | 10.4320 | 30.94 | 0.2998 | 0.0000 | 0.7002 |
| PDTrim            | 0.35   | 9.1562  | 14.92 | 0.3499 | 0.0000 | 0.6501 |
| PDTrim            | 0.4    | 8.6740  | 8.87  | 0.3999 | 0.0000 | 0.6001 |
| PDTrim            | 0.45   | 8.7232  | 9.49  | 0.4500 | 0.0000 | 0.5500 |
| PDTrim            | 0.5    | 8.7088  | 9.31  | 0.5000 | 0.0000 | 0.5000 |
| PDTrim            | 0.55   | 8.4783  | 6.41  | 0.5498 | 0.0000 | 0.4502 |
| PDTrim            | 0.6    | 8.7604  | 9.96  | 0.5999 | 0.0000 | 0.4001 |
| PDTrim            | 0.65   | 8.5612  | 7.46  | 0.6499 | 0.0000 | 0.3501 |
| PDTrim            | 0.7    | 8.7197  | 9.45  | 0.7000 | 0.0000 | 0.3000 |
| PDTrim            | 0.75   | 8.5129  | 6.85  | 0.7500 | 0.0000 | 0.2500 |
| SWS_Original      | 0.25   | 11.2239 | 40.88 | 0.2500 | 0.0000 | 0.7500 |
| SWS_Original      | 0.3    | 10.5287 | 32.15 | 0.2998 | 0.0000 | 0.7002 |
| SWS_Original      | 0.35   | 10.1410 | 27.28 | 0.3499 | 0.0000 | 0.6501 |
| SWS_Original      | 0.4    | 9.7578  | 22.47 | 0.3999 | 0.0000 | 0.6001 |
| SWS_Original      | 0.45   | 9.4327  | 18.39 | 0.4500 | 0.0000 | 0.5500 |
| SWS_Original      | 0.5    | 9.1500  | 14.85 | 0.5000 | 0.0000 | 0.5000 |
| SWS_Original      | 0.55   | 8.8388  | 10.94 | 0.5498 | 0.0000 | 0.4502 |
| SWS_Original      | 0.6    | 8.7429  | 9.74  | 0.5999 | 0.0000 | 0.4001 |
| SWS_Original      | 0.65   | 8.8862  | 11.54 | 0.6499 | 0.0000 | 0.3501 |
| SWS_Original      | 0.7    | 8.9376  | 12.18 | 0.7000 | 0.0000 | 0.3000 |
| SWS_Original      | 0.75   | 8.7537  | 9.87  | 0.7500 | 0.0000 | 0.2500 |
| SWS_ValueAware    | 0.25   | 11.2239 | 40.88 | 0.2500 | 0.0000 | 0.7500 |
| SWS_ValueAware    | 0.3    | 10.5287 | 32.15 | 0.2998 | 0.0000 | 0.7002 |
| SWS_ValueAware    | 0.35   | 10.1410 | 27.28 | 0.3499 | 0.0000 | 0.6501 |
| SWS_ValueAware    | 0.4    | 9.7578  | 22.47 | 0.3999 | 0.0000 | 0.6001 |
| SWS_ValueAware    | 0.45   | 9.4327  | 18.39 | 0.4500 | 0.0000 | 0.5500 |
| SWS_ValueAware    | 0.5    | 9.1500  | 14.85 | 0.5000 | 0.0000 | 0.5000 |
| SWS_ValueAware    | 0.55   | 8.8388  | 10.94 | 0.5498 | 0.0000 | 0.4502 |
| SWS_ValueAware    | 0.6    | 8.7429  | 9.74  | 0.5999 | 0.0000 | 0.4001 |
| SWS_ValueAware    | 0.65   | 8.8862  | 11.54 | 0.6499 | 0.0000 | 0.3501 |
| SWS_ValueAware    | 0.7    | 8.9376  | 12.18 | 0.7000 | 0.0000 | 0.3000 |
| SWS_ValueAware    | 0.75   | 8.7537  | 9.87  | 0.7500 | 0.0000 | 0.2500 |
| SpectrumKV_Greedy | 0.25   | 8.2133  | 3.09  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Greedy | 0.3    | 8.1852  | 2.74  | 0.0000 | 0.2000 | 0.8000 |
| SpectrumKV_Greedy | 0.35   | 8.1937  | 2.84  | 0.0000 | 0.3999 | 0.6001 |
| SpectrumKV_Greedy | 0.4    | 8.0919  | 1.56  | 0.0000 | 0.5999 | 0.4001 |

|                        |      |        |       |        |        |        |
|------------------------|------|--------|-------|--------|--------|--------|
| SpectrumKV_Greedy      | 0.45 | 8.0243 | 0.72  | 0.0000 | 0.7998 | 0.2002 |
| SpectrumKV_Greedy      | 0.5  | 7.9781 | 0.14  | 0.0000 | 1.0000 | 0.0000 |
| SpectrumKV_Greedy      | 0.55 | 7.9653 | -0.02 | 0.1001 | 0.8999 | 0.0000 |
| SpectrumKV_Greedy      | 0.6  | 7.9606 | -0.08 | 0.2000 | 0.8000 | 0.0000 |
| SpectrumKV_Greedy      | 0.65 | 7.9699 | 0.03  | 0.3000 | 0.7000 | 0.0000 |
| SpectrumKV_Greedy      | 0.7  | 7.9608 | -0.08 | 0.3999 | 0.6001 | 0.0000 |
| SpectrumKV_Greedy      | 0.75 | 7.9675 | 0.00  | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_Balanced    | 0.25 | 8.2133 | 3.09  | 0.0000 | 0.0000 | 1.0000 |
| SpectrumKV_Balanced    | 0.3  | 8.1619 | 2.44  | 0.0398 | 0.0798 | 0.8804 |
| SpectrumKV_Balanced    | 0.35 | 8.1139 | 1.84  | 0.0798 | 0.1599 | 0.7603 |
| SpectrumKV_Balanced    | 0.4  | 8.1607 | 2.43  | 0.1199 | 0.2400 | 0.6401 |
| SpectrumKV_Balanced    | 0.45 | 8.0530 | 1.08  | 0.1599 | 0.3198 | 0.5203 |
| SpectrumKV_Balanced    | 0.5  | 8.0649 | 1.23  | 0.2000 | 0.3999 | 0.4001 |
| SpectrumKV_Balanced    | 0.55 | 8.0290 | 0.78  | 0.2400 | 0.4800 | 0.2800 |
| SpectrumKV_Balanced    | 0.6  | 8.0003 | 0.42  | 0.2798 | 0.5598 | 0.1604 |
| SpectrumKV_Balanced    | 0.65 | 7.9664 | -0.01 | 0.3198 | 0.6399 | 0.0403 |
| SpectrumKV_Balanced    | 0.7  | 7.9611 | -0.08 | 0.3999 | 0.5999 | 0.0002 |
| SpectrumKV_Balanced    | 0.75 | 7.9675 | 0.00  | 0.5000 | 0.5000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.25 | 8.2419 | 3.45  | 0.0010 | 0.0000 | 0.9990 |
| SpectrumKV_SinkProtect | 0.3  | 8.1195 | 1.91  | 0.0010 | 0.1990 | 0.8000 |
| SpectrumKV_SinkProtect | 0.35 | 8.0832 | 1.46  | 0.0010 | 0.3989 | 0.6001 |
| SpectrumKV_SinkProtect | 0.4  | 8.0264 | 0.74  | 0.0010 | 0.5989 | 0.4001 |
| SpectrumKV_SinkProtect | 0.45 | 7.9835 | 0.20  | 0.0010 | 0.7988 | 0.2002 |
| SpectrumKV_SinkProtect | 0.5  | 7.9514 | -0.20 | 0.0010 | 0.9990 | 0.0000 |
| SpectrumKV_SinkProtect | 0.55 | 7.9653 | -0.02 | 0.1001 | 0.8999 | 0.0000 |
| SpectrumKV_SinkProtect | 0.6  | 7.9606 | -0.08 | 0.2000 | 0.8000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.65 | 7.9699 | 0.03  | 0.3000 | 0.7000 | 0.0000 |
| SpectrumKV_SinkProtect | 0.7  | 7.9608 | -0.08 | 0.3999 | 0.6001 | 0.0000 |
| SpectrumKV_SinkProtect | 0.75 | 7.9675 | 0.00  | 0.5000 | 0.5000 | 0.0000 |
| Random_Tier            | 0.25 | 8.2133 | 3.09  | 0.0000 | 0.0000 | 1.0000 |
| Random_Tier            | 0.3  | 8.2315 | 3.32  | 0.0000 | 0.2000 | 0.8000 |
| Random_Tier            | 0.35 | 8.0375 | 0.88  | 0.0000 | 0.3999 | 0.6001 |
| Random_Tier            | 0.4  | 8.1223 | 1.95  | 0.0000 | 0.5999 | 0.4001 |
| Random_Tier            | 0.45 | 8.0706 | 1.30  | 0.0000 | 0.7998 | 0.2002 |
| Random_Tier            | 0.5  | 7.9781 | 0.14  | 0.0000 | 1.0000 | 0.0000 |
| Random_Tier            | 0.55 | 7.9724 | 0.07  | 0.1001 | 0.8999 | 0.0000 |
| Random_Tier            | 0.6  | 7.9568 | -0.13 | 0.2000 | 0.8000 | 0.0000 |
| Random_Tier            | 0.65 | 7.9448 | -0.28 | 0.3000 | 0.7000 | 0.0000 |
| Random_Tier            | 0.7  | 7.9462 | -0.26 | 0.3999 | 0.6001 | 0.0000 |
| Random_Tier            | 0.75 | 7.9508 | -0.21 | 0.5000 | 0.5000 | 0.0000 |

RQ9: TTFT Latency Savings（首token延迟节省）

数据来源：experiment\_log\_v2.md（GPU端到端实测，非JSON文件）

| Model      | Seq | Full TTFT(ms) | b=0.5 TTFT(ms) | TTFT ↓ | Full TPS | b=0.5 TPS | TPSΔ |
|------------|-----|---------------|----------------|--------|----------|-----------|------|
| Qwen-7B    | 2K  | 391           | 154            | -61%   | 49.2     | 43.5      | -11% |
| Qwen-7B    | 4K  | 630           | 296            | -53%   | 42.4     | 43.2      | +2%  |
| Mistral-7B | 2K  | 274           | 116            | -58%   | 51.9     | 50.1      | -3%  |
| Mistral-7B | 4K  | 497           | 233            | -53%   | 47.1     | 49.1      | +4%  |
| Mistral-7B | 8K  | 1211          | 502            | -59%   | 44.1     | 47.0      | +7%  |
| Gemma-9B   | 2K  | 269           | 135            | -50%   | 38.2     | 38.9      | +2%  |
| Gemma-9B   | 4K  | 709           | 270            | -62%   | 28.8     | 38.2      | +33% |
| Gemma-9B   | 8K  | 1664          | 713            | -57%   | 27.3     | 28.8      | +5%  |

RQ10: Context Length Impact（上下文长度影响）

数据来源：exp1\_ppl\_qwen7b.json, exp1\_ppl\_mistral7b.json, exp1\_ppl\_gemma9b.json

SpectrumKV\_Greedy b=0.5 跨序列长度对比

| Model      | seq=2048 PPL | seq=2048 Δ% | seq=4096 PPL | seq=4096 Δ% | seq=8192 PPL | seq=8192 Δ% |
|------------|--------------|-------------|--------------|-------------|--------------|-------------|
| Qwen-7B    | 7.1805       | 1.97        | 6.1554       | 2.45        | 6.3949       | 2.90        |
| Mistral-7B | 7.9517       | -0.06       | 6.2301       | -0.13       | 6.3998       | -0.29       |
| Gemma-9B   | 11.1548      | -0.44       | 7.9781       | 0.14        | 未测试          | 未测试         |

SpectrumKV\_SinkProtect b=0.5 跨序列长度对比

| Model      | seq=2048 PPL | seq=2048 Δ% | seq=4096 PPL | seq=4096 Δ% | seq=8192 PPL | seq=8192 Δ% |
|------------|--------------|-------------|--------------|-------------|--------------|-------------|
| Qwen-7B    | 7.1397       | 1.39        | 6.1144       | 1.76        | 6.3722       | 2.54        |
| Mistral-7B | 7.9492       | -0.10       | 6.2397       | 0.03        | 6.4136       | -0.08       |
| Gemma-9B   | 11.1514      | -0.47       | 7.9514       | -0.20       | 未测试          | 未测试         |

PDTrim b=0.5 跨序列长度对比

| Model      | seq=2048 PPL | seq=2048 Δ% | seq=4096 PPL | seq=4096 Δ% | seq=8192 PPL | seq=8192 Δ% |
|------------|--------------|-------------|--------------|-------------|--------------|-------------|
| Qwen-7B    | 8.8614       | 25.85       | 6.9987       | 16.48       | 6.6644       | 7.24        |
| Mistral-7B | 9.7132       | 22.07       | 7.0919       | 13.69       | 6.9286       | 7.95        |
| Gemma-9B   | 15.1960      | 35.63       | 8.7088       | 9.31        | 未测试          | 未测试         |

FP16基线PPL（不同序列长度）

| Model      | seq=2048 | seq=4096 | seq=8192 |
|------------|----------|----------|----------|
| Qwen-7B    | 7.0415   | 6.0084   | 6.2145   |
| Mistral-7B | 7.9568   | 6.2379   | 6.4185   |
| Gemma-9B   | 11.2042  | 7.9672   | 未测试      |

RQ11: NIAH Retrieval Preservation（NIAH检索保持）

数据来源：exp2\_niah\_qwen7b.json, exp2\_niah\_mistral7b.json, exp2\_niah\_gemma9b.json, exp2\_spectrumkv\_mistral7b.json, exp2\_spectrumkv\_gemma9b.json

Qwen-7B NIAH（seq\_len=4096）

数据来源：exp2\_niah\_qwen7b.json

Full\_FP16 budget=0.0

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |

|      |     |
|------|-----|
| 0.25 | 1.0 |
| 0.30 | 1.0 |
| 0.35 | 1.0 |
| 0.40 | 1.0 |
| 0.45 | 1.0 |
| 0.50 | 1.0 |
| 0.55 | 1.0 |
| 0.60 | 1.0 |
| 0.65 | 1.0 |
| 0.70 | 1.0 |
| 0.75 | 1.0 |
| 0.80 | 1.0 |
| 0.85 | 1.0 |
| 0.90 | 1.0 |
| 0.95 | 1.0 |

PDTrim budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

PDTrim budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

PDTrim budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

PDTrim budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |

|      |     |
|------|-----|
| 0.70 | 1.0 |
| 0.75 | 1.0 |
| 0.80 | 1.0 |
| 0.85 | 1.0 |
| 0.90 | 1.0 |
| 0.95 | 1.0 |

PDTrim budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

Random\_Tier budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.2            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.2            |
| 0.85  | 0.0            |
| 0.90  | 0.0            |
| 0.95  | 0.0            |

Random\_Tier budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.2            |
| 0.85  | 0.0            |
| 0.90  | 0.0            |
| 0.95  | 0.0            |

Random\_Tier budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

Random\_Tier budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
|-------|----------------|

|      |     |
|------|-----|
| 0.05 | 1.0 |
| 0.10 | 1.0 |
| 0.15 | 1.0 |
| 0.20 | 1.0 |
| 0.25 | 1.0 |
| 0.30 | 1.0 |
| 0.35 | 1.0 |
| 0.40 | 1.0 |
| 0.45 | 1.0 |
| 0.50 | 1.0 |
| 0.55 | 1.0 |
| 0.60 | 1.0 |
| 0.65 | 1.0 |
| 0.70 | 1.0 |
| 0.75 | 1.0 |
| 0.80 | 1.0 |
| 0.85 | 1.0 |
| 0.90 | 1.0 |
| 0.95 | 1.0 |

Random\_Tier budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 未测试            |
| 0.85  | 未测试            |
| 0.90  | 未测试            |
| 0.95  | 未测试            |

SWS budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SWS budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SWS budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 0.2            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |

|      |     |
|------|-----|
| 0.50 | 0.0 |
| 0.55 | 0.0 |
| 0.60 | 0.0 |
| 0.65 | 0.0 |
| 0.70 | 0.0 |
| 0.75 | 0.0 |
| 0.80 | 0.0 |
| 0.85 | 1.0 |
| 0.90 | 1.0 |
| 0.95 | 1.0 |

SWS budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 0.4            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SWS budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Adaptive budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Adaptive budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 0.2            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |

|      |     |
|------|-----|
| 0.95 | 1.0 |
|------|-----|

SpectrumKV\_Adaptive budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Adaptive budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Adaptive budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Balanced budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.2            |
| 0.10  | 0.2            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.2            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.2            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.0            |
| 0.85  | 0.0            |
| 0.90  | 0.0            |
| 0.95  | 0.0            |

SpectrumKV\_Balanced budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |

|      |     |
|------|-----|
| 0.30 | 0.0 |
| 0.35 | 0.0 |
| 0.40 | 0.0 |
| 0.45 | 0.0 |
| 0.50 | 0.0 |
| 0.55 | 0.0 |
| 0.60 | 0.0 |
| 0.65 | 0.0 |
| 0.70 | 0.0 |
| 0.75 | 0.0 |
| 0.80 | 0.0 |
| 0.85 | 0.0 |
| 0.90 | 0.0 |
| 0.95 | 0.0 |

SpectrumKV\_Balanced budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.0            |
| 0.85  | 0.0            |
| 0.90  | 0.0            |
| 0.95  | 0.0            |

SpectrumKV\_Balanced budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.0            |
| 0.85  | 0.0            |
| 0.90  | 0.0            |
| 0.95  | 0.0            |

SpectrumKV\_Balanced budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 0.4            |
| 0.90  | 0.4            |
| 0.95  | 0.4            |

SpectrumKV\_Greedy budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.2            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.2            |
| 0.45  | 0.0            |
| 0.50  | 0.2            |
| 0.55  | 0.2            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |



|      |     |
|------|-----|
| 0.75 | 0.0 |
| 0.80 | 0.0 |
| 0.85 | 0.0 |
| 0.90 | 0.0 |
| 0.95 | 0.0 |

SpectrumKV\_Greedy budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.0            |
| 0.85  | 0.0            |
| 0.90  | 0.0            |
| 0.95  | 0.0            |

SpectrumKV\_Greedy budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Greedy budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Greedy budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_SinkProtect budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.2            |

|      |     |
|------|-----|
| 0.10 | 0.0 |
| 0.15 | 0.0 |
| 0.20 | 0.0 |
| 0.25 | 0.0 |
| 0.30 | 0.0 |
| 0.35 | 0.0 |
| 0.40 | 0.2 |
| 0.45 | 0.0 |
| 0.50 | 0.2 |
| 0.55 | 0.0 |
| 0.60 | 0.0 |
| 0.65 | 0.0 |
| 0.70 | 0.0 |
| 0.75 | 0.0 |
| 0.80 | 0.0 |
| 0.85 | 0.0 |
| 0.90 | 0.0 |
| 0.95 | 0.0 |

SpectrumKV\_SinkProtect budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.10  | 0.0            |
| 0.15  | 0.0            |
| 0.20  | 0.0            |
| 0.25  | 0.0            |
| 0.30  | 0.0            |
| 0.35  | 0.0            |
| 0.40  | 0.0            |
| 0.45  | 0.0            |
| 0.50  | 0.0            |
| 0.55  | 0.0            |
| 0.60  | 0.0            |
| 0.65  | 0.0            |
| 0.70  | 0.0            |
| 0.75  | 0.0            |
| 0.80  | 0.0            |
| 0.85  | 0.0            |
| 0.90  | 0.0            |
| 0.95  | 0.0            |

SpectrumKV\_SinkProtect budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_SinkProtect budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |
| 0.55  | 1.0            |
| 0.60  | 1.0            |
| 0.65  | 1.0            |
| 0.70  | 1.0            |
| 0.75  | 1.0            |
| 0.80  | 1.0            |
| 0.85  | 1.0            |
| 0.90  | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_SinkProtect budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.10  | 1.0            |
| 0.15  | 1.0            |
| 0.20  | 1.0            |
| 0.25  | 1.0            |
| 0.30  | 1.0            |
| 0.35  | 1.0            |
| 0.40  | 1.0            |
| 0.45  | 1.0            |
| 0.50  | 1.0            |

|      |     |
|------|-----|
| 0.55 | 1.0 |
| 0.60 | 1.0 |
| 0.65 | 1.0 |
| 0.70 | 1.0 |
| 0.75 | 1.0 |
| 0.80 | 1.0 |
| 0.85 | 1.0 |
| 0.90 | 1.0 |
| 0.95 | 1.0 |

Qwen-7B NIAH Summary (avg across all depths)

| 方法                     | b=0.0  | b=0.3 | b=0.4 | b=0.5  | b=0.6  | b=0.7  |
|------------------------|--------|-------|-------|--------|--------|--------|
| Full_FP16              | 100.0% | —     | —     | —      | —      | —      |
| PDTrim                 | —      | 26.3% | 36.8% | 47.4%  | 57.9%  | 68.4%  |
| Random_Tier            | —      | 2.1%  | 1.1%  | 100.0% | 100.0% | 100.0% |
| SWS                    | —      | 21.1% | 31.6% | 43.2%  | 54.7%  | 68.4%  |
| SpectrumKV_Adaptive    | —      | 52.6% | 74.7% | 100.0% | 100.0% | 100.0% |
| SpectrumKV_Balanced    | —      | 4.2%  | 0.0%  | 0.0%   | 0.0%   | 90.5%  |
| SpectrumKV_Greedy      | —      | 4.2%  | 0.0%  | 100.0% | 100.0% | 100.0% |
| SpectrumKV_SinkProtect | —      | 3.2%  | 0.0%  | 100.0% | 100.0% | 100.0% |

Mistral-7B NIAH (seq\_len=4096)

数据来源: exp2\_niah\_mistral7b.json

Full\_FP16 budget=0.0

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

PDTrim budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.1   | 0.0            |
| 0.15  | 0.0            |
| 0.2   | 0.0            |
| 0.25  | 0.0            |
| 0.3   | 0.0            |
| 0.35  | 0.0            |
| 0.4   | 0.0            |
| 0.45  | 0.0            |
| 0.5   | 0.0            |
| 0.55  | 0.0            |
| 0.6   | 0.0            |
| 0.65  | 0.0            |
| 0.7   | 0.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

PDTrim budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.1   | 0.0            |
| 0.15  | 0.0            |
| 0.2   | 0.0            |
| 0.25  | 0.0            |
| 0.3   | 0.0            |
| 0.35  | 0.0            |
| 0.4   | 0.0            |
| 0.45  | 0.0            |
| 0.5   | 0.0            |
| 0.55  | 0.0            |
| 0.6   | 0.2            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

PDTrim budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.1   | 0.0            |
| 0.15  | 0.0            |
| 0.2   | 0.0            |

|      |     |
|------|-----|
| 0.25 | 0.0 |
| 0.3  | 0.0 |
| 0.35 | 0.0 |
| 0.4  | 0.0 |
| 0.45 | 0.0 |
| 0.5  | 0.0 |
| 0.55 | 1.0 |
| 0.6  | 1.0 |
| 0.65 | 0.8 |
| 0.7  | 0.8 |
| 0.75 | 0.8 |
| 0.8  | 1.0 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

PDTrim budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.1   | 0.0            |
| 0.15  | 0.0            |
| 0.2   | 0.0            |
| 0.25  | 0.0            |
| 0.3   | 0.0            |
| 0.35  | 0.0            |
| 0.4   | 0.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

PDTrim budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.0            |
| 0.1   | 0.0            |
| 0.15  | 0.0            |
| 0.2   | 0.0            |
| 0.25  | 0.0            |
| 0.3   | 0.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

Random\_Tier budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 0.8            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

Random\_Tier budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |

|      |     |
|------|-----|
| 0.7  | 0.8 |
| 0.75 | 0.8 |
| 0.8  | 1.0 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

Random\_Tier budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

Random\_Tier budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

Random\_Tier budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 未测试            |
| 0.6   | 未测试            |
| 0.65  | 未测试            |
| 0.7   | 未测试            |
| 0.75  | 未测试            |
| 0.8   | 未测试            |
| 0.85  | 未测试            |
| 0.9   | 未测试            |
| 0.95  | 未测试            |

SWS budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 0.0            |
| 0.15  | 0.0            |
| 0.2   | 0.0            |
| 0.25  | 0.0            |
| 0.3   | 0.0            |
| 0.35  | 0.0            |
| 0.4   | 0.0            |
| 0.45  | 0.0            |
| 0.5   | 0.0            |
| 0.55  | 0.0            |
| 0.6   | 0.0            |
| 0.65  | 0.0            |
| 0.7   | 0.0            |
| 0.75  | 0.0            |
| 0.8   | 0.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SWS budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
|-------|----------------|

|      |     |
|------|-----|
| 0.05 | 0.8 |
| 0.1  | 1.0 |
| 0.15 | 0.8 |
| 0.2  | 0.0 |
| 0.25 | 0.0 |
| 0.3  | 0.0 |
| 0.35 | 0.0 |
| 0.4  | 0.0 |
| 0.45 | 0.0 |
| 0.5  | 0.0 |
| 0.55 | 0.0 |
| 0.6  | 0.0 |
| 0.65 | 0.0 |
| 0.7  | 0.0 |
| 0.75 | 0.0 |
| 0.8  | 0.0 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

SWS budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 0.0            |
| 0.35  | 0.0            |
| 0.4   | 0.0            |
| 0.45  | 0.0            |
| 0.5   | 0.0            |
| 0.55  | 0.0            |
| 0.6   | 0.0            |
| 0.65  | 0.0            |
| 0.7   | 0.0            |
| 0.75  | 0.0            |
| 0.8   | 0.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SWS budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 0.2            |
| 0.45  | 0.0            |
| 0.5   | 0.0            |
| 0.55  | 0.0            |
| 0.6   | 0.0            |
| 0.65  | 0.0            |
| 0.7   | 0.0            |
| 0.75  | 0.0            |
| 0.8   | 0.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SWS budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 0.0            |
| 0.6   | 0.0            |
| 0.65  | 0.0            |
| 0.7   | 0.0            |
| 0.75  | 0.0            |
| 0.8   | 0.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Balanced budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |

|      |     |
|------|-----|
| 0.5  | 0.8 |
| 0.55 | 1.0 |
| 0.6  | 1.0 |
| 0.65 | 0.8 |
| 0.7  | 0.8 |
| 0.75 | 0.8 |
| 0.8  | 1.0 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

SpectrumKV\_Balanced budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Balanced budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Balanced budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Balanced budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |

|      |     |
|------|-----|
| 0.95 | 1.0 |
|------|-----|

SpectrumKV\_Greedy budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Greedy budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Greedy budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Greedy budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Greedy budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |



|      |     |
|------|-----|
| 0.3  | 1.0 |
| 0.35 | 0.8 |
| 0.4  | 1.0 |
| 0.45 | 0.8 |
| 0.5  | 0.8 |
| 0.55 | 1.0 |
| 0.6  | 1.0 |
| 0.65 | 0.8 |
| 0.7  | 0.8 |
| 0.75 | 0.8 |
| 0.8  | 1.0 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

SpectrumKV\_SinkProtect budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_SinkProtect budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_SinkProtect budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_SinkProtect budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |

|      |     |
|------|-----|
| 0.75 | 0.8 |
| 0.8  | 1.0 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

SpectrumKV\_SinkProtect budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.8            |
| 0.1   | 1.0            |
| 0.15  | 0.8            |
| 0.2   | 0.8            |
| 0.25  | 0.8            |
| 0.3   | 1.0            |
| 0.35  | 0.8            |
| 0.4   | 1.0            |
| 0.45  | 0.8            |
| 0.5   | 0.8            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 0.8            |
| 0.7   | 0.8            |
| 0.75  | 0.8            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

Mistral-7B NIAH Summary (avg across all depths)

| 方法                     | b=0.0 | b=0.3 | b=0.4 | b=0.5 | b=0.6 | b=0.7 |
|------------------------|-------|-------|-------|-------|-------|-------|
| Full_FP16              | 89.5% | —     | —     | —     | —     | —     |
| PDTrim                 | —     | 26.3% | 34.7% | 44.2% | 52.6% | 62.1% |
| Random_Tier            | —     | 88.4% | 89.5% | 89.5% | 89.5% | 86.0% |
| SWS                    | —     | 20.0% | 29.5% | 37.9% | 48.4% | 61.1% |
| SpectrumKV_Balanced    | —     | 89.5% | 89.5% | 89.5% | 89.5% | 89.5% |
| SpectrumKV_Greedy      | —     | 89.5% | 89.5% | 89.5% | 89.5% | 89.5% |
| SpectrumKV_SinkProtect | —     | 89.5% | 89.5% | 89.5% | 89.5% | 89.5% |

Gemma-9B NIAH (seq\_len=4096)

数据来源: exp2\_niah\_gemma9b.json

Full\_FP16 budget=0.0

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

PDTrim budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.2            |
| 0.1   | 0.2            |
| 0.15  | 0.2            |
| 0.2   | 0.2            |
| 0.25  | 0.2            |
| 0.3   | 0.2            |
| 0.35  | 0.2            |
| 0.4   | 0.2            |
| 0.45  | 0.2            |
| 0.5   | 0.2            |
| 0.55  | 0.2            |
| 0.6   | 0.2            |
| 0.65  | 0.2            |
| 0.7   | 0.2            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

PDTrim budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.2            |
| 0.1   | 0.2            |
| 0.15  | 0.2            |
| 0.2   | 0.2            |
| 0.25  | 0.2            |
| 0.3   | 0.2            |
| 0.35  | 0.2            |
| 0.4   | 0.2            |
| 0.45  | 0.2            |

|      |     |
|------|-----|
| 0.5  | 0.2 |
| 0.55 | 0.2 |
| 0.6  | 0.2 |
| 0.65 | 1.0 |
| 0.7  | 1.0 |
| 0.75 | 1.0 |
| 0.8  | 1.0 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

PDTrim budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.2            |
| 0.1   | 0.2            |
| 0.15  | 0.2            |
| 0.2   | 0.2            |
| 0.25  | 0.2            |
| 0.3   | 0.2            |
| 0.35  | 0.2            |
| 0.4   | 0.2            |
| 0.45  | 0.2            |
| 0.5   | 0.2            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

PDTrim budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.2            |
| 0.1   | 0.2            |
| 0.15  | 0.2            |
| 0.2   | 0.2            |
| 0.25  | 0.2            |
| 0.3   | 0.2            |
| 0.35  | 0.2            |
| 0.4   | 0.2            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

PDTrim budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 0.2            |
| 0.1   | 0.2            |
| 0.15  | 0.2            |
| 0.2   | 0.2            |
| 0.25  | 0.2            |
| 0.3   | 0.2            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

Random\_Tier budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |

|      |     |
|------|-----|
| 0.95 | 1.0 |
|------|-----|

Random\_Tier budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

Random\_Tier budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

Random\_Tier budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

Random\_Tier budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SWS budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 0.2            |
| 0.15  | 0.2            |
| 0.2   | 0.2            |
| 0.25  | 0.2            |

|      |     |
|------|-----|
| 0.3  | 0.2 |
| 0.35 | 0.2 |
| 0.4  | 0.2 |
| 0.45 | 0.2 |
| 0.5  | 0.2 |
| 0.55 | 0.2 |
| 0.6  | 0.0 |
| 0.65 | 0.2 |
| 0.7  | 0.2 |
| 0.75 | 0.2 |
| 0.8  | 0.2 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

SWS budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 0.2            |
| 0.25  | 0.2            |
| 0.3   | 0.2            |
| 0.35  | 0.2            |
| 0.4   | 0.2            |
| 0.45  | 0.2            |
| 0.5   | 0.2            |
| 0.55  | 0.2            |
| 0.6   | 0.0            |
| 0.65  | 0.2            |
| 0.7   | 0.2            |
| 0.75  | 0.2            |
| 0.8   | 0.2            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SWS budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 0.2            |
| 0.35  | 0.2            |
| 0.4   | 0.2            |
| 0.45  | 0.2            |
| 0.5   | 0.2            |
| 0.55  | 0.2            |
| 0.6   | 0.0            |
| 0.65  | 0.2            |
| 0.7   | 0.2            |
| 0.75  | 0.2            |
| 0.8   | 0.2            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SWS budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 0.2            |
| 0.45  | 0.2            |
| 0.5   | 0.2            |
| 0.55  | 0.0            |
| 0.6   | 0.0            |
| 0.65  | 0.2            |
| 0.7   | 0.2            |
| 0.75  | 0.2            |
| 0.8   | 0.2            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SWS budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 0.0            |
| 0.6   | 0.0            |
| 0.65  | 0.2            |
| 0.7   | 0.2            |

|      |     |
|------|-----|
| 0.75 | 0.2 |
| 0.8  | 0.2 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

SpectrumKV\_Adaptive budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 0.2            |
| 0.45  | 0.2            |
| 0.5   | 0.2            |
| 0.55  | 0.0            |
| 0.6   | 0.0            |
| 0.65  | 0.2            |
| 0.7   | 0.2            |
| 0.75  | 0.2            |
| 0.8   | 0.2            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Adaptive budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 0.6            |
| 0.65  | 0.2            |
| 0.7   | 0.2            |
| 0.75  | 0.2            |
| 0.8   | 0.2            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Adaptive budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Adaptive budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Adaptive budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |

|      |     |
|------|-----|
| 0.1  | 1.0 |
| 0.15 | 1.0 |
| 0.2  | 1.0 |
| 0.25 | 1.0 |
| 0.3  | 1.0 |
| 0.35 | 1.0 |
| 0.4  | 1.0 |
| 0.45 | 1.0 |
| 0.5  | 1.0 |
| 0.55 | 1.0 |
| 0.6  | 1.0 |
| 0.65 | 1.0 |
| 0.7  | 1.0 |
| 0.75 | 1.0 |
| 0.8  | 1.0 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

SpectrumKV\_Balanced budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Balanced budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Balanced budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Balanced budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |

|      |     |
|------|-----|
| 0.55 | 1.0 |
| 0.6  | 1.0 |
| 0.65 | 1.0 |
| 0.7  | 1.0 |
| 0.75 | 1.0 |
| 0.8  | 1.0 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

SpectrumKV\_Balanced budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Greedy budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Greedy budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Greedy budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |



SpectrumKV\_Greedy budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_Greedy budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_SinkProtect budget=0.3

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_SinkProtect budget=0.4

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_SinkProtect budget=0.5

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |

|      |     |
|------|-----|
| 0.35 | 1.0 |
| 0.4  | 1.0 |
| 0.45 | 1.0 |
| 0.5  | 1.0 |
| 0.55 | 1.0 |
| 0.6  | 1.0 |
| 0.65 | 1.0 |
| 0.7  | 1.0 |
| 0.75 | 1.0 |
| 0.8  | 1.0 |
| 0.85 | 1.0 |
| 0.9  | 1.0 |
| 0.95 | 1.0 |

SpectrumKV\_SinkProtect budget=0.6

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

SpectrumKV\_SinkProtect budget=0.7

| Depth | Retrieval Rate |
|-------|----------------|
| 0.05  | 1.0            |
| 0.1   | 1.0            |
| 0.15  | 1.0            |
| 0.2   | 1.0            |
| 0.25  | 1.0            |
| 0.3   | 1.0            |
| 0.35  | 1.0            |
| 0.4   | 1.0            |
| 0.45  | 1.0            |
| 0.5   | 1.0            |
| 0.55  | 1.0            |
| 0.6   | 1.0            |
| 0.65  | 1.0            |
| 0.7   | 1.0            |
| 0.75  | 1.0            |
| 0.8   | 1.0            |
| 0.85  | 1.0            |
| 0.9   | 1.0            |
| 0.95  | 1.0            |

Gemma-9B NIAH Summary (avg across all depths)

| 方法                     | b=0.0  | b=0.3  | b=0.4  | b=0.5  | b=0.6  | b=0.7  |
|------------------------|--------|--------|--------|--------|--------|--------|
| Full_FP16              | 100.0% | —      | —      | —      | —      | —      |
| PDTrim                 | —      | 41.1%  | 49.5%  | 57.9%  | 66.3%  | 74.7%  |
| Random_Tier            | —      | 100.0% | 100.0% | 100.0% | 100.0% | 100.0% |
| SWS                    | —      | 35.8%  | 44.2%  | 52.6%  | 60.0%  | 72.6%  |
| SpectrumKV_Adaptive    | —      | 60.0%  | 81.1%  | 100.0% | 100.0% | 100.0% |
| SpectrumKV_Balanced    | —      | 100.0% | 100.0% | 100.0% | 100.0% | 100.0% |
| SpectrumKV_Greedy      | —      | 100.0% | 100.0% | 100.0% | 100.0% | 100.0% |
| SpectrumKV_SinkProtect | —      | 100.0% | 100.0% | 100.0% | 100.0% | 100.0% |

SpectrumKV Adaptive v2 Probe验证数据

数据来源：exp2\_spectrumkv\_mistral7b.json,exp2\_spectrumkv\_gemma9b.json

Mistral-7B SpectrumKV\_Adaptive\_v2

- int4\_tolerant: True
- tier\_choice: 3-tier
- probe\_successes: 3

budget=0.3

| Depth | Success/Total | Rate  |
|-------|---------------|-------|
| 0.05  | 5/5           | 1.000 |
| 0.1   | 5/5           | 1.000 |
| 0.15  | 5/5           | 1.000 |
| 0.2   | 5/5           | 1.000 |
| 0.25  | 5/5           | 1.000 |
| 0.3   | 5/5           | 1.000 |
| 0.35  | 5/5           | 1.000 |
| 0.4   | 5/5           | 1.000 |
| 0.45  | 5/5           | 1.000 |
| 0.5   | 5/5           | 1.000 |
| 0.55  | 5/5           | 1.000 |
| 0.6   | 5/5           | 1.000 |
| 0.65  | 5/5           | 1.000 |
| 0.7   | 5/5           | 1.000 |
| 0.75  | 5/5           | 1.000 |
| 0.8   | 5/5           | 1.000 |
| 0.85  | 5/5           | 1.000 |
| 0.9   | 5/5           | 1.000 |

|      |     |       |
|------|-----|-------|
| 0.95 | 5/5 | 1.000 |
|------|-----|-------|

budget=0.4

| Depth | Success/Total | Rate  |
|-------|---------------|-------|
| 0.05  | 5/5           | 1.000 |
| 0.1   | 5/5           | 1.000 |
| 0.15  | 5/5           | 1.000 |
| 0.2   | 5/5           | 1.000 |
| 0.25  | 5/5           | 1.000 |
| 0.3   | 5/5           | 1.000 |
| 0.35  | 5/5           | 1.000 |
| 0.4   | 5/5           | 1.000 |
| 0.45  | 5/5           | 1.000 |
| 0.5   | 5/5           | 1.000 |
| 0.55  | 5/5           | 1.000 |
| 0.6   | 5/5           | 1.000 |
| 0.65  | 5/5           | 1.000 |
| 0.7   | 5/5           | 1.000 |
| 0.75  | 5/5           | 1.000 |
| 0.8   | 5/5           | 1.000 |
| 0.85  | 5/5           | 1.000 |
| 0.9   | 5/5           | 1.000 |
| 0.95  | 5/5           | 1.000 |

budget=0.5

| Depth | Success/Total | Rate  |
|-------|---------------|-------|
| 0.05  | 5/5           | 1.000 |
| 0.1   | 5/5           | 1.000 |
| 0.15  | 5/5           | 1.000 |
| 0.2   | 5/5           | 1.000 |
| 0.25  | 5/5           | 1.000 |
| 0.3   | 5/5           | 1.000 |
| 0.35  | 5/5           | 1.000 |
| 0.4   | 5/5           | 1.000 |
| 0.45  | 5/5           | 1.000 |
| 0.5   | 5/5           | 1.000 |
| 0.55  | 5/5           | 1.000 |
| 0.6   | 5/5           | 1.000 |
| 0.65  | 5/5           | 1.000 |
| 0.7   | 5/5           | 1.000 |
| 0.75  | 5/5           | 1.000 |
| 0.8   | 5/5           | 1.000 |
| 0.85  | 5/5           | 1.000 |
| 0.9   | 5/5           | 1.000 |
| 0.95  | 5/5           | 1.000 |

budget=0.6

| Depth | Success/Total | Rate  |
|-------|---------------|-------|
| 0.05  | 5/5           | 1.000 |
| 0.1   | 5/5           | 1.000 |
| 0.15  | 5/5           | 1.000 |
| 0.2   | 5/5           | 1.000 |
| 0.25  | 5/5           | 1.000 |
| 0.3   | 5/5           | 1.000 |
| 0.35  | 5/5           | 1.000 |
| 0.4   | 5/5           | 1.000 |
| 0.45  | 5/5           | 1.000 |
| 0.5   | 5/5           | 1.000 |
| 0.55  | 5/5           | 1.000 |
| 0.6   | 5/5           | 1.000 |
| 0.65  | 5/5           | 1.000 |
| 0.7   | 5/5           | 1.000 |
| 0.75  | 5/5           | 1.000 |
| 0.8   | 5/5           | 1.000 |
| 0.85  | 5/5           | 1.000 |
| 0.9   | 5/5           | 1.000 |
| 0.95  | 5/5           | 1.000 |

budget=0.7

| Depth | Success/Total | Rate  |
|-------|---------------|-------|
| 0.05  | 5/5           | 1.000 |
| 0.1   | 5/5           | 1.000 |
| 0.15  | 5/5           | 1.000 |
| 0.2   | 5/5           | 1.000 |
| 0.25  | 5/5           | 1.000 |
| 0.3   | 5/5           | 1.000 |
| 0.35  | 5/5           | 1.000 |
| 0.4   | 5/5           | 1.000 |
| 0.45  | 5/5           | 1.000 |
| 0.5   | 5/5           | 1.000 |
| 0.55  | 5/5           | 1.000 |
| 0.6   | 5/5           | 1.000 |
| 0.65  | 5/5           | 1.000 |
| 0.7   | 5/5           | 1.000 |
| 0.75  | 5/5           | 1.000 |
| 0.8   | 5/5           | 1.000 |
| 0.85  | 5/5           | 1.000 |
| 0.9   | 5/5           | 1.000 |
| 0.95  | 5/5           | 1.000 |

Gemma-9B SpectrumKV\_Adaptive\_v2

- int4\_tolerant: True
- tier\_choice: 3-tier
- probe\_successes: 3

budget=0.3

| Depth | Success/Total | Rate |
|-------|---------------|------|
|       |               |      |

|      |     |       |
|------|-----|-------|
| 0.05 | 5/5 | 1.000 |
| 0.1  | 5/5 | 1.000 |
| 0.15 | 5/5 | 1.000 |
| 0.2  | 5/5 | 1.000 |
| 0.25 | 5/5 | 1.000 |
| 0.3  | 5/5 | 1.000 |
| 0.35 | 5/5 | 1.000 |
| 0.4  | 5/5 | 1.000 |
| 0.45 | 5/5 | 1.000 |
| 0.5  | 5/5 | 1.000 |
| 0.55 | 5/5 | 1.000 |
| 0.6  | 5/5 | 1.000 |
| 0.65 | 5/5 | 1.000 |
| 0.7  | 5/5 | 1.000 |
| 0.75 | 5/5 | 1.000 |
| 0.8  | 5/5 | 1.000 |
| 0.85 | 5/5 | 1.000 |
| 0.9  | 5/5 | 1.000 |
| 0.95 | 5/5 | 1.000 |

budget=0.4

| Depth | Success/Total | Rate  |
|-------|---------------|-------|
| 0.05  | 5/5           | 1.000 |
| 0.1   | 5/5           | 1.000 |
| 0.15  | 5/5           | 1.000 |
| 0.2   | 5/5           | 1.000 |
| 0.25  | 5/5           | 1.000 |
| 0.3   | 5/5           | 1.000 |
| 0.35  | 5/5           | 1.000 |
| 0.4   | 5/5           | 1.000 |
| 0.45  | 5/5           | 1.000 |
| 0.5   | 5/5           | 1.000 |
| 0.55  | 5/5           | 1.000 |
| 0.6   | 5/5           | 1.000 |
| 0.65  | 5/5           | 1.000 |
| 0.7   | 5/5           | 1.000 |
| 0.75  | 5/5           | 1.000 |
| 0.8   | 5/5           | 1.000 |
| 0.85  | 5/5           | 1.000 |
| 0.9   | 5/5           | 1.000 |
| 0.95  | 5/5           | 1.000 |

budget=0.5

| Depth | Success/Total | Rate  |
|-------|---------------|-------|
| 0.05  | 5/5           | 1.000 |
| 0.1   | 5/5           | 1.000 |
| 0.15  | 5/5           | 1.000 |
| 0.2   | 5/5           | 1.000 |
| 0.25  | 5/5           | 1.000 |
| 0.3   | 5/5           | 1.000 |
| 0.35  | 5/5           | 1.000 |
| 0.4   | 5/5           | 1.000 |
| 0.45  | 5/5           | 1.000 |
| 0.5   | 5/5           | 1.000 |
| 0.55  | 5/5           | 1.000 |
| 0.6   | 5/5           | 1.000 |
| 0.65  | 5/5           | 1.000 |
| 0.7   | 5/5           | 1.000 |
| 0.75  | 5/5           | 1.000 |
| 0.8   | 5/5           | 1.000 |
| 0.85  | 5/5           | 1.000 |
| 0.9   | 5/5           | 1.000 |
| 0.95  | 5/5           | 1.000 |

budget=0.6

| Depth | Success/Total | Rate  |
|-------|---------------|-------|
| 0.05  | 5/5           | 1.000 |
| 0.1   | 5/5           | 1.000 |
| 0.15  | 5/5           | 1.000 |
| 0.2   | 5/5           | 1.000 |
| 0.25  | 5/5           | 1.000 |
| 0.3   | 5/5           | 1.000 |
| 0.35  | 5/5           | 1.000 |
| 0.4   | 5/5           | 1.000 |
| 0.45  | 5/5           | 1.000 |
| 0.5   | 5/5           | 1.000 |
| 0.55  | 5/5           | 1.000 |
| 0.6   | 5/5           | 1.000 |
| 0.65  | 5/5           | 1.000 |
| 0.7   | 5/5           | 1.000 |
| 0.75  | 5/5           | 1.000 |
| 0.8   | 5/5           | 1.000 |
| 0.85  | 5/5           | 1.000 |
| 0.9   | 5/5           | 1.000 |
| 0.95  | 5/5           | 1.000 |

budget=0.7

| Depth | Success/Total | Rate  |
|-------|---------------|-------|
| 0.05  | 5/5           | 1.000 |
| 0.1   | 5/5           | 1.000 |
| 0.15  | 5/5           | 1.000 |
| 0.2   | 5/5           | 1.000 |
| 0.25  | 5/5           | 1.000 |
| 0.3   | 5/5           | 1.000 |
| 0.35  | 5/5           | 1.000 |
| 0.4   | 5/5           | 1.000 |
| 0.45  | 5/5           | 1.000 |

|      |     |       |
|------|-----|-------|
| 0.5  | 5/5 | 1.000 |
| 0.55 | 5/5 | 1.000 |
| 0.6  | 5/5 | 1.000 |
| 0.65 | 5/5 | 1.000 |
| 0.7  | 5/5 | 1.000 |
| 0.75 | 5/5 | 1.000 |
| 0.8  | 5/5 | 1.000 |
| 0.85 | 5/5 | 1.000 |
| 0.9  | 5/5 | 1.000 |
| 0.95 | 5/5 | 1.000 |

RQ12: PDTrim First-Ratio Sensitivity (PDTrim首层比例敏感性)

| 数据来源: experiment_log_v2.md (Exp1 PDTrim子集) |         |            |          |
|--------------------------------------------|---------|------------|----------|
| First-ratio                                | Qwen Δ% | Mistral Δ% | Gemma Δ% |
| 0.1                                        | +18.2%  | +45.3%     | +49.1%   |
| 0.3                                        | +14.1%  | +12.8%     | +18.7%   |
| 0.5 (default)                              | +25.8%  | +22.1%     | +35.6%   |
| 0.7                                        | +16.3%  | +10.5%     | +8.9%    |
| 0.9                                        | +19.7%  | +15.2%     | +3.8%    |

RQ13: Decay Rate Impact (衰减率影响)

| 数据来源: experiment_log_v2.md (仿真v3精扫扫描数据, 非GPU实测) |            |        |        |        |  |
|-------------------------------------------------|------------|--------|--------|--------|--|
| Model                                           | Decay Rate | b=0.3  | b=0.5  | b=0.7  |  |
| Mistral (attn×decay)                            | dr=0.001   | +36.7% | +11.7% | +1.9%  |  |
| Mistral (attn×decay)                            | dr=0.005   | +27.5% | +0.6%  | -10.7% |  |
| Mistral (attn×decay)                            | dr=0.01    | +24.7% | -2.0%  | -12.3% |  |
| Gemma (attn×decay)                              | dr=0.001   | +14.1% | -0.2%  | -5.3%  |  |
| Gemma (attn×decay)                              | dr=0.005   | +13.9% | -12.7% | -15.2% |  |
| Gemma (attn×decay)                              | dr=0.01    | +13.4% | -13.2% | -18.1% |  |
| Qwen (kv-norm×decay)                            | dr=0.001   | +38.4% | +9.6%  | -2.0%  |  |
| Qwen (kv-norm×decay)                            | dr=0.005   | +44.5% | +12.5% | -1.3%  |  |
| Qwen (kv-norm×decay)                            | dr=0.01    | +45.7% | +12.5% | -1.3%  |  |

RQ14: Multi-Seed/Task Robustness (多seed/任务鲁棒性)

|                                                |  |  |  |
|------------------------------------------------|--|--|--|
| 数据来源: experiment_log_v2.md (Exp1多seed/多task配置) |  |  |  |
|------------------------------------------------|--|--|--|

7-seed CI95 (b=0.5)

| Strategy    | Qwen CI95 | Mistral CI95 | Gemma CI95 |
|-------------|-----------|--------------|------------|
| Random_Tier | ± 22%     | ± 265%       | ± 75%      |
| PDTrim      | 0         | 0            | 0          |
| SKV Greedy  | 0         | 0            | 0          |

Multi-task (b=0.5)

| Model   | Task     | PDTrim Δ% | SKV Δ% |
|---------|----------|-----------|--------|
| Mistral | WikiText | +22.1%    | -0.1%  |
| Mistral | Code     | +15.7%    | +6.9%  |
| Mistral | Science  | +152.8%   | +33.9% |
| Gemma   | WikiText | +35.6%    | -0.4%  |
| Gemma   | Code     | +11.4%    | +8.0%  |
| Gemma   | Science  | +36.0%    | +32.7% |
| Qwen    | WikiText | +25.8%    | +2.0%  |
| Qwen    | Code     | +8.1%     | +5.5%  |
| Qwen    | Science  | +35.4%    | +30.5% |

RQ15: Adaptive Precision Strategy (自适应精度策略)

|                                                                                          |  |  |  |
|------------------------------------------------------------------------------------------|--|--|--|
| 数据来源: exp2_spectrumkv_mistral7b.json, exp2_spectrumkv_gemma9b.json, experiment_log_v2.md |  |  |  |
|------------------------------------------------------------------------------------------|--|--|--|

Probe结果

| Model      | Probe successes | INT4 tolerant | Tier choice             |
|------------|-----------------|---------------|-------------------------|
| Qwen-7B    | 0/3             | No            | 2-tier (FP16+INT8)      |
| Mistral-7B | 3/3             | Yes           | 3-tier (FP16+INT8+INT4) |
| Gemma-9B   | 3/3             | Yes           | 3-tier (FP16+INT8+INT4) |

NIAH at b=0.3

| Model   | Fixed 3-tier   | Adaptive       | PDTrim |
|---------|----------------|----------------|--------|
| Qwen    | 0%             | 52.6% (2-tier) | 26.3%  |
| Mistral | 89.5% (3-tier) | 100% (3-tier)  | 26.3%  |
| Gemma   | 100% (3-tier)  | 100% (3-tier)  | 41.1%  |

RQ16: Per-Layer Quantization Error (逐层量化误差)

|                                                                                                    |  |  |  |
|----------------------------------------------------------------------------------------------------|--|--|--|
| 数据来源: exp3_quant_error_qwen7b.json, exp3_quant_error_mistral7b.json, exp3_quant_error_gemma9b.json |  |  |  |
|----------------------------------------------------------------------------------------------------|--|--|--|

Qwen-7B 逐层量化误差

- seq\_len: 2048
- num\_samples: 3
- num\_layers: 28
- sink\_count: 4
- sensitivity\_threshold: 0.15
- token\_sample\_step: 8
- timestamp: 2026-06-04 17:57:47

Per-Layer Quantization Error

| Layer | INT8 Key rel_L2      | INT8 Key max_abs    | INT8 Key cos_sim   | INT8 Val rel_L2       | INT8 Val max_abs     | INT8 Val cos_sim   | INT4 Key rel_L2     | INT4 Key max_abs   | INT4 Key cos_sim   | INT4 Val rel_L2     | INT4 Val max_abs    | INT4 Val cos_sim |
|-------|----------------------|---------------------|--------------------|-----------------------|----------------------|--------------------|---------------------|--------------------|--------------------|---------------------|---------------------|------------------|
| 0     | 0.010838510368935954 | 1.0                 | 0.9999414285024008 | 0.01119542072397188   | 0.013020833333333334 | 0.9999372959136963 | 0.10262856312704582 | 12.3125            | 0.9947418967882792 | 0.1940965894753737  | 0.20670572916666666 | 0.98161075       |
| 1     | 0.01128078098166145  | 0.375               | 0.999364217122396  | 0.009029734713441602  | 0.0078125            | 0.9999592900276184 | 0.14928985650900675 | 4.75               | 0.988815426826477  | 0.1611308279748379  | 0.12727864583333334 | 0.98725998       |
| 2     | 0.01337696673198241  | 0.1328125           | 0.9999105135599772 | 0.009770003766714714  | 0.0159912109375      | 0.9999522964159647 | 0.2249275415991967  | 2.3645833333333335 | 0.9755614995956421 | 0.1748434564830591  | 0.291015625         | 0.98505795       |
| 3     | 0.01737932743492863  | 0.3125              | 0.9998491009076437 | 0.008153543576580614  | 0.015625             | 0.9999667604764303 | 0.23861616876641026 | 5.260416666666667  | 0.9717976649602255 | 0.14479327592593738 | 0.19954427083333334 | 0.98967732       |
| 4     | 0.011878670634759676 | 0.0625              | 0.9999294281005859 | 0.011159058068625603  | 0.0234375            | 0.9999377528826395 | 0.2130419530562885  | 0.88671875         | 0.9780000646909078 | 0.1984642479276004  | 0.3951822916666667  | 0.98079286       |
| 5     | 0.01397972622647625  | 0.0703125           | 0.9999023278554281 | 0.009463563659140629  | 0.0224609375         | 0.9999552766482035 | 0.2503970587343884  | 1.2578125          | 0.9699203570683798 | 0.16923848149232704 | 0.3971354166666667  | 0.98597321       |
| 6     | 0.012784472218320903 | 0.06998697916666667 | 0.9999182422955831 | 0.009087245581588072  | 0.021321614583333332 | 0.9999586939811707 | 0.22855546031841026 | 1.1640625          | 0.9747212727864584 | 0.16233000036503603 | 0.3854166666666667  | 0.98707801       |
| 7     | 0.013603542442802522 | 0.06412760416666667 | 0.9999074935913086 | 0.0071195273641021725 | 0.028645833333333332 | 0.9999746680259705 | 0.24498992798040944 | 1.1380208333333333 | 0.9712090293566386 | 0.12531695397499068 | 0.4290364583333333  | 0.99223796       |
| 8     | 0.011955287882336746 | 0.0625              | 0.9999285538991293 | 0.00931016661393863   | 0.02799479166666668  | 0.9999567071596781 | 0.21556439071754321 | 1.1223958333333333 | 0.9775662422180176 | 0.16638302156640705 | 0.4830729166666667  | 0.98643461       |

|    |                      |                     |                    |                       |                     |                     |                      |                    |                    |                     |                    |            |
|----|----------------------|---------------------|--------------------|-----------------------|---------------------|---------------------|----------------------|--------------------|--------------------|---------------------|--------------------|------------|
| 9  | 0.012881449706406042 | 0.0693359375        | 0.9999170502026876 | 0.00684914494879807   | 0.02473958333333332 | 0.9999765753746033  | 0.2319145087469612   | 1.2682291666666667 | 0.9741342663764954 | 0.12023154186294761 | 0.3919270833333333 | 0.99284958 |
| 10 | 0.01378904262158752  | 0.0625              | 0.9999049305915833 | 0.00790430543239018   | 0.02034505208333332 | 0.9999687870343527  | 0.248132172889615034 | 1.0898583333333333 | 0.9705716371536255 | 0.14009766713281321 | 0.3463541666666667 | 0.99032755 |
| 11 | 0.0121718853612212   | 0.0625              | 0.9999191761016846 | 0.008987567034024264  | 0.0234375           | 0.9999596277877221  | 0.2292125137890568   | 1.0729166666666667 | 0.9747029344240824 | 0.16038066356012456 | 0.388671875        | 0.98737343 |
| 12 | 0.01105944183549948  | 0.08919270833333333 | 0.9999388655026754 | 0.0076897956652137    | 0.022216796875      | 0.9999704162279764  | 0.19795811225671275  | 1.3125             | 0.9809310436248779 | 0.13575907012678923 | 0.384765625        | 0.99090351 |
| 13 | 0.0170758357159727   | 0.1875              | 0.9998542070388794 | 0.0075912336506292885 | 0.0234375           | 0.9999712109565735  | 0.25557966894484535  | 3.203125           | 0.9678070147832235 | 0.13417023092691055 | 0.39453125         | 0.99111378 |
| 14 | 0.011001654161769394 | 0.07861328125       | 0.9999395211537679 | 0.008353589544518926  | 0.03125             | 0.9999650915463766  | 0.19730008481144154  | 1.3958333333333333 | 0.9810690879821777 | 0.1483733553174941  | 0.400390625        | 0.98916648 |
| 15 | 0.013053667027162444 | 0.125               | 0.9999147256215414 | 0.00862563903468324   | 0.02473958333333332 | 0.9999628067016602  | 0.2277978499317006   | 2.0                | 0.9747335513432821 | 0.15369726088500632 | 0.4329427083333333 | 0.98838678 |
| 16 | 0.011193248157172894 | 0.08268229166666667 | 0.999937375386556  | 0.007716260309685617  | 0.03125             | 0.9999702572822571  | 0.1999725899214028   | 1.5260416666666667 | 0.9805418451627096 | 0.13669881390899946 | 0.4623958333333333 | 0.99078198 |
| 17 | 0.01018292430981959  | 0.06575520833333333 | 0.9999481836954752 | 0.007627289382004954  | 0.03125             | 0.9999709129333496  | 0.18242999289947706  | 1.0052083333333333 | 0.983751555474345  | 0.13526120207808442 | 0.41015625         | 0.99097061 |
| 18 | 0.01201434373008962  | 0.07063802083333333 | 0.9999277591705322 | 0.007407968263558014  | 0.09145833333333333 | 0.9999726017316183  | 0.21553672132610271  | 1.1875             | 0.977527409121195  | 0.13103341312011282 | 1.2890625          | 0.99152155 |
| 19 | 0.018634177089156323 | 0.21875             | 0.9998263915379842 | 0.007602853453642442  | 0.03125             | 0.9999710321426392  | 0.2501830231050708   | 3.5364583333333335 | 0.968186215400066  | 0.13460127697731009 | 0.478515625        | 0.99105701 |
| 20 | 0.01050071995165998  | 0.0625              | 0.9999448458353678 | 0.0073896273687530795 | 0.03125             | 0.9999727010726929  | 0.18801723695843195  | 0.9791666666666666 | 0.9827547272046407 | 0.13053609020046286 | 0.4869791666666667 | 0.99158521 |
| 21 | 0.01290902463173418  | 0.07421875          | 0.999916692574819  | 0.007146695237494211  | 0.03645833333333336 | 0.9999744693438212  | 0.23257763386425245  | 1.2864583333333333 | 0.9739675521805086 | 0.1258348261805026  | 0.5911458333333334 | 0.99217234 |
| 22 | 0.011473560912498806 | 0.0625              | 0.9999342560768127 | 0.007065692314973008  | 0.03548170833333336 | 0.9999750852584839  | 0.20513077135793778  | 1.1536458333333333 | 0.9795591831207275 | 0.1242763933899469  | 0.6197916666666666 | 0.99236428 |
| 23 | 0.0121743167608357   | 0.07291666666666667 | 0.9999184608459473 | 0.007686940921028044  | 0.0625              | 0.99997046758326212 | 0.23004838340065273  | 1.28125            | 0.9744954506556193 | 0.1361693916172622  | 0.9778645833333334 | 0.99085471 |
| 24 | 0.01427901433809712  | 0.0732421875        | 0.999880959525469  | 0.007437806792354194  | 0.0625              | 0.9999723633130392  | 0.25676513265342643  | 1.2786458333333333 | 0.9684377312660217 | 0.13149509757520149 | 0.9361979166666666 | 0.99146016 |
| 25 | 0.013827251611995967 | 0.07421875          | 0.9999043742815653 | 0.007417999651565373  | 0.10172526041666667 | 0.9999724825223287  | 0.24754965256914552  | 1.28125            | 0.9705677032477073 | 0.1309998779561864  | 1.7630283333333333 | 0.99152811 |
| 26 | 0.012751642553431095 | 0.06705729166666667 | 0.9999187191327413 | 0.00689294279014543   | 0.125               | 0.9999763568242391  | 0.22762505679700876  | 1.2213541666666667 | 0.9749536116917928 | 0.12105376523700258 | 1.8776041666666667 | 0.99275177 |
| 27 | 0.011449116881101779 | 2.0                 | 0.9999345342318217 | 0.007021981208610715  | 0.2838541666666667  | 0.999975343545278   | 0.09678420708194683  | 29.833333333333332 | 0.9955188830693563 | 0.12316871175967231 | 5.125              | 0.99250108 |

Per-Position Quantization Error

| Position Group | INT8 Key rel_I2      | INT8 Key max_abs     | INT8 Key cos_sim   | INT8 Val rel_I2      | INT8 Val max_abs     | INT8 Val cos_sim   | INT4 Key rel_I2     | INT4 Key max_abs   | INT4 Key cos_sim   | INT4 Val rel_I2     | INT4 Val max_abs    | INT4 Val |
|----------------|----------------------|----------------------|--------------------|----------------------|----------------------|--------------------|---------------------|--------------------|--------------------|---------------------|---------------------|----------|
| sink           | 0.017040837269560728 | 0.14299437747381272  | 0.9998484750588735 | 0.021066411441386646 | 0.018825904216750393 | 0.9997647291138059 | 0.16986574461978707 | 2.4955096640402363 | 0.9849757211548942 | 0.1935849165557419  | 0.32559266766267164 | 0.980505 |
| middle         | 0.01276936313857728  | 0.153081661920271103 | 0.9999163200179625 | 0.008009610747341337 | 0.0156651072398519   | 0.999966151866303  | 0.21340983173659725 | 2.7115254097354677 | 0.9771264078883397 | 0.14469721676691993 | 0.2843554452466878  | 0.989355 |
| recent         | 0.012709339301132591 | 0.1537569921050944   | 0.999917078681372  | 0.007969789373422768 | 0.015667306872054036 | 0.9999670177670844 | 0.21287685718821778 | 2.714464255823539  | 0.9772227111344154 | 0.14410404014486874 | 0.2844353172511899  | 0.989455 |

Per-Importance Tier Quantization Error

| Importance Tier | INT8 Key rel_I2      | INT8 Key max_abs     | INT8 Key cos_sim   | INT8 Val rel_I2      | INT8 Val max_abs     | INT8 Val cos_sim   | INT4 Key rel_I2     | INT4 Key max_abs   | INT4 Key cos_sim   | INT4 Val rel_I2     | INT4 Val max_abs    | INT4 Val |
|-----------------|----------------------|----------------------|--------------------|----------------------|----------------------|--------------------|---------------------|--------------------|--------------------|---------------------|---------------------|----------|
| high            | 0.01302311408649304  | 0.1532294329460668   | 0.9999123945866042 | 0.008555694627473836 | 0.015820651815395353 | 0.9999576708694704 | 0.21229072432960408 | 2.711137152709861  | 0.977272367666638  | 0.1458564116845108  | 0.2864780656300705  | 0.989    |
| medium          | 0.012733335165875728 | 0.153231149956744376 | 0.9999167502652503 | 0.00798758262221755  | 0.01549664817831416  | 0.9999668465500258 | 0.2129765248248783  | 2.7082631885675363 | 0.9772105091565999 | 0.14440030256030517 | 0.281359265712095   | 0.989    |
| low             | 0.012757739654140978 | 0.1531663441730908   | 0.999916504157669  | 0.00801282856915532  | 0.01570286468374371  | 0.9999665992806466 | 0.21325892339005958 | 2.7131967032319112 | 0.9771584056716714 | 0.14473072315581725 | 0.28502721342589277 | 0.989    |

KV Comparison (Key vs Value全局误差)

| KV Type | INT8 rel_I2          | INT8 max_abs        | INT8 cos_sim       | INT4 rel_I2         | INT4 max_abs        | INT4 cos_sim       |
|---------|----------------------|---------------------|--------------------|---------------------|---------------------|--------------------|
| key     | 0.012879985795028989 | 0.2053164527529762  | 0.9999150598332995 | 0.21387591997111302 | 3.11351376488009526 | 0.9770456091279075 |
| value   | 0.008167985609755995 | 0.04536220005580357 | 0.9999659763915199 | 0.14465842360587833 | 0.7382696242595523  | 0.9894926370609374 |

Per-Head Quantization Error (每层多头平均±标准差)

| Layer | INT8 Key rel_I2 μ    | INT8 Key rel_I2 σ     | INT8 Key max_abs μ   | INT8 Key max_abs σ    | INT8 Key cos_sim μ | INT8 Key cos_sim σ     | INT4 Key rel_I2 μ   | INT4 Key rel_I2 σ    | INT4 Key max_abs μ  | INT4 Key max_abs σ   | INT4 Key cos_sim μ | INT4 Key cos_sim σ     |
|-------|----------------------|-----------------------|----------------------|-----------------------|--------------------|------------------------|---------------------|----------------------|---------------------|----------------------|--------------------|------------------------|
| 0     | 0.01296008079366079  | 0.0037612188079467644 | 0.4914800673723221   | 0.12275476789898952   | 0.9999651536345482 | 6.254120092826797e-05  | 0.12947509943864366 | 0.0553749711695659   | 8.898438215255737   | 2.2089866488366754   | 0.9904478266835213 | 0.00010000000000000000 |
| 1     | 0.009666710439542568 | 0.0012240825406553213 | 0.11377153359353542  | 0.08608175885129715   | 0.9999581277370453 | 3.275545040138742e-06  | 0.1592226826919641  | 0.012905621551302078 | 2.0667550414800644  | 1.561371007856503    | 0.9874491095542908 | 0.00010000000000000000 |
| 2     | 0.01175323039481728  | 0.002858807743692071  | 0.07663903385400772  | 0.03379970181809406   | 0.9999226704239845 | 2.830978962817594e-05  | 0.20657027322494978 | 0.047797945032381606 | 1.3922990411520004  | 0.6103379047442364   | 0.9783293157815933 | 0.00010000000000000000 |
| 3     | 0.01614238575786785  | 0.0031618918505490466 | 0.1499260887503624   | 0.09114589340961654   | 0.9998692125082016 | 4.491247310733282e-05  | 0.25546292436900417 | 0.04117165303988416  | 2.724888503551483   | 1.649075152068884    | 0.9675961434841156 | 0.00010000000000000000 |
| 4     | 0.011745129488093602 | 0.0018604844452505    | 0.045406341552734375 | 0.0032422395695427837 | 0.9999243691563606 | 2.2279571955742928e-05 | 0.21238861193384956 | 0.03409897259620008  | 0.8246372640132904  | 0.05924711391267958  | 0.9776450544595718 | 0.00010000000000000000 |
| 5     | 0.013466924290031115 | 0.0027478407467779005 | 0.0519980796044445   | 0.01254443609898095   | 0.9999015852808952 | 3.543939050318294e-05  | 0.24313208738352693 | 0.04773178582771447  | 0.9471261203289032  | 0.22863478818344976  | 0.9706695973873138 | 0.00010000000000000000 |
| 6     | 0.01250422835811659  | 0.0017795812328196636 | 0.05222402326762676  | 0.010686437434892766  | 0.9999160841107368 | 2.1673019242491522e-05 | 0.22541373820881958 | 0.031128331931139067 | 0.9215960055589676  | 0.15563297383733918  | 0.975015439093113  | 0.00010000000000000000 |
| 7     | 0.013743288518185922 | 0.001828649015190865  | 0.058239069767296314 | 0.002671563906453997  | 0.9999014586210251 | 2.157837867646147e-05  | 0.24879130916915    | 0.0332552216409722   | 1.0643135607242584  | 0.052100804742573864 | 0.969944640994072  | 0.00010000000000000000 |
| 8     | 0.01147764399042696  | 0.0024679939147521125 | 0.050256417132914066 | 0.008724098301202948  | 0.999926820397377  | 2.7603237750819784e-05 | 0.20848276686548103 | 0.04513228655256238  | 0.9079241156578064  | 0.15106798852576434  | 0.9781659469008446 | 0.00010000000000000000 |
| 9     | 0.012754757065843532 | 0.0016551859493389828 | 0.06192425638437271  | 0.005644040788316733  | 0.9999126642942429 | 2.0137611393699564e-05 | 0.23126767761392497 | 0.02967271526900958  | 1.121512234210968   | 0.10350839386861482  | 0.973943904042244  | 0.00010000000000000000 |
| 10    | 0.01364610043746997  | 0.0014618407662605245 | 0.056694433093070984 | 0.00488214931206664   | 0.9999027484243713 | 1.922843438529532e-05  | 0.24676497733986985 | 0.026090655685577654 | 1.037039626644516   | 0.09191254871855215  | 0.9706465005874634 | 0.00010000000000000000 |
| 11    | 0.012757147159094942 | 0.0009039483105706648 | 0.05138777755200863  | 0.005251426116899108  | 0.9999131187796593 | 1.1434034429432513e-05 | 0.23112146305182393 | 0.01634947273444377  | 0.9344307333230972  | 0.09181415805373754  | 0.9742257595062256 | 0.00010000000000000000 |
| 12    | 0.0104205609850529   | 0.0013406337138235746 | 0.054863397032022476 | 0.017709731335440237  | 0.9999400824308395 | 1.2280392250804252e-05 | 0.18860980960949783 | 0.02403585382724568  | 0.95703122019167776 | 0.2654531843010617   | 0.9824159815907478 | 0.00010000000000000000 |
| 13    | 0.014192356309245728 | 0.003421854722705401  | 0.09882118180394173  | 0.04981082435654971   | 0.9998963698744774 | 4.211714493056699e-05  | 0.2342400485523921  | 0.03396087444692428  | 1.779575765132904   | 0.8922752354003132   | 0.9726373627781868 | 0.00010000000000000000 |
| 14    | 0.010690145774930854 | 0.001896437385353604  | 0.05954195559024811  | 0.011580010087141416  | 0.9999369457364082 | 1.9602486046035647e-05 | 0.193537238323765   | 0.03400173995090616  | 1.076171875         | 0.19879919721104472  | 0.9813037142157555 | 0.00010000000000000000 |
| 15    | 0.01114663400652684  | 0.003035288997726396  | 0.060550276190042496 | 0.030053749832685415  | 0.9999303445219994 | 3.082983540761156e-05  | 0.19932533587364298 | 0.05078158243845007  | 1.0953543335199356  | 0.5358757759045661   | 0.9794665202498436 | 0.00010000000000000000 |
| 16    | 0.01068320995617273  | 0.0017786548691189042 | 0.06696036830544472  | 0.015358714662013771  | 0.9999368786811829 | 1.7906232679301267e-05 | 0.19263775991461735 | 0.031373820629689055 | 1.1954519748687744  | 0.2780728438972307   | 0.9815095006214142 | 0.00010000000000000000 |
| 17    | 0.00996837055581776  | 0.0016068525712907519 | 0.0546335068202734   | 0.00799935662804424   | 0.9999434649944305 | 1.635960206010764e-05  | 0.1808409012472727  | 0.0291706364931781   | 0.9259207546710968  | 0.08292313118956135  | 0.9836837947368622 | 0.00010000000000000000 |
| 18    | 0.011697188692527629 | 0.0021595640476319146 | 0.0541636526616573   | 0.007987487745981178  | 0.9999260231852531 | 2.5348805654198213e-05 | 0.21169635657356484 | 0.03863908047442972  | 0.9845145046710968  | 0.14629498630864123  | 0.9777055755257607 | 0.00010000000000000000 |
| 19    | 0.013668400806174433 | 0.00440655115179555   | 0.09880700521171093  | 0.05918145245016537   | 0.9998990744352341 | 6.137272170290339e-05  | 0.21772854443112966 | 0.03520929868670057  | 1.7341657131910324  | 1.0799313842339102   | 0.9761552810668945 | 0.00010000000000000000 |
| 20    | 0.010452498250902945 | 0.0007692250340555172 | 0.0517597533762455   | 0.005522782401680717  | 0.9999390244483948 | 8.607042792408322e-06  | 0.18936726447348914 | 0.01412110251368358  | 0.932896226644516   | 0.09763301011461026  | 0.982451319694519  | 0.00010000000000000000 |
| 21    | 0.012958068653196268 | 0.0014343794995057245 | 0.06441660411655903  | 0.006519519910096765  | 0.9999119341373444 | 1.9035310463916853e-05 | 0.2346705560332999  | 0.025916688589112013 | 1.170619398355484   | 0.11754891085359762  | 0.9732683151960373 | 0.00010000000000000000 |
| 22    | 0.011333243036498127 | 0.0014645086896474212 | 0.05060621271878481  | 0.005968710838749921  | 0.9999310672283173 | 1.4839611490939693e-05 | 0.2042423306589132  | 0.02622738604611563  | 1.0161830484867096  | 0.10736935183878926  | 0.9794538244055064 | 0.00010000000000000000 |
| 23    | 0.012556216073080311 | 0.0019465145647900948 | 0.05981831066310406  | 0.007457602421667781  | 0.9999157786369324 | 2.4076587800499526e-05 | 0.22766829367649483 | 0.03480362510518756  | 1.0828683227300644  | 0.13300227626542127  | 0.9745339304208755 | 0.00010000000000000000 |
| 24    | 0.014045151852476612 | 0.0017921902797710214 | 0.060895755887031555 | 0.00836515926462982   | 0.999898225069046  | 2.387422733576814e-05  | 0.25344855271118403 | 0.03264675282846326  | 1.1070730984210968  | 0.1528086611760057   | 0.9688132628798485 | 0.00010000000000000000 |
| 25    | 0.013149377188791632 | 0.0021846625449377295 | 0.0602315841242671   | 0.008704151333778193  | 0.9999074712395668 | 2.757590930322806e-05  | 0.23665496236380987 | 0.039656406956186344 | 1.0946568101644516  | 0.158546462917817    | 0.9723895564675331 | 0.00010000000000000000 |
| 26    | 0.012645971104943487 | 0.0018983467985734053 | 0.05577194970101118  | 0.008445295943507968  | 0.9999145120382309 | 2.4400618467599833e-05 | 0.2271373353055288  | 0.0339964671232765   | 1.0133230984210968  | 0.15615244028521882  | 0.9745943158563614 | 0.00010000000000000000 |
| 27    | 0.011664688325126684 | 0.0002949826138948371 | 0.045458219293504953 | 0.6890261287056627    | 0.9999425485730171 | 2.6708242701871315e-05 | 0.18186128850946062 | 0.04950852620059615  | 8.243582427501678   | 12.49934139579438    | 0.9827740862965584 | 0.00010000000000000000 |

|    |                     |                     |                     |
|----|---------------------|---------------------|---------------------|
| 6  | 0.22855546031841026 | 0.16233000036503603 | 0.22855546031841026 |
| 7  | 0.24498992798040944 | 0.12531695397499068 | 0.24498992798040944 |
| 8  | 0.21556439071754321 | 0.16638302156640705 | 0.21556439071754321 |
| 9  | 0.2319145087469612  | 0.12023154186294761 | 0.2319145087469612  |
| 10 | 0.24813172889615034 | 0.14009766713281321 | 0.24813172889615034 |
| 11 | 0.2292125137890568  | 0.16038066356012456 | 0.2292125137890568  |
| 12 | 0.19795811225671275 | 0.13575907102678923 | 0.19795811225671275 |
| 13 | 0.25557966894484535 | 0.13417023092691055 | 0.25557966894484535 |
| 14 | 0.19730008481144154 | 0.1483733553174941  | 0.19730008481144154 |
| 15 | 0.2277978499317006  | 0.15369726088500632 | 0.2277978499317006  |
| 16 | 0.19997255899214028 | 0.13669881390989946 | 0.19997255899214028 |
| 17 | 0.18242999289947706 | 0.13526120207808442 | 0.18242999289947706 |
| 18 | 0.21553671232610271 | 0.13103341312011282 | 0.21553671232610271 |
| 19 | 0.2501830231050708  | 0.13460127697731009 | 0.2501830231050708  |
| 20 | 0.18801723695843195 | 0.13053609020046286 | 0.18801723695843195 |
| 21 | 0.23257763386425245 | 0.1258348261805026  | 0.23257763386425245 |
| 22 | 0.20513077135793778 | 0.12427663933899469 | 0.20513077135793778 |
| 23 | 0.23004838340065273 | 0.1361693916172622  | 0.23004838340065273 |
| 24 | 0.25676513265342643 | 0.1314950977520149  | 0.25676513265342643 |
| 25 | 0.24754965256914552 | 0.13099998779561864 | 0.24754965256914552 |
| 26 | 0.22762505679700876 | 0.12105376523700258 | 0.22762505679700876 |

Importance-Error Correlation

int8: pearson\_r=0.0 - high: 0.0 - medium: 0.0 - low: 0.0  
int4: pearson\_r=0.0 - high: 0.0 - medium: 0.0 - low: 0.0

Mistral-7B 逐层量化误差

- seq\_len: 2048
- num\_samples: 3
- num\_layers: 36
- sink\_count: 4
- sensitivity\_threshold: 0.15
- token\_sample\_step: 8
- timestamp: 2026-06-04 19:04:14

Per-Layer Quantization Error

| Layer | INT8 Key rel_L2       | INT8 Key max_abs     | INT8 Key cos_sim   | INT8 Val rel_L2       | INT8 Val max_abs     | INT8 Val cos_sim   | INT4 Key rel_L2      | INT4 Key max_abs   | INT4 Key cos_sim   | INT4 Val rel_L2      | INT4 Val max_abs    | INT4 Val |
|-------|-----------------------|----------------------|--------------------|-----------------------|----------------------|--------------------|----------------------|--------------------|--------------------|----------------------|---------------------|----------|
| 0     | 0.008079455927427558  | 0.041666666666666664 | 0.9999673962593079 | 0.012359802449087623  | 0.001953125          | 0.9999236067136129 | 0.1453050275917731   | 0.6653645833333334 | 0.9895923137664795 | 0.19917138497788     | 0.0255126953125     | 0.980441 |
| 1     | 0.0074803569402428745 | 0.052083333333333336 | 0.9999720056851705 | 0.009626738091482767  | 0.00390625           | 0.9999536871910095 | 0.130847445444434648 | 0.6705729166666666 | 0.9915528496106466 | 0.1717996079476797   | 0.06551106770833333 | 0.985545 |
| 2     | 0.0124812777439896424 | 0.0625               | 0.9999221563339233 | 0.009141699260185524  | 0.009756525          | 0.9999581972757975 | 0.12414189507620264  | 1.1302083333333333 | 0.9759079846700033 | 0.1630812136020677   | 0.17024739583333334 | 0.986954 |
| 3     | 0.010599606963263635  | 0.0625               | 0.9999438325564066 | 0.008191798694980027  | 0.0078125            | 0.9999664227167765 | 0.1898642182237497   | 1.0065104166666667 | 0.9824726780255636 | 0.14548031771234185  | 0.1298828125        | 0.989587 |
| 4     | 0.011578396939305813  | 0.0625               | 0.9999329845110575 | 0.00780260185855854   | 0.009684244791666666 | 0.9999695618947347 | 0.20795429932338313  | 0.9790464838345846 | 0.9790464838345846 | 0.1383347431602303   | 0.1640625           | 0.990565 |
| 5     | 0.011604271307994178  | 0.0625               | 0.9999326666196188 | 0.008023857203630514  | 0.01171875           | 0.9999678333600363 | 0.20867731585610372  | 1.0859375          | 0.9788947701454163 | 0.12204345693463677  | 0.1640625           | 0.992635 |
| 6     | 0.01212395809006121   | 0.08333333333333333  | 0.9999265074729919 | 0.008097845677251726  | 0.0111083984375      | 0.9999672969182333 | 0.21752719436037316  | 1.2838541666666667 | 0.9771188894907633 | 0.14367773732155686  | 0.18619791666666666 | 0.989831 |
| 7     | 0.011207398585897338  | 0.078125             | 0.9999371965726217 | 0.00792453834280628   | 0.014616666666666666 | 0.9999686479568481 | 0.20057160107768798  | 1.2734375          | 0.9804451664288839 | 0.14036719625003918  | 0.17610677083333334 | 0.990290 |
| 8     | 0.01208228693612238   | 0.0625               | 0.9999270439147949 | 0.0069778732603826025 | 0.009404104166666666 | 0.9999757210413615 | 0.21737788432989788  | 1.09375            | 0.977163294951121  | 0.12204345693463677  | 0.1640625           | 0.992635 |
| 9     | 0.012499055989678145  | 0.0703125            | 0.9999218781789144 | 0.00811063976925946   | 0.011067708333333334 | 0.999967118104299  | 0.2248484100316297   | 1.2265625          | 0.9756291111310323 | 0.143797764736826    | 0.18294270833333334 | 0.989810 |
| 10    | 0.011478683490209555  | 0.0693359375         | 0.999938984711965  | 0.0075175114401729289 | 0.0107421875         | 0.9999717473983765 | 0.19816535991998666  | 1.2369791666666667 | 0.9809154470761617 | 0.1325375862531404   | 0.1826171875        | 0.991325 |
| 11    | 0.011314673414537908  | 0.0703125            | 0.9999360044797262 | 0.008605617064701934  | 0.013020833333333334 | 0.9999630451202393 | 0.203128580637171376 | 1.1796875          | 0.980020654996237  | 0.1532773080274995   | 0.20084635416666666 | 0.988452 |
| 12    | 0.011397407200248963  | 0.07552083333333333  | 0.9999350706736246 | 0.007641717476377443  | 0.013020833333333334 | 0.999970813592275  | 0.20500387631040184  | 1.2239583333333333 | 0.9796182115872701 | 0.13518095617055523  | 0.19889322916666666 | 0.990984 |
| 13    | 0.011478683490209555  | 0.084635416666666667 | 0.9999341368675232 | 0.008288921845865885  | 0.015625             | 0.9999657074610392 | 0.20519879820113843  | 1.4713541666666667 | 0.9795516133308411 | 0.14742562667072004  | 0.2373046875        | 0.983030 |
| 14    | 0.013067408689909556  | 0.08072916666666667  | 0.9999146858851115 | 0.008077380739863258  | 0.015625             | 0.9999673763910929 | 0.2484196126860419   | 1.4270833333333333 | 0.9734167456626892 | 0.1434142120425166   | 0.24544270833333334 | 0.989877 |
| 15    | 0.010640314679605814  | 0.07649739583333333  | 0.9999433954556783 | 0.00749117746555025   | 0.0166015625         | 0.9999719262123108 | 0.19050654395378522  | 1.2317708333333333 | 0.982323169708252  | 0.13243724074234034  | 0.2955729166666667  | 0.991342 |
| 16    | 0.011274019383589637  | 0.078125             | 0.9999364614486694 | 0.007610484188772001  | 0.015625             | 0.9999710321426392 | 0.20080652118854564  | 1.34375            | 0.9803897397138876 | 0.134668782079821084 | 0.2643229166666667  | 0.991045 |
| 17    | 0.011947366192897343  | 0.078125             | 0.9999286731084188 | 0.00685586729661872   | 0.015625             | 0.9999765356381735 | 0.21497846132013065  | 1.3723958333333333 | 0.9776379863421122 | 0.1201639966466534   | 0.2584635416666667  | 0.992851 |
| 18    | 0.01133100186809736   | 0.07552083333333333  | 0.9999379913012186 | 0.0071563978973392325 | 0.015625             | 0.9999743898709615 | 0.1995429107691364   | 1.265625           | 0.9806432525316874 | 0.12609188815637573  | 0.24934895833333334 | 0.992142 |
| 19    | 0.0115874771216089    | 0.069010416666666667 | 0.9999328255653381 | 0.007409761073921572  | 0.015950520833333332 | 0.9999725818634033 | 0.20845532548026982  | 1.2265625          | 0.9789491295814514 | 0.131049079386394    | 0.283203125         | 0.991520 |
| 20    | 0.011851586403386585  | 0.08333333333333333  | 0.9999297857284546 | 0.007510061718568582  | 0.0234375            | 0.9999718070030212 | 0.2122857660995157   | 1.4010416666666667 | 0.9781676530838013 | 0.1325601524957849   | 0.3795572916666667  | 0.991326 |
| 21    | 0.012287521775074793  | 0.078125             | 0.9999245405197144 | 0.00756863196623294   | 0.016927083333333332 | 0.999971389705078  | 0.2205380876834758   | 1.265625           | 0.9765006307877231 | 0.13377081936287705  | 0.3001302083333333  | 0.991160 |
| 22    | 0.011378255815888519  | 0.08333333333333333  | 0.9999352892239889 | 0.007798477693693702  | 0.023600260416666668 | 0.9999695817629496 | 0.2046766489154034   | 1.3020833333333333 | 0.9796702265739441 | 0.13797386075540863  | 0.396484375         | 0.990614 |
| 23    | 0.01246712186900366   | 0.08984375           | 0.9999223152796427 | 0.007460774728766266  | 0.01953125           | 0.9999671844991049 | 0.22356769634846196  | 1.5494791666666667 | 0.9758732318878174 | 0.1316595800285527   | 0.345703125         | 0.991440 |
| 24    | 0.012388920387841377  | 0.078125             | 0.9999232490857443 | 0.0077939554337446105 | 0.022135416666666668 | 0.9999696413675944 | 0.2214916681526136   | 1.2552083333333333 | 0.976263145605723  | 0.137973064459987    | 0.3776041666666667  | 0.990611 |
| 25    | 0.013107042456814289  | 0.078125             | 0.9999144494433085 | 0.007037359944952769  | 0.03125              | 0.9999752243359884 | 0.23566992414566337  | 1.3229166666666667 | 0.9732633630434672 | 0.12387936373980014  | 0.3723958333333333  | 0.992412 |
| 26    | 0.012810726903100483  | 0.08072916666666667  | 0.9999179244044143 | 0.006756685387207082  | 0.03125              | 0.9999771281289266 | 0.23081800339331377  | 1.3515625          | 0.9747315127309164 | 0.118091880835712    | 0.4453125           | 0.993095 |
| 27    | 0.011721580724048032  | 0.078125             | 0.9999313354921188 | 0.007146735429391845  | 0.023600260416666668 | 0.9999744892120361 | 0.21076065163746036  | 1.2890625          | 0.9784980018933614 | 0.12575811534648307  | 0.3528645833333333  | 0.992184 |
| 28    | 0.012455960368490367  | 0.069010416666666667 | 0.9999224742253622 | 0.0074770138598144855 | 0.022135416666666668 | 0.9999720454216003 | 0.22435289085248714  | 1.2239583333333333 | 0.9757319092750549 | 0.13201653855517917  | 0.3717447916666667  | 0.991394 |
| 29    | 0.011644645477620633  | 0.0703125            | 0.9999322096506754 | 0.006912000008191578  | 0.028645833333333332 | 0.9999761382738749 | 0.20956430818849023  | 1.2786458333333333 | 0.9787387053171793 | 0.1213529745772865   | 0.4173177083333333  | 0.992711 |
| 30    | 0.012982634185644638  | 0.078125             | 0.9999157587687174 | 0.008603032129588817  | 0.03125              | 0.9999629855155945 | 0.23155843297801548  | 1.3255208333333333 | 0.9740554292996725 | 0.15298073962591618  | 0.4947916666666667  | 0.988490 |
| 31    | 0.011448849580515915  | 0.07671875           | 0.9999344944953918 | 0.007011300908075052  | 0.032552083333333336 | 0.9999754627545675 | 0.2057654883026122   | 1.3645833333333333 | 0.979474147160848  | 0.12312731428203819  | 0.5442708333333334  | 0.992496 |
| 32    | 0.01245848540861965   | 0.093175             | 0.9999223947525024 | 0.007667582262321451  | 0.03515625           | 0.9999705950419108 | 0.21807284680160885  | 1.4869791666666667 | 0.9768872769793811 | 0.1356481484315173   | 0.6067708333333334  | 0.990924 |
| 33    | 0.011252062384102085  | 0.078125             | 0.9999367197354635 | 0.006872499706162288  | 0.04296875           | 0.999976376692454  | 0.2017180534339378   | 1.3463541666666667 | 0.9802477161089579 | 0.12066689720160374  | 0.75390625          | 0.992791 |
| 34    | 0.011334754794225489  | 0.0703125            | 0.9999357461929321 | 0.006818389502776229  | 0.040364583333333336 | 0.9999767541885376 | 0.2025570923982705   | 1.2161458333333333 | 0.980086922455688  | 0.11949027815087905  | 0.6861979166666666  | 0.992931 |
| 35    | 0.00758283783871636   | 0.046875             | 0.9999712705612183 | 0.00672265187623716   | 0.03125              | 0.9999774495760599 | 0.1337932567884545   | 0.7213541666666666 | 0.9911649425824484 | 0.11785103850715355  | 0.5455729166666666  | 0.993121 |

Per-Position Quantization Error

| Position Group | INT8 Key rel_L2      | INT8 Key max_abs     | INT8 Key cos_sim   | INT8 Val rel_L2       | INT8 Val max_abs      | INT8 Val cos_sim   | INT4 Key rel_L2     | INT4 Key max_abs   | INT4 Key cos_sim   | INT4 Val rel_L2     | INT4 Val max_abs    | INT4   |
|----------------|----------------------|----------------------|--------------------|-----------------------|-----------------------|--------------------|---------------------|--------------------|--------------------|---------------------|---------------------|--------|
| sink           | 0.01154160004907138  | 0.018968010907647787 | 0.999928840332561  | 0.013596074089662383  | 0.003797355852805561  | 0.9998874956810916 | 0.17690047071983342 | 0.3428580511223387 | 0.9843344191710154 | 0.20214399716277753 | 0.06699084911357474 | 0.9775 |
| middle         | 0.011319117871170696 | 0.0482501863117168   | 0.9999349760939958 | 0.0075674870169043975 | 0.008559092704104345  | 0.9999706410520867 | 0.20477115710917656 | 0.875886572612391  | 0.9793774578057516 | 0.1367436811693662  | 0.1553749237505558  | 0.9900 |
| recent         | 0.011324359540367408 | 0.049431292985916205 | 0.9999349078434145 | 0.007576088293035744  | 0.0085608075047829951 | 0.9999705696335206 | 0.20465132639595439 | 0.8972508034475168 | 0.9793655773619817 | 0.13690685401759567 | 0.1556157213386992  | 0.9900 |

|    |                      |                        |                      |                       |                    |                        |                      |                      |                    |                     |                    |       |
|----|----------------------|------------------------|----------------------|-----------------------|--------------------|------------------------|----------------------|----------------------|--------------------|---------------------|--------------------|-------|
| 3  | 0.010264686482673091 | 0.0016409023774359285  | 0.04282144736498594  | 0.007758562387077046  | 0.9999418705701828 | 1.7837573788829257e-05 | 0.186270377980001    | 0.029802185232409777 | 0.7790527194738388 | 0.13883450284567225 | 0.98277697712183   | 0.005 |
| 4  | 0.011219077660917554 | 0.0020157677562263917  | 0.045615379232913256 | 0.006878496284407122  | 0.9999302625656128 | 2.1314203435053274e-05 | 0.20311244480724736  | 0.03613286564005986  | 0.8327287882566452 | 0.1252665098129533  | 0.9794440232217312 | 0.006 |
| 5  | 0.01135560078439372  | 0.0017006438210190272  | 0.047879207879304886 | 0.00603315791938642   | 0.9999293982982635 | 1.8396218277175863e-05 | 0.20575614513693163  | 0.030573640757109243 | 0.8714076206088066 | 0.118179428607773   | 0.9790939688682556 | 0.005 |
| 6  | 0.011773216123201952 | 0.0016854408894205724  | 0.050450123846530914 | 0.008271594192771698  | 0.9999256357550621 | 1.9930478567616593e-05 | 0.21323355480326964  | 0.02936865775658949  | 0.9132951945066452 | 0.14988946198315722 | 0.97763916477561   | 0.006 |
| 7  | 0.010863459653275235 | 0.001926113816974108   | 0.052211152389645576 | 0.012048929530372092  | 0.9999340921640396 | 2.2335190406672497e-05 | 0.19645988192073416  | 0.03459850288663791  | 0.9482421651482582 | 0.21981925587630358 | 0.9807319417595863 | 0.007 |
| 8  | 0.011769274806152336 | 0.001944640300566594   | 0.04728974401950836  | 0.008387714052894568  | 0.9999241232872009 | 2.3412707123275535e-05 | 0.21355373750795936  | 0.03524914975033784  | 0.8572823405265808 | 0.1533290721297216  | 0.9774473235011101 | 0.007 |
| 9  | 0.012224740607081053 | 0.002076290164392784   | 0.05285740131512284  | 0.009107306952811233  | 0.9999188967049122 | 2.539879840388607e-05  | 0.2214617778061748   | 0.03742076792010978  | 0.963065005838871  | 0.16534380792371992 | 0.9757969155907631 | 0.007 |
| 10 | 0.010934914041293405 | 0.00161771133097917516 | 0.053300525061786175 | 0.008339256399334737  | 0.9999336041510105 | 1.8645616074874474e-05 | 0.19823685475905878  | 0.02918161685161983  | 0.9716099202632904 | 0.15011562191742212 | 0.9805633686482906 | 0.005 |
| 11 | 0.010955059001383223 | 0.0019570324191782516  | 0.05153191462159157  | 0.00951742871733462   | 0.9999335817992687 | 2.2657809416622537e-05 | 0.19847627724941447  | 0.03507695073290366  | 0.9342215210199356 | 0.1738343494349735  | 0.980399876832962  | 0.007 |
| 12 | 0.011147596608836772 | 0.0011137812507041756  | 0.055054083466529846 | 0.00898208955936707   | 0.9999324381351471 | 1.1655604083104061e-05 | 0.20220962778252086  | 0.020002509469687388 | 0.999770355427742  | 0.1661920731867382  | 0.9800025671720505 | 0.003 |
| 13 | 0.011262205370298624 | 0.0013926061967149028  | 0.06253650924190879  | 0.008832778028356334  | 0.999931339174509  | 1.5748680047661394e-05 | 0.20317055408547585  | 0.02525600382694219  | 1.1303013637661934 | 0.15908153419266785 | 0.9797025509178638 | 0.005 |
| 14 | 0.012787884088593928 | 0.002211020102711022   | 0.065239820163697    | 0.008051205321206407  | 0.9999120198190212 | 2.7837032029625362e-05 | 0.23122357230800025  | 0.03959416590152003  | 1.1811174750328064 | 0.14661478517386156 | 0.9735741764307022 | 0.008 |
| 15 | 0.010400169202529359 | 0.0007571864642342197  | 0.055268543772399426 | 0.008257685083776023  | 0.9999407231807709 | 7.569027190195394e-06  | 0.18832918070792215  | 0.013822808134640917 | 0.999965138733387  | 0.1496860401218294  | 0.982649028301239  | 0.002 |
| 16 | 0.011236547287994532 | 0.0010675245303896316  | 0.06576202437281609  | 0.0061993589660478444 | 0.9999327808618546 | 1.1790717404829e-05    | 0.20191501865633016  | 0.01917714579955167  | 1.1973004937171936 | 0.11508006353939676 | 0.980044286698103  | 0.003 |
| 17 | 0.011967638880899021 | 0.0016775557971527117  | 0.0648010652512312   | 0.006299299102863823  | 0.999921690672636  | 2.059981082806347e-05  | 0.21700202000257035  | 0.03037577822427683  | 1.1864536702632904 | 0.11198141220596529 | 0.9768793061375618 | 0.006 |
| 18 | 0.010824892912093948 | 0.001063219566495291   | 0.0580307194031775   | 0.005905392246234583  | 0.9999364912509918 | 1.1005735784808951e-05 | 0.19587849128698714  | 0.0189113540612817   | 1.053989939391613  | 0.10223626552171722 | 0.9811970852315426 | 0.003 |
| 19 | 0.011305512740382884 | 0.0013007960706511769  | 0.055860626976937056 | 0.007682660963315055  | 0.9999312944710255 | 1.4773202675138624e-05 | 0.204997373736038062 | 0.02348983183733397  | 1.018123636021614  | 0.13722024681994185 | 0.9794194288551807 | 0.004 |
| 20 | 0.011704081564601287 | 0.0018200355638907947  | 0.063338475599885    | 0.007952836104498831  | 0.9999260827898979 | 2.1349384864416556e-05 | 0.21141649032217438  | 0.03311702214420013  | 1.147774875164032  | 0.14220009799838884 | 0.9779131300747395 | 0.006 |
| 21 | 0.01208475724537865  | 0.002179055722399725   | 0.0599805558818579   | 0.005956206420230388  | 0.9999210499227047 | 2.794865995291207e-05  | 0.21837802583830207  | 0.03932056012590964  | 1.0910992920398712 | 0.11042402734753859 | 0.9763640314340591 | 0.008 |
| 22 | 0.0112669529380296   | 0.0017324178923545513  | 0.0621164683252573   | 0.006074400157481668  | 0.9999307692050934 | 1.9878450628020157e-05 | 0.20445806023720633  | 0.031541655165557306 | 1.126255601644516  | 0.1103351115706084  | 0.9793090894818306 | 0.006 |
| 23 | 0.011764691313944184 | 0.003049656203264369   | 0.057655283249914646 | 0.011469225255683234  | 0.9999224990606308 | 3.862399665340662e-05  | 0.21256777049039693  | 0.054792764180725616 | 1.0467703714966774 | 0.2123113385749947  | 0.9769062250852585 | 0.011 |
| 24 | 0.011706779589158771 | 0.0027786378791815228  | 0.05340239964425564  | 0.00982382257990143   | 0.999923050403595  | 3.4026683309813906e-05 | 0.21122261228698136  | 0.049363877749718608 | 0.9672502726316452 | 0.1778194197378453  | 0.9773561470210552 | 0.010 |
| 25 | 0.01261403203045603  | 0.0024891708604916976  | 0.06097652018070221  | 0.008956151582737578  | 0.9999141469597816 | 2.9680037517173725e-05 | 0.22824337366278802  | 0.04521540656726992  | 1.1033063975976752 | 0.1639527414370169  | 0.9740577898919582 | 0.009 |
| 26 | 0.012325959497483555 | 0.0022302595756282357  | 0.062233248725533485 | 0.009422666831836695  | 0.9999175779521465 | 2.619195392443746e-05  | 0.22356397572571451  | 0.040412728852751746 | 1.133161261677742  | 0.17268510479869895 | 0.9752695336937904 | 0.008 |
| 27 | 0.011312510179608373 | 0.002674187053818371   | 0.05367535538971424  | 0.00975624410120483   | 0.9999276846647263 | 3.2762130286568036e-05 | 0.20527330163157864  | 0.04857138718082593  | 0.9772251769900322 | 0.17583973266885014 | 0.9786393009126186 | 0.010 |
| 28 | 0.012240607358817022 | 0.0014520877050987046  | 0.05759505135938525  | 0.005866358829638504  | 0.9999184682965279 | 1.725810359918779e-05  | 0.22194030009668703  | 0.026296529438901102 | 1.045131171105484  | 0.10523816548355318 | 0.975977666760445  | 0.005 |
| 29 | 0.01117553070256675  | 0.001797090149637172   | 0.054624080657958984 | 0.008870555983452842  | 0.9999310411512852 | 1.9714235907486682e-05 | 0.2027319837333691   | 0.032723218745125744 | 0.994595146463871  | 0.1636899978436617  | 0.9796410724520683 | 0.006 |
| 30 | 0.012382049681737228 | 0.0034902262620723386  | 0.05551723297685385  | 0.011579597761087903  | 0.9999133460223675 | 4.455844486441417e-05  | 0.2230704748912808   | 0.06135272461186662  | 1.006033755838871  | 0.2076145202528367  | 0.9743212834000587 | 0.013 |
| 31 | 0.011084436048023491 | 0.0015481831442792175  | 0.0582487927749753   | 0.0103966991970534    | 0.9999323152005672 | 1.8101187860261464e-05 | 0.2009616975521344   | 0.028102067435901813 | 1.0560128390789032 | 0.18774812316357223 | 0.9800851307809353 | 0.005 |
| 32 | 0.012028437098922333 | 0.002414110615651322   | 0.05871867388486862  | 0.011560647890421936  | 0.9999210685491562 | 2.9187495198736658e-05 | 0.2143274246562546   | 0.03581767730465704  | 1.0669642835855484 | 0.20717909525741213 | 0.9771157093346119 | 0.007 |
| 33 | 0.010845616691858198 | 0.0020487868849887007  | 0.055975005496293306 | 0.010664087116478376  | 0.9999342747032642 | 2.5064388490185496e-05 | 0.1962881114747288   | 0.03723933077895121  | 1.0163225382566452 | 0.19530446752354674 | 0.9807193987071514 | 0.007 |
| 34 | 0.011028292404661478 | 0.001883915307278841   | 0.05251762829720974  | 0.007163352919577667  | 0.9999324418604374 | 2.186375605548501e-05  | 0.19908698394235738  | 0.034089274452147524 | 0.9548397369244194 | 0.13259272274730263 | 0.9802867881953716 | 0.006 |
| 35 | 0.007323399119336404 | 0.0005744505744122417  | 0.03316274844110012  | 0.004091907201139799  | 0.9999653957784176 | 4.320872665737637e-06  | 0.1327005571116005   | 0.010343755976690693 | 0.600237175822258  | 0.07201197382912826 | 0.9912631884217262 | 0.001 |

Sensitive Layers (INT4量化敏感层, threshold=0.15)

| Layer | INT4 Key rel_L2     | INT4 Value rel_L2   | Max rel_L2          |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.1453050275917731  | 0.19917138497788    | 0.19917138497788    |
| 1     | 0.13084744544434648 | 0.1717996079476797  | 0.1717996079476797  |
| 2     | 0.22414189507620264 | 0.1630812136020677  | 0.22414189507620264 |
| 3     | 0.1898642182237497  | 0.14548031771234185 | 0.1898642182237497  |
| 4     | 0.20795429932338313 | 0.1383347431602303  | 0.20795429932338313 |
| 5     | 0.20867731585610372 | 0.14232100435355205 | 0.20867731585610372 |
| 6     | 0.21752719436037316 | 0.14367773732155686 | 0.21752719436037316 |
| 7     | 0.20057160107768798 | 0.14036719625003918 | 0.20057160107768798 |
| 8     | 0.21737788432989788 | 0.12204345693463677 | 0.21737788432989788 |
| 9     | 0.2248484100316297  | 0.1437977644736826  | 0.2248484100316297  |
| 10    | 0.19816535591998666 | 0.1325375862513404  | 0.19816535591998666 |
| 11    | 0.20312858063717376 | 0.15327730802724995 | 0.20312858063717376 |
| 12    | 0.20500387631040184 | 0.13518095617055523 | 0.20500387631040184 |
| 13    | 0.20519879820113843 | 0.14742562667072004 | 0.20519879820113843 |
| 14    | 0.2348196126860419  | 0.1434142120425166  | 0.2348196126860419  |
| 15    | 0.19050564395378522 | 0.13243724074234034 | 0.19050564395378522 |
| 16    | 0.20080652118854694 | 0.13468782079821084 | 0.20080652118854694 |
| 17    | 0.21497846132013065 | 0.12016396966446534 | 0.21497846132013065 |
| 18    | 0.1995429107691364  | 0.12609188815637573 | 0.1995429107691364  |
| 19    | 0.20845532548026982 | 0.131049079386394   | 0.20845532548026982 |
| 20    | 0.21228857660957157 | 0.1325601524957849  | 0.21228857660957157 |
| 21    | 0.2205380876834758  | 0.13377081936287705 | 0.2205380876834758  |
| 22    | 0.2046766489154034  | 0.13797386075540863 | 0.2046766489154034  |
| 23    | 0.22356769634846196 | 0.1316595800285527  | 0.22356769634846196 |
| 24    | 0.22149166814526136 | 0.137937064459987   | 0.22149166814526136 |
| 25    | 0.23566992414656637 | 0.12387963673980014 | 0.23566992414656637 |
| 26    | 0.23081800333931377 | 0.1180918090835712  | 0.23081800333931377 |
| 27    | 0.21076065163746036 | 0.12575811534648307 | 0.21076065163746036 |
| 28    | 0.22435289085248714 | 0.13201653855517917 | 0.22435289085248714 |
| 29    | 0.20956430818849023 | 0.12135297457772865 | 0.20956430818849023 |
| 30    | 0.23155843297801548 | 0.15298073962591618 | 0.23155843297801548 |
| 31    | 0.2057654883026122  | 0.12317231428203819 | 0.2057654883026122  |
| 32    | 0.21807284680160885 | 0.1356481484315173  | 0.21807284680160885 |
| 33    | 0.20171805343393978 | 0.12066689720160374 | 0.20171805343393978 |







Exp4: Layer Budget Allocation (层级预算分配)

数据来源：exp4\_layer\_budget\_all.json, exp4\_layer\_budget\_mistral7b.json, exp4\_layer\_budget\_qwen7b.json

exp4\_layer\_budget\_all.json

beta=0.0, budget=0.3, model=mistral7b

- PPL: 8.8336
- PPL Δ%: 5.3
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.3    |
| 1     | 0.3    |
| 2     | 0.3    |
| 3     | 0.3    |
| 4     | 0.3    |
| 5     | 0.3    |
| 6     | 0.3    |
| 7     | 0.3    |
| 8     | 0.3    |
| 9     | 0.3    |
| 10    | 0.3    |
| 11    | 0.3    |
| 12    | 0.3    |
| 13    | 0.3    |
| 14    | 0.3    |
| 15    | 0.3    |
| 16    | 0.3    |
| 17    | 0.3    |
| 18    | 0.3    |
| 19    | 0.3    |
| 20    | 0.3    |
| 21    | 0.3    |
| 22    | 0.3    |
| 23    | 0.3    |
| 24    | 0.3    |
| 25    | 0.3    |
| 26    | 0.3    |
| 27    | 0.3    |
| 28    | 0.3    |
| 29    | 0.3    |
| 30    | 0.3    |
| 31    | 0.3    |
| 32    | 0.3    |
| 33    | 0.3    |
| 34    | 0.3    |
| 35    | 0.3    |

Tier Fractions Per Layer:

| Layer | FP16 | INT8   | INT4   |
|-------|------|--------|--------|
| 0     | 0.0  | 0.1997 | 0.8003 |
| 1     | 0.0  | 0.1997 | 0.8003 |
| 2     | 0.0  | 0.1997 | 0.8003 |
| 3     | 0.0  | 0.1997 | 0.8003 |
| 4     | 0.0  | 0.1997 | 0.8003 |
| 5     | 0.0  | 0.1997 | 0.8003 |
| 6     | 0.0  | 0.1997 | 0.8003 |
| 7     | 0.0  | 0.1997 | 0.8003 |
| 8     | 0.0  | 0.1997 | 0.8003 |
| 9     | 0.0  | 0.1997 | 0.8003 |
| 10    | 0.0  | 0.1997 | 0.8003 |
| 11    | 0.0  | 0.1997 | 0.8003 |
| 12    | 0.0  | 0.1997 | 0.8003 |
| 13    | 0.0  | 0.1997 | 0.8003 |
| 14    | 0.0  | 0.1997 | 0.8003 |
| 15    | 0.0  | 0.1997 | 0.8003 |
| 16    | 0.0  | 0.1997 | 0.8003 |
| 17    | 0.0  | 0.1997 | 0.8003 |
| 18    | 0.0  | 0.1997 | 0.8003 |
| 19    | 0.0  | 0.1997 | 0.8003 |
| 20    | 0.0  | 0.1997 | 0.8003 |
| 21    | 0.0  | 0.1997 | 0.8003 |
| 22    | 0.0  | 0.1997 | 0.8003 |
| 23    | 0.0  | 0.1997 | 0.8003 |
| 24    | 0.0  | 0.1997 | 0.8003 |
| 25    | 0.0  | 0.1997 | 0.8003 |
| 26    | 0.0  | 0.1997 | 0.8003 |
| 27    | 0.0  | 0.1997 | 0.8003 |
| 28    | 0.0  | 0.1997 | 0.8003 |
| 29    | 0.0  | 0.1997 | 0.8003 |
| 30    | 0.0  | 0.1997 | 0.8003 |
| 31    | 0.0  | 0.1997 | 0.8003 |
| 32    | 0.0  | 0.1997 | 0.8003 |
| 33    | 0.0  | 0.1997 | 0.8003 |
| 34    | 0.0  | 0.1997 | 0.8003 |
| 35    | 0.0  | 0.1997 | 0.8003 |

beta=0.0, budget=0.5, model=mistral7b

- PPL: 8.3766
- PPL Δ%: -0.14
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
|-------|--------|

|    |     |
|----|-----|
| 0  | 0.5 |
| 1  | 0.5 |
| 2  | 0.5 |
| 3  | 0.5 |
| 4  | 0.5 |
| 5  | 0.5 |
| 6  | 0.5 |
| 7  | 0.5 |
| 8  | 0.5 |
| 9  | 0.5 |
| 10 | 0.5 |
| 11 | 0.5 |
| 12 | 0.5 |
| 13 | 0.5 |
| 14 | 0.5 |
| 15 | 0.5 |
| 16 | 0.5 |
| 17 | 0.5 |
| 18 | 0.5 |
| 19 | 0.5 |
| 20 | 0.5 |
| 21 | 0.5 |
| 22 | 0.5 |
| 23 | 0.5 |
| 24 | 0.5 |
| 25 | 0.5 |
| 26 | 0.5 |
| 27 | 0.5 |
| 28 | 0.5 |
| 29 | 0.5 |
| 30 | 0.5 |
| 31 | 0.5 |
| 32 | 0.5 |
| 33 | 0.5 |
| 34 | 0.5 |
| 35 | 0.5 |

Tier Fractions Per Layer:

| Layer | FP16 | INT8 | INT4 |
|-------|------|------|------|
| 0     | 0.0  | 1.0  | 0.0  |
| 1     | 0.0  | 1.0  | 0.0  |
| 2     | 0.0  | 1.0  | 0.0  |
| 3     | 0.0  | 1.0  | 0.0  |
| 4     | 0.0  | 1.0  | 0.0  |
| 5     | 0.0  | 1.0  | 0.0  |
| 6     | 0.0  | 1.0  | 0.0  |
| 7     | 0.0  | 1.0  | 0.0  |
| 8     | 0.0  | 1.0  | 0.0  |
| 9     | 0.0  | 1.0  | 0.0  |
| 10    | 0.0  | 1.0  | 0.0  |
| 11    | 0.0  | 1.0  | 0.0  |
| 12    | 0.0  | 1.0  | 0.0  |
| 13    | 0.0  | 1.0  | 0.0  |
| 14    | 0.0  | 1.0  | 0.0  |
| 15    | 0.0  | 1.0  | 0.0  |
| 16    | 0.0  | 1.0  | 0.0  |
| 17    | 0.0  | 1.0  | 0.0  |
| 18    | 0.0  | 1.0  | 0.0  |
| 19    | 0.0  | 1.0  | 0.0  |
| 20    | 0.0  | 1.0  | 0.0  |
| 21    | 0.0  | 1.0  | 0.0  |
| 22    | 0.0  | 1.0  | 0.0  |
| 23    | 0.0  | 1.0  | 0.0  |
| 24    | 0.0  | 1.0  | 0.0  |
| 25    | 0.0  | 1.0  | 0.0  |
| 26    | 0.0  | 1.0  | 0.0  |
| 27    | 0.0  | 1.0  | 0.0  |
| 28    | 0.0  | 1.0  | 0.0  |
| 29    | 0.0  | 1.0  | 0.0  |
| 30    | 0.0  | 1.0  | 0.0  |
| 31    | 0.0  | 1.0  | 0.0  |
| 32    | 0.0  | 1.0  | 0.0  |
| 33    | 0.0  | 1.0  | 0.0  |
| 34    | 0.0  | 1.0  | 0.0  |
| 35    | 0.0  | 1.0  | 0.0  |

beta=0.0, budget=0.7, model=mistral7b

- PPL: 8.3878
- PPL Δ%: -0.01
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.7    |
| 1     | 0.7    |
| 2     | 0.7    |
| 3     | 0.7    |
| 4     | 0.7    |
| 5     | 0.7    |
| 6     | 0.7    |
| 7     | 0.7    |
| 8     | 0.7    |
| 9     | 0.7    |
| 10    | 0.7    |
| 11    | 0.7    |
| 12    | 0.7    |
| 13    | 0.7    |

|    |     |
|----|-----|
| 14 | 0.7 |
| 15 | 0.7 |
| 16 | 0.7 |
| 17 | 0.7 |
| 18 | 0.7 |
| 19 | 0.7 |
| 20 | 0.7 |
| 21 | 0.7 |
| 22 | 0.7 |
| 23 | 0.7 |
| 24 | 0.7 |
| 25 | 0.7 |
| 26 | 0.7 |
| 27 | 0.7 |
| 28 | 0.7 |
| 29 | 0.7 |
| 30 | 0.7 |
| 31 | 0.7 |
| 32 | 0.7 |
| 33 | 0.7 |
| 34 | 0.7 |
| 35 | 0.7 |

Tier Fractions Per Layer:

| Layer | FP16   | INT8   | INT4 |
|-------|--------|--------|------|
| 0     | 0.3999 | 0.6001 | 0.0  |
| 1     | 0.3999 | 0.6001 | 0.0  |
| 2     | 0.3999 | 0.6001 | 0.0  |
| 3     | 0.3999 | 0.6001 | 0.0  |
| 4     | 0.3999 | 0.6001 | 0.0  |
| 5     | 0.3999 | 0.6001 | 0.0  |
| 6     | 0.3999 | 0.6001 | 0.0  |
| 7     | 0.3999 | 0.6001 | 0.0  |
| 8     | 0.3999 | 0.6001 | 0.0  |
| 9     | 0.3999 | 0.6001 | 0.0  |
| 10    | 0.3999 | 0.6001 | 0.0  |
| 11    | 0.3999 | 0.6001 | 0.0  |
| 12    | 0.3999 | 0.6001 | 0.0  |
| 13    | 0.3999 | 0.6001 | 0.0  |
| 14    | 0.3999 | 0.6001 | 0.0  |
| 15    | 0.3999 | 0.6001 | 0.0  |
| 16    | 0.3999 | 0.6001 | 0.0  |
| 17    | 0.3999 | 0.6001 | 0.0  |
| 18    | 0.3999 | 0.6001 | 0.0  |
| 19    | 0.3999 | 0.6001 | 0.0  |
| 20    | 0.3999 | 0.6001 | 0.0  |
| 21    | 0.3999 | 0.6001 | 0.0  |
| 22    | 0.3999 | 0.6001 | 0.0  |
| 23    | 0.3999 | 0.6001 | 0.0  |
| 24    | 0.3999 | 0.6001 | 0.0  |
| 25    | 0.3999 | 0.6001 | 0.0  |
| 26    | 0.3999 | 0.6001 | 0.0  |
| 27    | 0.3999 | 0.6001 | 0.0  |
| 28    | 0.3999 | 0.6001 | 0.0  |
| 29    | 0.3999 | 0.6001 | 0.0  |
| 30    | 0.3999 | 0.6001 | 0.0  |
| 31    | 0.3999 | 0.6001 | 0.0  |
| 32    | 0.3999 | 0.6001 | 0.0  |
| 33    | 0.3999 | 0.6001 | 0.0  |
| 34    | 0.3999 | 0.6001 | 0.0  |
| 35    | 0.3999 | 0.6001 | 0.0  |

beta=0.01, budget=0.3, model=mistral7b

- PPL: 8.809
- PPL Δ%: 5.01
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |
| 13    | 0.255  |
| 14    | 0.265  |
| 15    | 0.275  |
| 16    | 0.285  |
| 17    | 0.295  |
| 18    | 0.305  |
| 19    | 0.315  |
| 20    | 0.325  |
| 21    | 0.335  |
| 22    | 0.345  |
| 23    | 0.355  |
| 24    | 0.365  |
| 25    | 0.375  |
| 26    | 0.385  |
| 27    | 0.395  |

|    |       |
|----|-------|
| 28 | 0.405 |
| 29 | 0.415 |
| 30 | 0.425 |
| 31 | 0.435 |
| 32 | 0.445 |
| 33 | 0.455 |
| 34 | 0.465 |
| 35 | 0.475 |

Tier Fractions Per Layer:

| Layer | FP16 | INT8                 | INT4                 |
|-------|------|----------------------|----------------------|
| 0     | 0.0  | 0.0                  | 1.0                  |
| 1     | 0.0  | 0.0                  | 1.0                  |
| 2     | 0.0  | 0.0                  | 1.0                  |
| 3     | 0.0  | 0.0                  | 1.0                  |
| 4     | 0.0  | 0.0                  | 1.0                  |
| 5     | 0.0  | 0.0                  | 1.0                  |
| 6     | 0.0  | 0.0                  | 1.0                  |
| 7     | 0.0  | 0.0                  | 1.0                  |
| 8     | 0.0  | 0.0                  | 1.0                  |
| 9     | 0.0  | 0.0                  | 1.0                  |
| 10    | 0.0  | 0.0                  | 1.0                  |
| 11    | 0.0  | 0.0                  | 1.0                  |
| 12    | 0.0  | 0.0                  | 1.0                  |
| 13    | 0.0  | 0.0195               | 0.9805               |
| 14    | 0.0  | 0.059600000000000001 | 0.9404               |
| 15    | 0.0  | 0.099599999999999998 | 0.9004               |
| 16    | 0.0  | 0.1396               | 0.86039999999999999  |
| 17    | 0.0  | 0.1797               | 0.8203               |
| 18    | 0.0  | 0.2197               | 0.7803               |
| 19    | 0.0  | 0.2598               | 0.740200000000000001 |
| 20    | 0.0  | 0.2998               | 0.7002               |
| 21    | 0.0  | 0.339800000000000005 | 0.6602               |
| 22    | 0.0  | 0.379899999999999996 | 0.6201               |
| 23    | 0.0  | 0.4199               | 0.5801               |
| 24    | 0.0  | 0.46                 | 0.54                 |
| 25    | 0.0  | 0.5                  | 0.5                  |
| 26    | 0.0  | 0.5396               | 0.4604               |
| 27    | 0.0  | 0.5796               | 0.420400000000000005 |
| 28    | 0.0  | 0.6196               | 0.3804               |
| 29    | 0.0  | 0.6597               | 0.3403               |
| 30    | 0.0  | 0.6997               | 0.3003               |
| 31    | 0.0  | 0.7397               | 0.2603               |
| 32    | 0.0  | 0.779800000000000002 | 0.220200000000000003 |
| 33    | 0.0  | 0.8198               | 0.1802               |
| 34    | 0.0  | 0.8599               | 0.1401               |
| 35    | 0.0  | 0.8999               | 0.100100000000000001 |

beta=0.01, budget=0.5, model=mistral7b

- PPL: 8.5589
- PPL Δ%: 2.03
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.325  |
| 1     | 0.335  |
| 2     | 0.345  |
| 3     | 0.355  |
| 4     | 0.365  |
| 5     | 0.375  |
| 6     | 0.385  |
| 7     | 0.395  |
| 8     | 0.405  |
| 9     | 0.415  |
| 10    | 0.425  |
| 11    | 0.435  |
| 12    | 0.445  |
| 13    | 0.455  |
| 14    | 0.465  |
| 15    | 0.475  |
| 16    | 0.485  |
| 17    | 0.495  |
| 18    | 0.505  |
| 19    | 0.515  |
| 20    | 0.525  |
| 21    | 0.535  |
| 22    | 0.545  |
| 23    | 0.555  |
| 24    | 0.565  |
| 25    | 0.575  |
| 26    | 0.585  |
| 27    | 0.595  |
| 28    | 0.605  |
| 29    | 0.615  |
| 30    | 0.625  |
| 31    | 0.635  |
| 32    | 0.645  |
| 33    | 0.655  |
| 34    | 0.665  |
| 35    | 0.675  |

Tier Fractions Per Layer:

| Layer | FP16 | INT8                 | INT4   |
|-------|------|----------------------|--------|
| 0     | 0.0  | 0.2998               | 0.7002 |
| 1     | 0.0  | 0.339800000000000005 | 0.6602 |
| 2     | 0.0  | 0.379899999999999996 | 0.6201 |

|    |                      |                    |                     |
|----|----------------------|--------------------|---------------------|
| 3  | 0.0                  | 0.4199             | 0.5801              |
| 4  | 0.0                  | 0.46               | 0.54                |
| 5  | 0.0                  | 0.5                | 0.5                 |
| 6  | 0.0                  | 0.5396             | 0.4604              |
| 7  | 0.0                  | 0.5796             | 0.4204000000000005  |
| 8  | 0.0                  | 0.6196             | 0.3804              |
| 9  | 0.0                  | 0.6597             | 0.3403              |
| 10 | 0.0                  | 0.6997             | 0.3003              |
| 11 | 0.0                  | 0.7397             | 0.2603              |
| 12 | 0.0                  | 0.7798000000000002 | 0.2202000000000003  |
| 13 | 0.0                  | 0.8198             | 0.1802              |
| 14 | 0.0                  | 0.8599             | 0.1401              |
| 15 | 0.0                  | 0.8999             | 0.10010000000000001 |
| 16 | 0.0                  | 0.9399000000000001 | 0.06010000000000001 |
| 17 | 0.0                  | 0.98               | 0.02                |
| 18 | 0.0098               | 0.9902000000000001 | 0.0                 |
| 19 | 0.029800000000000004 | 0.9701999999999998 | 0.0                 |
| 20 | 0.04979999999999999  | 0.9502             | 0.0                 |
| 21 | 0.0698               | 0.9302             | 0.0                 |
| 22 | 0.0898               | 0.9102             | 0.0                 |
| 23 | 0.1099               | 0.8901             | 0.0                 |
| 24 | 0.1299               | 0.8701             | 0.0                 |
| 25 | 0.1499               | 0.8501             | 0.0                 |
| 26 | 0.16990000000000002  | 0.8301             | 0.0                 |
| 27 | 0.18990000000000004  | 0.8100999999999999 | 0.0                 |
| 28 | 0.21                 | 0.79               | 0.0                 |
| 29 | 0.23                 | 0.77               | 0.0                 |
| 30 | 0.25                 | 0.75               | 0.0                 |
| 31 | 0.27                 | 0.73               | 0.0                 |
| 32 | 0.29                 | 0.71               | 0.0                 |
| 33 | 0.3101               | 0.6899000000000001 | 0.0                 |
| 34 | 0.3301               | 0.6699             | 0.0                 |
| 35 | 0.3501               | 0.6499             | 0.0                 |

beta=0.01, budget=0.7, model=mistral7b

- PPL: 8.3804
- PPL Δ%: -0.1
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.525  |
| 1     | 0.535  |
| 2     | 0.545  |
| 3     | 0.555  |
| 4     | 0.565  |
| 5     | 0.575  |
| 6     | 0.585  |
| 7     | 0.595  |
| 8     | 0.605  |
| 9     | 0.615  |
| 10    | 0.625  |
| 11    | 0.635  |
| 12    | 0.645  |
| 13    | 0.655  |
| 14    | 0.665  |
| 15    | 0.675  |
| 16    | 0.685  |
| 17    | 0.695  |
| 18    | 0.705  |
| 19    | 0.715  |
| 20    | 0.725  |
| 21    | 0.735  |
| 22    | 0.745  |
| 23    | 0.755  |
| 24    | 0.765  |
| 25    | 0.775  |
| 26    | 0.785  |
| 27    | 0.795  |
| 28    | 0.805  |
| 29    | 0.815  |
| 30    | 0.825  |
| 31    | 0.835  |
| 32    | 0.845  |
| 33    | 0.855  |
| 34    | 0.865  |
| 35    | 0.875  |

Tier Fractions Per Layer:

| Layer | FP16                | INT8               | INT4 |
|-------|---------------------|--------------------|------|
| 0     | 0.04979999999999999 | 0.9502             | 0.0  |
| 1     | 0.0698              | 0.9302             | 0.0  |
| 2     | 0.0898              | 0.9102             | 0.0  |
| 3     | 0.1099              | 0.8901             | 0.0  |
| 4     | 0.1299              | 0.8701             | 0.0  |
| 5     | 0.1499              | 0.8501             | 0.0  |
| 6     | 0.16990000000000002 | 0.8301             | 0.0  |
| 7     | 0.18990000000000004 | 0.8100999999999999 | 0.0  |
| 8     | 0.21                | 0.79               | 0.0  |
| 9     | 0.23                | 0.77               | 0.0  |
| 10    | 0.25                | 0.75               | 0.0  |
| 11    | 0.27                | 0.73               | 0.0  |
| 12    | 0.29                | 0.71               | 0.0  |
| 13    | 0.3101              | 0.6899000000000001 | 0.0  |
| 14    | 0.3301              | 0.6699             | 0.0  |
| 15    | 0.3501              | 0.6499             | 0.0  |
| 16    | 0.37010000000000004 | 0.6299             | 0.0  |

|    |                     |                     |     |
|----|---------------------|---------------------|-----|
| 17 | 0.39010000000000006 | 0.6099              | 0.0 |
| 18 | 0.41019999999999995 | 0.5898              | 0.0 |
| 19 | 0.43019999999999997 | 0.5698              | 0.0 |
| 20 | 0.4502              | 0.5498              | 0.0 |
| 21 | 0.4702              | 0.5298              | 0.0 |
| 22 | 0.4902              | 0.5098              | 0.0 |
| 23 | 0.5098              | 0.4902              | 0.0 |
| 24 | 0.5298              | 0.4702              | 0.0 |
| 25 | 0.5498              | 0.4502              | 0.0 |
| 26 | 0.5698              | 0.43019999999999997 | 0.0 |
| 27 | 0.5898              | 0.41019999999999995 | 0.0 |
| 28 | 0.6099              | 0.39010000000000006 | 0.0 |
| 29 | 0.6299              | 0.37010000000000004 | 0.0 |
| 30 | 0.6499              | 0.3501              | 0.0 |
| 31 | 0.6699              | 0.3301              | 0.0 |
| 32 | 0.68990000000000001 | 0.3101              | 0.0 |
| 33 | 0.71                | 0.29                | 0.0 |
| 34 | 0.73                | 0.27                | 0.0 |
| 35 | 0.75                | 0.25                | 0.0 |

beta=0.02, budget=0.3, model=mistral7b

- PPL: 8.8028
- PPL Δ%: 4.94
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |
| 13    | 0.25   |
| 14    | 0.25   |
| 15    | 0.25   |
| 16    | 0.27   |
| 17    | 0.29   |
| 18    | 0.31   |
| 19    | 0.33   |
| 20    | 0.35   |
| 21    | 0.37   |
| 22    | 0.39   |
| 23    | 0.41   |
| 24    | 0.43   |
| 25    | 0.45   |
| 26    | 0.47   |
| 27    | 0.49   |
| 28    | 0.51   |
| 29    | 0.53   |
| 30    | 0.55   |
| 31    | 0.57   |
| 32    | 0.59   |
| 33    | 0.61   |
| 34    | 0.63   |
| 35    | 0.65   |

Tier Fractions Per Layer:

| Layer | FP16                 | INT8                | INT4                |
|-------|----------------------|---------------------|---------------------|
| 0     | 0.0                  | 0.0                 | 1.0                 |
| 1     | 0.0                  | 0.0                 | 1.0                 |
| 2     | 0.0                  | 0.0                 | 1.0                 |
| 3     | 0.0                  | 0.0                 | 1.0                 |
| 4     | 0.0                  | 0.0                 | 1.0                 |
| 5     | 0.0                  | 0.0                 | 1.0                 |
| 6     | 0.0                  | 0.0                 | 1.0                 |
| 7     | 0.0                  | 0.0                 | 1.0                 |
| 8     | 0.0                  | 0.0                 | 1.0                 |
| 9     | 0.0                  | 0.0                 | 1.0                 |
| 10    | 0.0                  | 0.0                 | 1.0                 |
| 11    | 0.0                  | 0.0                 | 1.0                 |
| 12    | 0.0                  | 0.0                 | 1.0                 |
| 13    | 0.0                  | 0.0                 | 1.0                 |
| 14    | 0.0                  | 0.0                 | 1.0                 |
| 15    | 0.0                  | 0.0                 | 1.0                 |
| 16    | 0.0                  | 0.0796              | 0.9204              |
| 17    | 0.0                  | 0.1597              | 0.8403              |
| 18    | 0.0                  | 0.2397              | 0.7603              |
| 19    | 0.0                  | 0.3198              | 0.6802              |
| 20    | 0.0                  | 0.3999              | 0.6001              |
| 21    | 0.0                  | 0.48                | 0.52                |
| 22    | 0.0                  | 0.5596              | 0.44040000000000007 |
| 23    | 0.0                  | 0.6396              | 0.3604              |
| 24    | 0.0                  | 0.7197              | 0.2803              |
| 25    | 0.0                  | 0.7998              | 0.20020000000000002 |
| 26    | 0.0                  | 0.8799              | 0.1201              |
| 27    | 0.0                  | 0.96                | 0.04                |
| 28    | 0.02                 | 0.98                | 0.0                 |
| 29    | 0.060100000000000001 | 0.93990000000000001 | 0.0                 |
| 30    | 0.100100000000000001 | 0.8999              | 0.0                 |



|    |                     |                    |     |
|----|---------------------|--------------------|-----|
| 31 | 0.1401              | 0.8599             | 0.0 |
| 32 | 0.1802              | 0.8198             | 0.0 |
| 33 | 0.22020000000000003 | 0.7798000000000002 | 0.0 |
| 34 | 0.2598              | 0.7402000000000001 | 0.0 |
| 35 | 0.2998              | 0.7002             | 0.0 |

beta=0.02, budget=0.5, model=mistral7b

- PPL: 8.6255
- PPL Δ%: 2.82
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.27   |
| 7     | 0.29   |
| 8     | 0.31   |
| 9     | 0.33   |
| 10    | 0.35   |
| 11    | 0.37   |
| 12    | 0.39   |
| 13    | 0.41   |
| 14    | 0.43   |
| 15    | 0.45   |
| 16    | 0.47   |
| 17    | 0.49   |
| 18    | 0.51   |
| 19    | 0.53   |
| 20    | 0.55   |
| 21    | 0.57   |
| 22    | 0.59   |
| 23    | 0.61   |
| 24    | 0.63   |
| 25    | 0.65   |
| 26    | 0.67   |
| 27    | 0.69   |
| 28    | 0.71   |
| 29    | 0.73   |
| 30    | 0.75   |
| 31    | 0.77   |
| 32    | 0.79   |
| 33    | 0.81   |
| 34    | 0.83   |
| 35    | 0.85   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0796              | 0.9204              |
| 7     | 0.0                 | 0.1597              | 0.8403              |
| 8     | 0.0                 | 0.2397              | 0.7603              |
| 9     | 0.0                 | 0.3198              | 0.6802              |
| 10    | 0.0                 | 0.3999              | 0.6001              |
| 11    | 0.0                 | 0.48                | 0.52                |
| 12    | 0.0                 | 0.5596              | 0.44040000000000007 |
| 13    | 0.0                 | 0.6396              | 0.3604              |
| 14    | 0.0                 | 0.7197              | 0.2803              |
| 15    | 0.0                 | 0.7998              | 0.20020000000000002 |
| 16    | 0.0                 | 0.8799              | 0.1201              |
| 17    | 0.0                 | 0.96                | 0.04                |
| 18    | 0.02                | 0.98                | 0.0                 |
| 19    | 0.06010000000000001 | 0.9399000000000001  | 0.0                 |
| 20    | 0.10010000000000001 | 0.8999              | 0.0                 |
| 21    | 0.1401              | 0.8599              | 0.0                 |
| 22    | 0.1802              | 0.8198              | 0.0                 |
| 23    | 0.22020000000000003 | 0.7798000000000002  | 0.0                 |
| 24    | 0.2598              | 0.7402000000000001  | 0.0                 |
| 25    | 0.2998              | 0.7002              | 0.0                 |
| 26    | 0.33980000000000005 | 0.6602              | 0.0                 |
| 27    | 0.37989999999999996 | 0.6201              | 0.0                 |
| 28    | 0.4199              | 0.5801              | 0.0                 |
| 29    | 0.46                | 0.54                | 0.0                 |
| 30    | 0.5                 | 0.5                 | 0.0                 |
| 31    | 0.54                | 0.46                | 0.0                 |
| 32    | 0.5801              | 0.4199              | 0.0                 |
| 33    | 0.6201              | 0.37989999999999996 | 0.0                 |
| 34    | 0.6602              | 0.33980000000000005 | 0.0                 |
| 35    | 0.7002              | 0.2998              | 0.0                 |

beta=0.02, budget=0.7, model=mistral7b

- PPL: 8.4382
- PPL Δ%: 0.59
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

|  |  |
|--|--|
|  |  |
|--|--|

| Layer | Budget |
|-------|--------|
| 0     | 0.35   |
| 1     | 0.37   |
| 2     | 0.39   |
| 3     | 0.41   |
| 4     | 0.43   |
| 5     | 0.45   |
| 6     | 0.47   |
| 7     | 0.49   |
| 8     | 0.51   |
| 9     | 0.53   |
| 10    | 0.55   |
| 11    | 0.57   |
| 12    | 0.59   |
| 13    | 0.61   |
| 14    | 0.63   |
| 15    | 0.65   |
| 16    | 0.67   |
| 17    | 0.69   |
| 18    | 0.71   |
| 19    | 0.73   |
| 20    | 0.75   |
| 21    | 0.77   |
| 22    | 0.79   |
| 23    | 0.81   |
| 24    | 0.83   |
| 25    | 0.85   |
| 26    | 0.87   |
| 27    | 0.89   |
| 28    | 0.91   |
| 29    | 0.93   |
| 30    | 0.95   |
| 31    | 0.95   |
| 32    | 0.95   |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                 | INT8                 | INT4                |
|-------|----------------------|----------------------|---------------------|
| 0     | 0.0                  | 0.3999               | 0.6001              |
| 1     | 0.0                  | 0.48                 | 0.52                |
| 2     | 0.0                  | 0.5596               | 0.44040000000000007 |
| 3     | 0.0                  | 0.6396               | 0.3604              |
| 4     | 0.0                  | 0.7197               | 0.2803              |
| 5     | 0.0                  | 0.7998               | 0.20020000000000002 |
| 6     | 0.0                  | 0.8799               | 0.1201              |
| 7     | 0.0                  | 0.96                 | 0.04                |
| 8     | 0.02                 | 0.98                 | 0.0                 |
| 9     | 0.060100000000000001 | 0.93990000000000001  | 0.0                 |
| 10    | 0.100100000000000001 | 0.8999               | 0.0                 |
| 11    | 0.1401               | 0.8599               | 0.0                 |
| 12    | 0.1802               | 0.8198               | 0.0                 |
| 13    | 0.220200000000000003 | 0.77980000000000002  | 0.0                 |
| 14    | 0.2598               | 0.74020000000000001  | 0.0                 |
| 15    | 0.2998               | 0.7002               | 0.0                 |
| 16    | 0.339800000000000005 | 0.6602               | 0.0                 |
| 17    | 0.37989999999999996  | 0.6201               | 0.0                 |
| 18    | 0.4199               | 0.5801               | 0.0                 |
| 19    | 0.46                 | 0.54                 | 0.0                 |
| 20    | 0.5                  | 0.5                  | 0.0                 |
| 21    | 0.54                 | 0.46                 | 0.0                 |
| 22    | 0.5801               | 0.4199               | 0.0                 |
| 23    | 0.6201               | 0.37989999999999996  | 0.0                 |
| 24    | 0.6602               | 0.339800000000000005 | 0.0                 |
| 25    | 0.7002               | 0.2998               | 0.0                 |
| 26    | 0.74020000000000001  | 0.2598               | 0.0                 |
| 27    | 0.779800000000000002 | 0.220200000000000003 | 0.0                 |
| 28    | 0.8198               | 0.1802               | 0.0                 |
| 29    | 0.8599               | 0.1401               | 0.0                 |
| 30    | 0.8999               | 0.100100000000000001 | 0.0                 |
| 31    | 0.8999               | 0.100100000000000001 | 0.0                 |
| 32    | 0.8999               | 0.100100000000000001 | 0.0                 |
| 33    | 0.8999               | 0.100100000000000001 | 0.0                 |
| 34    | 0.8999               | 0.100100000000000001 | 0.0                 |
| 35    | 0.8999               | 0.100100000000000001 | 0.0                 |

beta=0.03, budget=0.3, model=mistral7b

- PPL: 8.7855
- PPL Δ%: 4.73
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |

|    |       |
|----|-------|
| 13 | 0.25  |
| 14 | 0.25  |
| 15 | 0.25  |
| 16 | 0.255 |
| 17 | 0.285 |
| 18 | 0.315 |
| 19 | 0.345 |
| 20 | 0.375 |
| 21 | 0.405 |
| 22 | 0.435 |
| 23 | 0.465 |
| 24 | 0.495 |
| 25 | 0.525 |
| 26 | 0.555 |
| 27 | 0.585 |
| 28 | 0.615 |
| 29 | 0.645 |
| 30 | 0.675 |
| 31 | 0.705 |
| 32 | 0.735 |
| 33 | 0.765 |
| 34 | 0.795 |
| 35 | 0.825 |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4               |
|-------|---------------------|---------------------|--------------------|
| 0     | 0.0                 | 0.0                 | 1.0                |
| 1     | 0.0                 | 0.0                 | 1.0                |
| 2     | 0.0                 | 0.0                 | 1.0                |
| 3     | 0.0                 | 0.0                 | 1.0                |
| 4     | 0.0                 | 0.0                 | 1.0                |
| 5     | 0.0                 | 0.0                 | 1.0                |
| 6     | 0.0                 | 0.0                 | 1.0                |
| 7     | 0.0                 | 0.0                 | 1.0                |
| 8     | 0.0                 | 0.0                 | 1.0                |
| 9     | 0.0                 | 0.0                 | 1.0                |
| 10    | 0.0                 | 0.0                 | 1.0                |
| 11    | 0.0                 | 0.0                 | 1.0                |
| 12    | 0.0                 | 0.0                 | 1.0                |
| 13    | 0.0                 | 0.0                 | 1.0                |
| 14    | 0.0                 | 0.0                 | 1.0                |
| 15    | 0.0                 | 0.0                 | 1.0                |
| 16    | 0.0                 | 0.0195              | 0.9805             |
| 17    | 0.0                 | 0.1396              | 0.8603999999999999 |
| 18    | 0.0                 | 0.2598              | 0.7402000000000001 |
| 19    | 0.0                 | 0.37989999999999996 | 0.6201             |
| 20    | 0.0                 | 0.5                 | 0.5                |
| 21    | 0.0                 | 0.6196              | 0.3804             |
| 22    | 0.0                 | 0.7397              | 0.2603             |
| 23    | 0.0                 | 0.8599              | 0.1401             |
| 24    | 0.0                 | 0.98                | 0.02               |
| 25    | 0.04979999999999999 | 0.9502              | 0.0                |
| 26    | 0.1099              | 0.8901              | 0.0                |
| 27    | 0.16990000000000002 | 0.8301              | 0.0                |
| 28    | 0.23                | 0.77                | 0.0                |
| 29    | 0.29                | 0.71                | 0.0                |
| 30    | 0.3501              | 0.6499              | 0.0                |
| 31    | 0.41019999999999995 | 0.5898              | 0.0                |
| 32    | 0.4702              | 0.5298              | 0.0                |
| 33    | 0.5298              | 0.4702              | 0.0                |
| 34    | 0.5898              | 0.41019999999999995 | 0.0                |
| 35    | 0.6499              | 0.3501              | 0.0                |

beta=0.03, budget=0.5, model=mistral7b

- PPL: 8.703
- PPL Δ%: 3.75
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.275  |
| 11    | 0.305  |
| 12    | 0.335  |
| 13    | 0.365  |
| 14    | 0.395  |
| 15    | 0.425  |
| 16    | 0.455  |
| 17    | 0.485  |
| 18    | 0.515  |
| 19    | 0.545  |
| 20    | 0.575  |
| 21    | 0.605  |
| 22    | 0.635  |
| 23    | 0.665  |
| 24    | 0.695  |
| 25    | 0.725  |
| 26    | 0.755  |

|    |       |
|----|-------|
| 27 | 0.785 |
| 28 | 0.815 |
| 29 | 0.845 |
| 30 | 0.875 |
| 31 | 0.905 |
| 32 | 0.935 |
| 33 | 0.95  |
| 34 | 0.95  |
| 35 | 0.95  |

Tier Fractions Per Layer:

| Layer | FP16                 | INT8                | INT4                |
|-------|----------------------|---------------------|---------------------|
| 0     | 0.0                  | 0.0                 | 1.0                 |
| 1     | 0.0                  | 0.0                 | 1.0                 |
| 2     | 0.0                  | 0.0                 | 1.0                 |
| 3     | 0.0                  | 0.0                 | 1.0                 |
| 4     | 0.0                  | 0.0                 | 1.0                 |
| 5     | 0.0                  | 0.0                 | 1.0                 |
| 6     | 0.0                  | 0.0                 | 1.0                 |
| 7     | 0.0                  | 0.0                 | 1.0                 |
| 8     | 0.0                  | 0.0                 | 1.0                 |
| 9     | 0.0                  | 0.0                 | 1.0                 |
| 10    | 0.0                  | 0.09959999999999998 | 0.9004              |
| 11    | 0.0                  | 0.2197              | 0.7803              |
| 12    | 0.0                  | 0.33980000000000005 | 0.6602              |
| 13    | 0.0                  | 0.46                | 0.54                |
| 14    | 0.0                  | 0.5796              | 0.42040000000000005 |
| 15    | 0.0                  | 0.6997              | 0.3003              |
| 16    | 0.0                  | 0.8198              | 0.1802              |
| 17    | 0.0                  | 0.9399000000000001  | 0.06010000000000001 |
| 18    | 0.029800000000000004 | 0.9701999999999998  | 0.0                 |
| 19    | 0.0898               | 0.9102              | 0.0                 |
| 20    | 0.1499               | 0.8501              | 0.0                 |
| 21    | 0.21                 | 0.79                | 0.0                 |
| 22    | 0.27                 | 0.73                | 0.0                 |
| 23    | 0.3301               | 0.6699              | 0.0                 |
| 24    | 0.39010000000000006  | 0.6099              | 0.0                 |
| 25    | 0.4502               | 0.5498              | 0.0                 |
| 26    | 0.5098               | 0.4902              | 0.0                 |
| 27    | 0.5698               | 0.4301999999999997  | 0.0                 |
| 28    | 0.6299               | 0.37010000000000004 | 0.0                 |
| 29    | 0.6899000000000001   | 0.3101              | 0.0                 |
| 30    | 0.75                 | 0.25                | 0.0                 |
| 31    | 0.8100999999999999   | 0.18990000000000004 | 0.0                 |
| 32    | 0.8701               | 0.1299              | 0.0                 |
| 33    | 0.8999               | 0.10010000000000001 | 0.0                 |
| 34    | 0.8999               | 0.10010000000000001 | 0.0                 |
| 35    | 0.8999               | 0.10010000000000001 | 0.0                 |

beta=0.03, budget=0.7, model=mistral7b

- PPL: 8.4844
- PPL Δ%: 1.14
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.265  |
| 4     | 0.295  |
| 5     | 0.325  |
| 6     | 0.355  |
| 7     | 0.385  |
| 8     | 0.415  |
| 9     | 0.445  |
| 10    | 0.475  |
| 11    | 0.505  |
| 12    | 0.535  |
| 13    | 0.565  |
| 14    | 0.595  |
| 15    | 0.625  |
| 16    | 0.655  |
| 17    | 0.685  |
| 18    | 0.715  |
| 19    | 0.745  |
| 20    | 0.775  |
| 21    | 0.805  |
| 22    | 0.835  |
| 23    | 0.865  |
| 24    | 0.895  |
| 25    | 0.925  |
| 26    | 0.95   |
| 27    | 0.95   |
| 28    | 0.95   |
| 29    | 0.95   |
| 30    | 0.95   |
| 31    | 0.95   |
| 32    | 0.95   |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16 | INT8 | INT4 |
|-------|------|------|------|
| 0     | 0.0  | 0.0  | 1.0  |
| 1     | 0.0  | 0.0  | 1.0  |

|    |                     |                     |                     |
|----|---------------------|---------------------|---------------------|
| 2  | 0.0                 | 0.0                 | 1.0                 |
| 3  | 0.0                 | 0.05960000000000001 | 0.9404              |
| 4  | 0.0                 | 0.1797              | 0.8203              |
| 5  | 0.0                 | 0.2998              | 0.7002              |
| 6  | 0.0                 | 0.4199              | 0.5801              |
| 7  | 0.0                 | 0.5396              | 0.4604              |
| 8  | 0.0                 | 0.6597              | 0.3403              |
| 9  | 0.0                 | 0.7798000000000002  | 0.2202000000000003  |
| 10 | 0.0                 | 0.8999              | 0.10010000000000001 |
| 11 | 0.0098              | 0.9902000000000001  | 0.0                 |
| 12 | 0.0698              | 0.9302              | 0.0                 |
| 13 | 0.1299              | 0.8701              | 0.0                 |
| 14 | 0.18990000000000004 | 0.8100999999999999  | 0.0                 |
| 15 | 0.25                | 0.75                | 0.0                 |
| 16 | 0.3101              | 0.6899000000000001  | 0.0                 |
| 17 | 0.37010000000000004 | 0.6299              | 0.0                 |
| 18 | 0.43019999999999997 | 0.5698              | 0.0                 |
| 19 | 0.4902              | 0.5098              | 0.0                 |
| 20 | 0.5498              | 0.4502              | 0.0                 |
| 21 | 0.6099              | 0.39010000000000006 | 0.0                 |
| 22 | 0.6699              | 0.3301              | 0.0                 |
| 23 | 0.73                | 0.27                | 0.0                 |
| 24 | 0.79                | 0.21                | 0.0                 |
| 25 | 0.8501              | 0.1499              | 0.0                 |
| 26 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 27 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 28 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 29 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 30 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 31 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 32 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 33 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 34 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35 | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.04, budget=0.3, model=mistral7b

- PPL: 8.7831
- PPL Δ%: 4.7
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |
| 13    | 0.25   |
| 14    | 0.25   |
| 15    | 0.25   |
| 16    | 0.25   |
| 17    | 0.28   |
| 18    | 0.32   |
| 19    | 0.36   |
| 20    | 0.4    |
| 21    | 0.44   |
| 22    | 0.48   |
| 23    | 0.52   |
| 24    | 0.56   |
| 25    | 0.6    |
| 26    | 0.64   |
| 27    | 0.68   |
| 28    | 0.72   |
| 29    | 0.76   |
| 30    | 0.8    |
| 31    | 0.84   |
| 32    | 0.88   |
| 33    | 0.92   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16 | INT8 | INT4 |
|-------|------|------|------|
| 0     | 0.0  | 0.0  | 1.0  |
| 1     | 0.0  | 0.0  | 1.0  |
| 2     | 0.0  | 0.0  | 1.0  |
| 3     | 0.0  | 0.0  | 1.0  |
| 4     | 0.0  | 0.0  | 1.0  |
| 5     | 0.0  | 0.0  | 1.0  |
| 6     | 0.0  | 0.0  | 1.0  |
| 7     | 0.0  | 0.0  | 1.0  |
| 8     | 0.0  | 0.0  | 1.0  |
| 9     | 0.0  | 0.0  | 1.0  |
| 10    | 0.0  | 0.0  | 1.0  |
| 11    | 0.0  | 0.0  | 1.0  |
| 12    | 0.0  | 0.0  | 1.0  |
| 13    | 0.0  | 0.0  | 1.0  |
| 14    | 0.0  | 0.0  | 1.0  |
| 15    | 0.0  | 0.0  | 1.0  |

|    |                     |                     |                     |
|----|---------------------|---------------------|---------------------|
| 16 | 0.0                 | 0.0                 | 1.0                 |
| 17 | 0.0                 | 0.1196              | 0.8804              |
| 18 | 0.0                 | 0.2798              | 0.7201999999999998  |
| 19 | 0.0                 | 0.4399              | 0.5601              |
| 20 | 0.0                 | 0.5996              | 0.40040000000000003 |
| 21 | 0.0                 | 0.7597999999999999  | 0.2402              |
| 22 | 0.0                 | 0.9199              | 0.0801              |
| 23 | 0.04                | 0.96                | 0.0                 |
| 24 | 0.1201              | 0.8799              | 0.0                 |
| 25 | 0.20020000000000002 | 0.7998              | 0.0                 |
| 26 | 0.2798              | 0.7201999999999998  | 0.0                 |
| 27 | 0.35989999999999994 | 0.6401              | 0.0                 |
| 28 | 0.4399              | 0.5601              | 0.0                 |
| 29 | 0.52                | 0.48                | 0.0                 |
| 30 | 0.6001              | 0.3999              | 0.0                 |
| 31 | 0.6802              | 0.3198              | 0.0                 |
| 32 | 0.7597999999999999  | 0.2402              | 0.0                 |
| 33 | 0.8398              | 0.1602              | 0.0                 |
| 34 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35 |                     | 0.10010000000000001 | 0.0                 |

beta=0.04, budget=0.5, model=mistral7b

- PPL: 8.7349
- PPL Δ%: 4.13
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.28   |
| 13    | 0.32   |
| 14    | 0.36   |
| 15    | 0.4    |
| 16    | 0.44   |
| 17    | 0.48   |
| 18    | 0.52   |
| 19    | 0.56   |
| 20    | 0.6    |
| 21    | 0.64   |
| 22    | 0.68   |
| 23    | 0.72   |
| 24    | 0.76   |
| 25    | 0.8    |
| 26    | 0.84   |
| 27    | 0.88   |
| 28    | 0.92   |
| 29    | 0.95   |
| 30    | 0.95   |
| 31    | 0.95   |
| 32    | 0.95   |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8               | INT4                |
|-------|---------------------|--------------------|---------------------|
| 0     | 0.0                 | 0.0                | 1.0                 |
| 1     | 0.0                 | 0.0                | 1.0                 |
| 2     | 0.0                 | 0.0                | 1.0                 |
| 3     | 0.0                 | 0.0                | 1.0                 |
| 4     | 0.0                 | 0.0                | 1.0                 |
| 5     | 0.0                 | 0.0                | 1.0                 |
| 6     | 0.0                 | 0.0                | 1.0                 |
| 7     | 0.0                 | 0.0                | 1.0                 |
| 8     | 0.0                 | 0.0                | 1.0                 |
| 9     | 0.0                 | 0.0                | 1.0                 |
| 10    | 0.0                 | 0.0                | 1.0                 |
| 11    | 0.0                 | 0.0                | 1.0                 |
| 12    | 0.0                 | 0.1196             | 0.8804              |
| 13    | 0.0                 | 0.2798             | 0.7201999999999998  |
| 14    | 0.0                 | 0.4399             | 0.5601              |
| 15    | 0.0                 | 0.5996             | 0.40040000000000003 |
| 16    | 0.0                 | 0.7597999999999999 | 0.2402              |
| 17    | 0.0                 | 0.9199             | 0.0801              |
| 18    | 0.04                | 0.96               | 0.0                 |
| 19    | 0.1201              | 0.8799             | 0.0                 |
| 20    | 0.20020000000000002 | 0.7998             | 0.0                 |
| 21    | 0.2798              | 0.7201999999999998 | 0.0                 |
| 22    | 0.35989999999999994 | 0.6401             | 0.0                 |
| 23    | 0.4399              | 0.5601             | 0.0                 |
| 24    | 0.52                | 0.48               | 0.0                 |
| 25    | 0.6001              | 0.3999             | 0.0                 |
| 26    | 0.6802              | 0.3198             | 0.0                 |
| 27    | 0.7597999999999999  | 0.2402             | 0.0                 |
| 28    | 0.8398              | 0.1602             | 0.0                 |
|       |                     |                    |                     |

|    |        |                     |     |
|----|--------|---------------------|-----|
| 29 | 0.8999 | 0.10010000000000001 | 0.0 |
| 30 | 0.8999 | 0.10010000000000001 | 0.0 |
| 31 | 0.8999 | 0.10010000000000001 | 0.0 |
| 32 | 0.8999 | 0.10010000000000001 | 0.0 |
| 33 | 0.8999 | 0.10010000000000001 | 0.0 |
| 34 | 0.8999 | 0.10010000000000001 | 0.0 |
| 35 | 0.8999 | 0.10010000000000001 | 0.0 |

beta=0.04, budget=0.7, model=mistral7b

- PPL: 8.5685
- PPL Δ%: 2.14
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.28   |
| 8     | 0.32   |
| 9     | 0.36   |
| 10    | 0.4    |
| 11    | 0.44   |
| 12    | 0.48   |
| 13    | 0.52   |
| 14    | 0.56   |
| 15    | 0.6    |
| 16    | 0.64   |
| 17    | 0.68   |
| 18    | 0.72   |
| 19    | 0.76   |
| 20    | 0.8    |
| 21    | 0.84   |
| 22    | 0.88   |
| 23    | 0.92   |
| 24    | 0.95   |
| 25    | 0.95   |
| 26    | 0.95   |
| 27    | 0.95   |
| 28    | 0.95   |
| 29    | 0.95   |
| 30    | 0.95   |
| 31    | 0.95   |
| 32    | 0.95   |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.1196              | 0.8804              |
| 8     | 0.0                 | 0.2798              | 0.7201999999999998  |
| 9     | 0.0                 | 0.4399              | 0.5601              |
| 10    | 0.0                 | 0.5996              | 0.40040000000000003 |
| 11    | 0.0                 | 0.7597999999999999  | 0.2402              |
| 12    | 0.0                 | 0.9199              | 0.0801              |
| 13    | 0.04                | 0.96                | 0.0                 |
| 14    | 0.1201              | 0.8799              | 0.0                 |
| 15    | 0.20020000000000002 | 0.7998              | 0.0                 |
| 16    | 0.2798              | 0.7201999999999998  | 0.0                 |
| 17    | 0.35989999999999994 | 0.6401              | 0.0                 |
| 18    | 0.4399              | 0.5601              | 0.0                 |
| 19    | 0.52                | 0.48                | 0.0                 |
| 20    | 0.6001              | 0.3999              | 0.0                 |
| 21    | 0.6802              | 0.3198              | 0.0                 |
| 22    | 0.7597999999999999  | 0.2402              | 0.0                 |
| 23    | 0.8398              | 0.1602              | 0.0                 |
| 24    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 25    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 26    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 27    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 28    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 29    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 30    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 31    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 32    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 33    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 34    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35    | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.05, budget=0.3, model=mistral7b

- PPL: 8.7787
- PPL Δ%: 4.65
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |
| 13    | 0.25   |
| 14    | 0.25   |
| 15    | 0.25   |
| 16    | 0.25   |
| 17    | 0.275  |
| 18    | 0.325  |
| 19    | 0.375  |
| 20    | 0.425  |
| 21    | 0.475  |
| 22    | 0.525  |
| 23    | 0.575  |
| 24    | 0.625  |
| 25    | 0.675  |
| 26    | 0.725  |
| 27    | 0.775  |
| 28    | 0.825  |
| 29    | 0.875  |
| 30    | 0.925  |
| 31    | 0.95   |
| 32    | 0.95   |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.0                 | 1.0                 |
| 8     | 0.0                 | 0.0                 | 1.0                 |
| 9     | 0.0                 | 0.0                 | 1.0                 |
| 10    | 0.0                 | 0.0                 | 1.0                 |
| 11    | 0.0                 | 0.0                 | 1.0                 |
| 12    | 0.0                 | 0.0                 | 1.0                 |
| 13    | 0.0                 | 0.0                 | 1.0                 |
| 14    | 0.0                 | 0.0                 | 1.0                 |
| 15    | 0.0                 | 0.0                 | 1.0                 |
| 16    | 0.0                 | 0.0                 | 1.0                 |
| 17    | 0.0                 | 0.09559999999999998 | 0.9004              |
| 18    | 0.0                 | 0.2998              | 0.7002              |
| 19    | 0.0                 | 0.5                 | 0.5                 |
| 20    | 0.0                 | 0.6997              | 0.3003              |
| 21    | 0.0                 | 0.8999              | 0.10010000000000001 |
| 22    | 0.04979999999999999 | 0.9502              | 0.0                 |
| 23    | 0.1499              | 0.8501              | 0.0                 |
| 24    | 0.25                | 0.75                | 0.0                 |
| 25    | 0.3501              | 0.6499              | 0.0                 |
| 26    | 0.4502              | 0.5498              | 0.0                 |
| 27    | 0.5498              | 0.4502              | 0.0                 |
| 28    | 0.6499              | 0.3501              | 0.0                 |
| 29    | 0.75                | 0.25                | 0.0                 |
| 30    | 0.8501              | 0.1499              | 0.0                 |
| 31    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 32    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 33    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 34    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35    | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.05, budget=0.5, model=mistral7b

- PPL: 8.7386
- PPL Δ%: 4.17
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |



|    |       |
|----|-------|
| 12 | 0.25  |
| 13 | 0.275 |
| 14 | 0.325 |
| 15 | 0.375 |
| 16 | 0.425 |
| 17 | 0.475 |
| 18 | 0.525 |
| 19 | 0.575 |
| 20 | 0.625 |
| 21 | 0.675 |
| 22 | 0.725 |
| 23 | 0.775 |
| 24 | 0.825 |
| 25 | 0.875 |
| 26 | 0.925 |
| 27 | 0.95  |
| 28 | 0.95  |
| 29 | 0.95  |
| 30 | 0.95  |
| 31 | 0.95  |
| 32 | 0.95  |
| 33 | 0.95  |
| 34 | 0.95  |
| 35 | 0.95  |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.0                 | 1.0                 |
| 8     | 0.0                 | 0.0                 | 1.0                 |
| 9     | 0.0                 | 0.0                 | 1.0                 |
| 10    | 0.0                 | 0.0                 | 1.0                 |
| 11    | 0.0                 | 0.0                 | 1.0                 |
| 12    | 0.0                 | 0.0                 | 1.0                 |
| 13    | 0.0                 | 0.09559999999999998 | 0.9004              |
| 14    | 0.0                 | 0.2998              | 0.7002              |
| 15    | 0.0                 | 0.5                 | 0.5                 |
| 16    | 0.0                 | 0.6997              | 0.3003              |
| 17    | 0.0                 | 0.8999              | 0.10010000000000001 |
| 18    | 0.04979999999999999 | 0.9502              | 0.0                 |
| 19    | 0.1499              | 0.8501              | 0.0                 |
| 20    | 0.25                | 0.75                | 0.0                 |
| 21    | 0.3501              | 0.6499              | 0.0                 |
| 22    | 0.4502              | 0.5498              | 0.0                 |
| 23    | 0.5498              | 0.4502              | 0.0                 |
| 24    | 0.6499              | 0.3501              | 0.0                 |
| 25    | 0.75                | 0.25                | 0.0                 |
| 26    | 0.8501              | 0.1499              | 0.0                 |
| 27    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 28    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 29    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 30    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 31    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 32    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 33    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 34    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35    | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.05, budget=0.7, model=mistral7b

- PPL: 8.6262
- PPL Δ%: 2.83
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.275  |
| 10    | 0.325  |
| 11    | 0.375  |
| 12    | 0.425  |
| 13    | 0.475  |
| 14    | 0.525  |
| 15    | 0.575  |
| 16    | 0.625  |
| 17    | 0.675  |
| 18    | 0.725  |
| 19    | 0.775  |
| 20    | 0.825  |
| 21    | 0.875  |
| 22    | 0.925  |
| 23    | 0.95   |
| 24    | 0.95   |
| 25    | 0.95   |

|    |      |
|----|------|
| 26 | 0.95 |
| 27 | 0.95 |
| 28 | 0.95 |
| 29 | 0.95 |
| 30 | 0.95 |
| 31 | 0.95 |
| 32 | 0.95 |
| 33 | 0.95 |
| 34 | 0.95 |
| 35 | 0.95 |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.0                 | 1.0                 |
| 8     | 0.0                 | 0.0                 | 1.0                 |
| 9     | 0.0                 | 0.09959999999999998 | 0.9004              |
| 10    | 0.0                 | 0.2998              | 0.7002              |
| 11    | 0.0                 | 0.4995              | 0.5005              |
| 12    | 0.0                 | 0.6997              | 0.3003              |
| 13    | 0.0                 | 0.8999              | 0.10010000000000001 |
| 14    | 0.04979999999999999 | 0.9502              | 0.0                 |
| 15    | 0.1499              | 0.8501              | 0.0                 |
| 16    | 0.25                | 0.75                | 0.0                 |
| 17    | 0.3501              | 0.6499              | 0.0                 |
| 18    | 0.4502              | 0.5498              | 0.0                 |
| 19    | 0.5498              | 0.4502              | 0.0                 |
| 20    | 0.6499              | 0.3501              | 0.0                 |
| 21    | 0.75                | 0.25                | 0.0                 |
| 22    | 0.8501              | 0.1499              | 0.0                 |
| 23    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 24    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 25    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 26    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 27    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 28    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 29    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 30    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 31    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 32    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 33    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 34    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35    | 0.8999              | 0.10010000000000001 | 0.0                 |

exp4\_layer\_budget\_mistral7b.json

beta=0.0, budget=0.3, model=mistral7b

- PPL: 8.8336
- PPL Δ%: 5.3
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.3    |
| 1     | 0.3    |
| 2     | 0.3    |
| 3     | 0.3    |
| 4     | 0.3    |
| 5     | 0.3    |
| 6     | 0.3    |
| 7     | 0.3    |
| 8     | 0.3    |
| 9     | 0.3    |
| 10    | 0.3    |
| 11    | 0.3    |
| 12    | 0.3    |
| 13    | 0.3    |
| 14    | 0.3    |
| 15    | 0.3    |
| 16    | 0.3    |
| 17    | 0.3    |
| 18    | 0.3    |
| 19    | 0.3    |
| 20    | 0.3    |
| 21    | 0.3    |
| 22    | 0.3    |
| 23    | 0.3    |
| 24    | 0.3    |
| 25    | 0.3    |
| 26    | 0.3    |
| 27    | 0.3    |
| 28    | 0.3    |
| 29    | 0.3    |
| 30    | 0.3    |
| 31    | 0.3    |
| 32    | 0.3    |
| 33    | 0.3    |
| 34    | 0.3    |
| 35    | 0.3    |

Tier Fractions Per Layer:

| Layer | FP16 | INT8 | INT4 |
|-------|------|------|------|
|-------|------|------|------|

|    |     |        |        |
|----|-----|--------|--------|
| 0  | 0.0 | 0.1997 | 0.8003 |
| 1  | 0.0 | 0.1997 | 0.8003 |
| 2  | 0.0 | 0.1997 | 0.8003 |
| 3  | 0.0 | 0.1997 | 0.8003 |
| 4  | 0.0 | 0.1997 | 0.8003 |
| 5  | 0.0 | 0.1997 | 0.8003 |
| 6  | 0.0 | 0.1997 | 0.8003 |
| 7  | 0.0 | 0.1997 | 0.8003 |
| 8  | 0.0 | 0.1997 | 0.8003 |
| 9  | 0.0 | 0.1997 | 0.8003 |
| 10 | 0.0 | 0.1997 | 0.8003 |
| 11 | 0.0 | 0.1997 | 0.8003 |
| 12 | 0.0 | 0.1997 | 0.8003 |
| 13 | 0.0 | 0.1997 | 0.8003 |
| 14 | 0.0 | 0.1997 | 0.8003 |
| 15 | 0.0 | 0.1997 | 0.8003 |
| 16 | 0.0 | 0.1997 | 0.8003 |
| 17 | 0.0 | 0.1997 | 0.8003 |
| 18 | 0.0 | 0.1997 | 0.8003 |
| 19 | 0.0 | 0.1997 | 0.8003 |
| 20 | 0.0 | 0.1997 | 0.8003 |
| 21 | 0.0 | 0.1997 | 0.8003 |
| 22 | 0.0 | 0.1997 | 0.8003 |
| 23 | 0.0 | 0.1997 | 0.8003 |
| 24 | 0.0 | 0.1997 | 0.8003 |
| 25 | 0.0 | 0.1997 | 0.8003 |
| 26 | 0.0 | 0.1997 | 0.8003 |
| 27 | 0.0 | 0.1997 | 0.8003 |
| 28 | 0.0 | 0.1997 | 0.8003 |
| 29 | 0.0 | 0.1997 | 0.8003 |
| 30 | 0.0 | 0.1997 | 0.8003 |
| 31 | 0.0 | 0.1997 | 0.8003 |
| 32 | 0.0 | 0.1997 | 0.8003 |
| 33 | 0.0 | 0.1997 | 0.8003 |
| 34 | 0.0 | 0.1997 | 0.8003 |
| 35 | 0.0 | 0.1997 | 0.8003 |

beta=0.0, budget=0.5, model=mistral7b

- PPL: 8.3766
- PPL Δ%: -0.14
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.5    |
| 1     | 0.5    |
| 2     | 0.5    |
| 3     | 0.5    |
| 4     | 0.5    |
| 5     | 0.5    |
| 6     | 0.5    |
| 7     | 0.5    |
| 8     | 0.5    |
| 9     | 0.5    |
| 10    | 0.5    |
| 11    | 0.5    |
| 12    | 0.5    |
| 13    | 0.5    |
| 14    | 0.5    |
| 15    | 0.5    |
| 16    | 0.5    |
| 17    | 0.5    |
| 18    | 0.5    |
| 19    | 0.5    |
| 20    | 0.5    |
| 21    | 0.5    |
| 22    | 0.5    |
| 23    | 0.5    |
| 24    | 0.5    |
| 25    | 0.5    |
| 26    | 0.5    |
| 27    | 0.5    |
| 28    | 0.5    |
| 29    | 0.5    |
| 30    | 0.5    |
| 31    | 0.5    |
| 32    | 0.5    |
| 33    | 0.5    |
| 34    | 0.5    |
| 35    | 0.5    |

Tier Fractions Per Layer:

| Layer | FP16 | INT8 | INT4 |
|-------|------|------|------|
| 0     | 0.0  | 1.0  | 0.0  |
| 1     | 0.0  | 1.0  | 0.0  |
| 2     | 0.0  | 1.0  | 0.0  |
| 3     | 0.0  | 1.0  | 0.0  |
| 4     | 0.0  | 1.0  | 0.0  |
| 5     | 0.0  | 1.0  | 0.0  |
| 6     | 0.0  | 1.0  | 0.0  |
| 7     | 0.0  | 1.0  | 0.0  |
| 8     | 0.0  | 1.0  | 0.0  |
| 9     | 0.0  | 1.0  | 0.0  |
| 10    | 0.0  | 1.0  | 0.0  |
| 11    | 0.0  | 1.0  | 0.0  |
| 12    | 0.0  | 1.0  | 0.0  |
| 13    | 0.0  | 1.0  | 0.0  |

|    |     |     |     |
|----|-----|-----|-----|
| 14 | 0.0 | 1.0 | 0.0 |
| 15 | 0.0 | 1.0 | 0.0 |
| 16 | 0.0 | 1.0 | 0.0 |
| 17 | 0.0 | 1.0 | 0.0 |
| 18 | 0.0 | 1.0 | 0.0 |
| 19 | 0.0 | 1.0 | 0.0 |
| 20 | 0.0 | 1.0 | 0.0 |
| 21 | 0.0 | 1.0 | 0.0 |
| 22 | 0.0 | 1.0 | 0.0 |
| 23 | 0.0 | 1.0 | 0.0 |
| 24 | 0.0 | 1.0 | 0.0 |
| 25 | 0.0 | 1.0 | 0.0 |
| 26 | 0.0 | 1.0 | 0.0 |
| 27 | 0.0 | 1.0 | 0.0 |
| 28 | 0.0 | 1.0 | 0.0 |
| 29 | 0.0 | 1.0 | 0.0 |
| 30 | 0.0 | 1.0 | 0.0 |
| 31 | 0.0 | 1.0 | 0.0 |
| 32 | 0.0 | 1.0 | 0.0 |
| 33 | 0.0 | 1.0 | 0.0 |
| 34 | 0.0 | 1.0 | 0.0 |
| 35 | 0.0 | 1.0 | 0.0 |

beta=0.0, budget=0.7, model=mistral7b

- PPL: 8.3878
- PPL Δ%: -0.01
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.7    |
| 1     | 0.7    |
| 2     | 0.7    |
| 3     | 0.7    |
| 4     | 0.7    |
| 5     | 0.7    |
| 6     | 0.7    |
| 7     | 0.7    |
| 8     | 0.7    |
| 9     | 0.7    |
| 10    | 0.7    |
| 11    | 0.7    |
| 12    | 0.7    |
| 13    | 0.7    |
| 14    | 0.7    |
| 15    | 0.7    |
| 16    | 0.7    |
| 17    | 0.7    |
| 18    | 0.7    |
| 19    | 0.7    |
| 20    | 0.7    |
| 21    | 0.7    |
| 22    | 0.7    |
| 23    | 0.7    |
| 24    | 0.7    |
| 25    | 0.7    |
| 26    | 0.7    |
| 27    | 0.7    |
| 28    | 0.7    |
| 29    | 0.7    |
| 30    | 0.7    |
| 31    | 0.7    |
| 32    | 0.7    |
| 33    | 0.7    |
| 34    | 0.7    |
| 35    | 0.7    |

Tier Fractions Per Layer:

| Layer | FP16   | INT8   | INT4 |
|-------|--------|--------|------|
| 0     | 0.3999 | 0.6001 | 0.0  |
| 1     | 0.3999 | 0.6001 | 0.0  |
| 2     | 0.3999 | 0.6001 | 0.0  |
| 3     | 0.3999 | 0.6001 | 0.0  |
| 4     | 0.3999 | 0.6001 | 0.0  |
| 5     | 0.3999 | 0.6001 | 0.0  |
| 6     | 0.3999 | 0.6001 | 0.0  |
| 7     | 0.3999 | 0.6001 | 0.0  |
| 8     | 0.3999 | 0.6001 | 0.0  |
| 9     | 0.3999 | 0.6001 | 0.0  |
| 10    | 0.3999 | 0.6001 | 0.0  |
| 11    | 0.3999 | 0.6001 | 0.0  |
| 12    | 0.3999 | 0.6001 | 0.0  |
| 13    | 0.3999 | 0.6001 | 0.0  |
| 14    | 0.3999 | 0.6001 | 0.0  |
| 15    | 0.3999 | 0.6001 | 0.0  |
| 16    | 0.3999 | 0.6001 | 0.0  |
| 17    | 0.3999 | 0.6001 | 0.0  |
| 18    | 0.3999 | 0.6001 | 0.0  |
| 19    | 0.3999 | 0.6001 | 0.0  |
| 20    | 0.3999 | 0.6001 | 0.0  |
| 21    | 0.3999 | 0.6001 | 0.0  |
| 22    | 0.3999 | 0.6001 | 0.0  |
| 23    | 0.3999 | 0.6001 | 0.0  |
| 24    | 0.3999 | 0.6001 | 0.0  |
| 25    | 0.3999 | 0.6001 | 0.0  |
| 26    | 0.3999 | 0.6001 | 0.0  |
| 27    | 0.3999 | 0.6001 | 0.0  |

|    |        |        |     |
|----|--------|--------|-----|
| 28 | 0.3999 | 0.6001 | 0.0 |
| 29 | 0.3999 | 0.6001 | 0.0 |
| 30 | 0.3999 | 0.6001 | 0.0 |
| 31 | 0.3999 | 0.6001 | 0.0 |
| 32 | 0.3999 | 0.6001 | 0.0 |
| 33 | 0.3999 | 0.6001 | 0.0 |
| 34 | 0.3999 | 0.6001 | 0.0 |
| 35 | 0.3999 | 0.6001 | 0.0 |

beta=0.01, budget=0.3, model=mistral7b

- PPL: 8.809
- PPL Δ%: 5.01
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |
| 13    | 0.255  |
| 14    | 0.265  |
| 15    | 0.275  |
| 16    | 0.285  |
| 17    | 0.295  |
| 18    | 0.305  |
| 19    | 0.315  |
| 20    | 0.325  |
| 21    | 0.335  |
| 22    | 0.345  |
| 23    | 0.355  |
| 24    | 0.365  |
| 25    | 0.375  |
| 26    | 0.385  |
| 27    | 0.395  |
| 28    | 0.405  |
| 29    | 0.415  |
| 30    | 0.425  |
| 31    | 0.435  |
| 32    | 0.445  |
| 33    | 0.455  |
| 34    | 0.465  |
| 35    | 0.475  |

Tier Fractions Per Layer:

| Layer | FP16 | INT8                | INT4                |
|-------|------|---------------------|---------------------|
| 0     | 0.0  | 0.0                 | 1.0                 |
| 1     | 0.0  | 0.0                 | 1.0                 |
| 2     | 0.0  | 0.0                 | 1.0                 |
| 3     | 0.0  | 0.0                 | 1.0                 |
| 4     | 0.0  | 0.0                 | 1.0                 |
| 5     | 0.0  | 0.0                 | 1.0                 |
| 6     | 0.0  | 0.0                 | 1.0                 |
| 7     | 0.0  | 0.0                 | 1.0                 |
| 8     | 0.0  | 0.0                 | 1.0                 |
| 9     | 0.0  | 0.0                 | 1.0                 |
| 10    | 0.0  | 0.0                 | 1.0                 |
| 11    | 0.0  | 0.0                 | 1.0                 |
| 12    | 0.0  | 0.0                 | 1.0                 |
| 13    | 0.0  | 0.0195              | 0.9805              |
| 14    | 0.0  | 0.05960000000000001 | 0.9404              |
| 15    | 0.0  | 0.09599999999999998 | 0.9004              |
| 16    | 0.0  | 0.1396              | 0.8603999999999999  |
| 17    | 0.0  | 0.1797              | 0.8203              |
| 18    | 0.0  | 0.2197              | 0.7803              |
| 19    | 0.0  | 0.2598              | 0.7402000000000001  |
| 20    | 0.0  | 0.2998              | 0.7002              |
| 21    | 0.0  | 0.33980000000000005 | 0.6602              |
| 22    | 0.0  | 0.37989999999999996 | 0.6201              |
| 23    | 0.0  | 0.4199              | 0.5801              |
| 24    | 0.0  | 0.46                | 0.54                |
| 25    | 0.0  | 0.5                 | 0.5                 |
| 26    | 0.0  | 0.5396              | 0.4604              |
| 27    | 0.0  | 0.5796              | 0.42040000000000005 |
| 28    | 0.0  | 0.6196              | 0.3804              |
| 29    | 0.0  | 0.6597              | 0.3403              |
| 30    | 0.0  | 0.6997              | 0.3003              |
| 31    | 0.0  | 0.7397              | 0.2603              |
| 32    | 0.0  | 0.7798000000000002  | 0.22020000000000003 |
| 33    | 0.0  | 0.8198              | 0.1802              |
| 34    | 0.0  | 0.8599              | 0.1401              |
| 35    | 0.0  | 0.8999              | 0.10010000000000001 |

beta=0.01, budget=0.5, model=mistral7b

- PPL: 8.5589
- PPL Δ%: 2.03
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048

- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.325  |
| 1     | 0.335  |
| 2     | 0.345  |
| 3     | 0.355  |
| 4     | 0.365  |
| 5     | 0.375  |
| 6     | 0.385  |
| 7     | 0.395  |
| 8     | 0.405  |
| 9     | 0.415  |
| 10    | 0.425  |
| 11    | 0.435  |
| 12    | 0.445  |
| 13    | 0.455  |
| 14    | 0.465  |
| 15    | 0.475  |
| 16    | 0.485  |
| 17    | 0.495  |
| 18    | 0.505  |
| 19    | 0.515  |
| 20    | 0.525  |
| 21    | 0.535  |
| 22    | 0.545  |
| 23    | 0.555  |
| 24    | 0.565  |
| 25    | 0.575  |
| 26    | 0.585  |
| 27    | 0.595  |
| 28    | 0.605  |
| 29    | 0.615  |
| 30    | 0.625  |
| 31    | 0.635  |
| 32    | 0.645  |
| 33    | 0.655  |
| 34    | 0.665  |
| 35    | 0.675  |

Tier Fractions Per Layer:

| Layer | FP16                 | INT8                | INT4                |
|-------|----------------------|---------------------|---------------------|
| 0     | 0.0                  | 0.2998              | 0.7002              |
| 1     | 0.0                  | 0.33980000000000005 | 0.6602              |
| 2     | 0.0                  | 0.37989999999999996 | 0.6201              |
| 3     | 0.0                  | 0.4199              | 0.5801              |
| 4     | 0.0                  | 0.46                | 0.54                |
| 5     | 0.0                  | 0.5                 | 0.5                 |
| 6     | 0.0                  | 0.5396              | 0.4604              |
| 7     | 0.0                  | 0.5796              | 0.42040000000000005 |
| 8     | 0.0                  | 0.6196              | 0.3804              |
| 9     | 0.0                  | 0.6597              | 0.3403              |
| 10    | 0.0                  | 0.6997              | 0.3003              |
| 11    | 0.0                  | 0.7397              | 0.2603              |
| 12    | 0.0                  | 0.77980000000000002 | 0.22020000000000003 |
| 13    | 0.0                  | 0.8198              | 0.1802              |
| 14    | 0.0                  | 0.8599              | 0.1401              |
| 15    | 0.0                  | 0.8999              | 0.10010000000000001 |
| 16    | 0.0                  | 0.93990000000000001 | 0.06010000000000001 |
| 17    | 0.0                  | 0.98                | 0.02                |
| 18    | 0.0098               | 0.99020000000000001 | 0.0                 |
| 19    | 0.029800000000000004 | 0.97019999999999998 | 0.0                 |
| 20    | 0.04979999999999999  | 0.9502              | 0.0                 |
| 21    | 0.0698               | 0.9302              | 0.0                 |
| 22    | 0.0898               | 0.9102              | 0.0                 |
| 23    | 0.1099               | 0.8901              | 0.0                 |
| 24    | 0.1299               | 0.8701              | 0.0                 |
| 25    | 0.1499               | 0.8501              | 0.0                 |
| 26    | 0.169900000000000002 | 0.8301              | 0.0                 |
| 27    | 0.189900000000000004 | 0.81009999999999999 | 0.0                 |
| 28    | 0.21                 | 0.79                | 0.0                 |
| 29    | 0.23                 | 0.77                | 0.0                 |
| 30    | 0.25                 | 0.75                | 0.0                 |
| 31    | 0.27                 | 0.73                | 0.0                 |
| 32    | 0.29                 | 0.71                | 0.0                 |
| 33    | 0.3101               | 0.68990000000000001 | 0.0                 |
| 34    | 0.3301               | 0.6699              | 0.0                 |
| 35    | 0.3501               | 0.6499              | 0.0                 |

beta=0.01, budget=0.7, model=mistral7b

- PPL: 8.3804
- PPL Δ%: -0.1
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.525  |
| 1     | 0.535  |
| 2     | 0.545  |
| 3     | 0.555  |
| 4     | 0.565  |
| 5     | 0.575  |
| 6     | 0.585  |
| 7     | 0.595  |
| 8     | 0.605  |
| 9     | 0.615  |

|    |       |
|----|-------|
| 10 | 0.625 |
| 11 | 0.635 |
| 12 | 0.645 |
| 13 | 0.655 |
| 14 | 0.665 |
| 15 | 0.675 |
| 16 | 0.685 |
| 17 | 0.695 |
| 18 | 0.705 |
| 19 | 0.715 |
| 20 | 0.725 |
| 21 | 0.735 |
| 22 | 0.745 |
| 23 | 0.755 |
| 24 | 0.765 |
| 25 | 0.775 |
| 26 | 0.785 |
| 27 | 0.795 |
| 28 | 0.805 |
| 29 | 0.815 |
| 30 | 0.825 |
| 31 | 0.835 |
| 32 | 0.845 |
| 33 | 0.855 |
| 34 | 0.865 |
| 35 | 0.875 |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4 |
|-------|---------------------|---------------------|------|
| 0     | 0.04979999999999999 | 0.9502              | 0.0  |
| 1     | 0.0698              | 0.9302              | 0.0  |
| 2     | 0.0898              | 0.9102              | 0.0  |
| 3     | 0.1099              | 0.8901              | 0.0  |
| 4     | 0.1299              | 0.8701              | 0.0  |
| 5     | 0.1499              | 0.8501              | 0.0  |
| 6     | 0.16990000000000002 | 0.8301              | 0.0  |
| 7     | 0.18990000000000004 | 0.8100999999999999  | 0.0  |
| 8     | 0.21                | 0.79                | 0.0  |
| 9     | 0.23                | 0.77                | 0.0  |
| 10    | 0.25                | 0.75                | 0.0  |
| 11    | 0.27                | 0.73                | 0.0  |
| 12    | 0.29                | 0.71                | 0.0  |
| 13    | 0.3101              | 0.6899000000000001  | 0.0  |
| 14    | 0.3301              | 0.6699              | 0.0  |
| 15    | 0.3501              | 0.6499              | 0.0  |
| 16    | 0.37010000000000004 | 0.6299              | 0.0  |
| 17    | 0.39010000000000006 | 0.6099              | 0.0  |
| 18    | 0.41019999999999995 | 0.5898              | 0.0  |
| 19    | 0.43019999999999997 | 0.5698              | 0.0  |
| 20    | 0.4502              | 0.5498              | 0.0  |
| 21    | 0.4702              | 0.5298              | 0.0  |
| 22    | 0.4902              | 0.5098              | 0.0  |
| 23    | 0.5098              | 0.4902              | 0.0  |
| 24    | 0.5298              | 0.4702              | 0.0  |
| 25    | 0.5498              | 0.4502              | 0.0  |
| 26    | 0.5698              | 0.43019999999999997 | 0.0  |
| 27    | 0.5898              | 0.41019999999999995 | 0.0  |
| 28    | 0.6099              | 0.39010000000000006 | 0.0  |
| 29    | 0.6299              | 0.37010000000000004 | 0.0  |
| 30    | 0.6499              | 0.3501              | 0.0  |
| 31    | 0.6699              | 0.3301              | 0.0  |
| 32    | 0.68990000000000001 | 0.3101              | 0.0  |
| 33    | 0.71                | 0.29                | 0.0  |
| 34    | 0.73                | 0.27                | 0.0  |
| 35    | 0.75                | 0.25                | 0.0  |

beta=0.02, budget=0.3, model=mistral7b

- PPL: 8.8028
- PPL Δ%: 4.94
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |
| 13    | 0.25   |
| 14    | 0.25   |
| 15    | 0.25   |
| 16    | 0.27   |
| 17    | 0.29   |
| 18    | 0.31   |
| 19    | 0.33   |
| 20    | 0.35   |
| 21    | 0.37   |
| 22    | 0.39   |
| 23    | 0.41   |

|    |      |
|----|------|
| 24 | 0.43 |
| 25 | 0.45 |
| 26 | 0.47 |
| 27 | 0.49 |
| 28 | 0.51 |
| 29 | 0.53 |
| 30 | 0.55 |
| 31 | 0.57 |
| 32 | 0.59 |
| 33 | 0.61 |
| 34 | 0.63 |
| 35 | 0.65 |

Tier Fractions Per Layer:

| Layer | FP16                 | INT8                | INT4                |
|-------|----------------------|---------------------|---------------------|
| 0     | 0.0                  | 0.0                 | 1.0                 |
| 1     | 0.0                  | 0.0                 | 1.0                 |
| 2     | 0.0                  | 0.0                 | 1.0                 |
| 3     | 0.0                  | 0.0                 | 1.0                 |
| 4     | 0.0                  | 0.0                 | 1.0                 |
| 5     | 0.0                  | 0.0                 | 1.0                 |
| 6     | 0.0                  | 0.0                 | 1.0                 |
| 7     | 0.0                  | 0.0                 | 1.0                 |
| 8     | 0.0                  | 0.0                 | 1.0                 |
| 9     | 0.0                  | 0.0                 | 1.0                 |
| 10    | 0.0                  | 0.0                 | 1.0                 |
| 11    | 0.0                  | 0.0                 | 1.0                 |
| 12    | 0.0                  | 0.0                 | 1.0                 |
| 13    | 0.0                  | 0.0                 | 1.0                 |
| 14    | 0.0                  | 0.0                 | 1.0                 |
| 15    | 0.0                  | 0.0                 | 1.0                 |
| 16    | 0.0                  | 0.0796              | 0.9204              |
| 17    | 0.0                  | 0.1597              | 0.8403              |
| 18    | 0.0                  | 0.2397              | 0.7603              |
| 19    | 0.0                  | 0.3198              | 0.6802              |
| 20    | 0.0                  | 0.3999              | 0.6001              |
| 21    | 0.0                  | 0.48                | 0.52                |
| 22    | 0.0                  | 0.5596              | 0.44040000000000007 |
| 23    | 0.0                  | 0.6396              | 0.3604              |
| 24    | 0.0                  | 0.7197              | 0.2803              |
| 25    | 0.0                  | 0.7998              | 0.20020000000000002 |
| 26    | 0.0                  | 0.8799              | 0.1201              |
| 27    | 0.0                  | 0.96                | 0.04                |
| 28    | 0.02                 | 0.98                | 0.0                 |
| 29    | 0.060100000000000001 | 0.93990000000000001 | 0.0                 |
| 30    | 0.100100000000000001 | 0.8999              | 0.0                 |
| 31    | 0.1401               | 0.8599              | 0.0                 |
| 32    | 0.1802               | 0.8198              | 0.0                 |
| 33    | 0.220200000000000003 | 0.77980000000000002 | 0.0                 |
| 34    | 0.2598               | 0.74020000000000001 | 0.0                 |
| 35    | 0.2998               | 0.7002              | 0.0                 |

beta=0.02, budget=0.5, model=mistral7b

- PPL: 8.6255
- PPL Δ%: 2.82
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.27   |
| 7     | 0.29   |
| 8     | 0.31   |
| 9     | 0.33   |
| 10    | 0.35   |
| 11    | 0.37   |
| 12    | 0.39   |
| 13    | 0.41   |
| 14    | 0.43   |
| 15    | 0.45   |
| 16    | 0.47   |
| 17    | 0.49   |
| 18    | 0.51   |
| 19    | 0.53   |
| 20    | 0.55   |
| 21    | 0.57   |
| 22    | 0.59   |
| 23    | 0.61   |
| 24    | 0.63   |
| 25    | 0.65   |
| 26    | 0.67   |
| 27    | 0.69   |
| 28    | 0.71   |
| 29    | 0.73   |
| 30    | 0.75   |
| 31    | 0.77   |
| 32    | 0.79   |
| 33    | 0.81   |
| 34    | 0.83   |
| 35    | 0.85   |

Tier Fractions Per Layer:

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|



| Layer | FP16                | INT8                | INT4               |
|-------|---------------------|---------------------|--------------------|
| 0     | 0.0                 | 0.0                 | 1.0                |
| 1     | 0.0                 | 0.0                 | 1.0                |
| 2     | 0.0                 | 0.0                 | 1.0                |
| 3     | 0.0                 | 0.0                 | 1.0                |
| 4     | 0.0                 | 0.0                 | 1.0                |
| 5     | 0.0                 | 0.0                 | 1.0                |
| 6     | 0.0                 | 0.0796              | 0.9204             |
| 7     | 0.0                 | 0.1597              | 0.8403             |
| 8     | 0.0                 | 0.2397              | 0.7603             |
| 9     | 0.0                 | 0.3198              | 0.6802             |
| 10    | 0.0                 | 0.3999              | 0.6001             |
| 11    | 0.0                 | 0.48                | 0.52               |
| 12    | 0.0                 | 0.5596              | 0.4404000000000007 |
| 13    | 0.0                 | 0.6396              | 0.3604             |
| 14    | 0.0                 | 0.7197              | 0.2803             |
| 15    | 0.0                 | 0.7998              | 0.2002000000000002 |
| 16    | 0.0                 | 0.8799              | 0.1201             |
| 17    | 0.0                 | 0.96                | 0.04               |
| 18    | 0.02                | 0.98                | 0.0                |
| 19    | 0.06010000000000001 | 0.9399000000000001  | 0.0                |
| 20    | 0.10010000000000001 | 0.8999              | 0.0                |
| 21    | 0.1401              | 0.8599              | 0.0                |
| 22    | 0.1802              | 0.8198              | 0.0                |
| 23    | 0.22020000000000003 | 0.7798000000000002  | 0.0                |
| 24    | 0.2598              | 0.7402000000000001  | 0.0                |
| 25    | 0.2998              | 0.7002              | 0.0                |
| 26    | 0.33980000000000005 | 0.6602              | 0.0                |
| 27    | 0.37989999999999996 | 0.6201              | 0.0                |
| 28    | 0.4199              | 0.5801              | 0.0                |
| 29    | 0.46                | 0.54                | 0.0                |
| 30    | 0.5                 | 0.5                 | 0.0                |
| 31    | 0.54                | 0.46                | 0.0                |
| 32    | 0.5801              | 0.4199              | 0.0                |
| 33    | 0.6201              | 0.37989999999999996 | 0.0                |
| 34    | 0.6602              | 0.33980000000000005 | 0.0                |
| 35    | 0.7002              | 0.2998              | 0.0                |

beta=0.02, budget=0.7, model=mistral7b

- PPL: 8.4382
- PPL Δ%: 0.59
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.35   |
| 1     | 0.37   |
| 2     | 0.39   |
| 3     | 0.41   |
| 4     | 0.43   |
| 5     | 0.45   |
| 6     | 0.47   |
| 7     | 0.49   |
| 8     | 0.51   |
| 9     | 0.53   |
| 10    | 0.55   |
| 11    | 0.57   |
| 12    | 0.59   |
| 13    | 0.61   |
| 14    | 0.63   |
| 15    | 0.65   |
| 16    | 0.67   |
| 17    | 0.69   |
| 18    | 0.71   |
| 19    | 0.73   |
| 20    | 0.75   |
| 21    | 0.77   |
| 22    | 0.79   |
| 23    | 0.81   |
| 24    | 0.83   |
| 25    | 0.85   |
| 26    | 0.87   |
| 27    | 0.89   |
| 28    | 0.91   |
| 29    | 0.93   |
| 30    | 0.95   |
| 31    | 0.95   |
| 32    | 0.95   |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8               | INT4               |
|-------|---------------------|--------------------|--------------------|
| 0     | 0.0                 | 0.3999             | 0.6001             |
| 1     | 0.0                 | 0.48               | 0.52               |
| 2     | 0.0                 | 0.5596             | 0.4404000000000007 |
| 3     | 0.0                 | 0.6396             | 0.3604             |
| 4     | 0.0                 | 0.7197             | 0.2803             |
| 5     | 0.0                 | 0.7998             | 0.2002000000000002 |
| 6     | 0.0                 | 0.8799             | 0.1201             |
| 7     | 0.0                 | 0.96               | 0.04               |
| 8     | 0.02                | 0.98               | 0.0                |
| 9     | 0.06010000000000001 | 0.9399000000000001 | 0.0                |
| 10    | 0.10010000000000001 | 0.8999             | 0.0                |
| 11    | 0.1401              | 0.8599             | 0.0                |
| 12    | 0.1802              | 0.8198             | 0.0                |

|    |                     |                     |     |
|----|---------------------|---------------------|-----|
| 13 | 0.22020000000000003 | 0.7798000000000002  | 0.0 |
| 14 | 0.2598              | 0.7402000000000001  | 0.0 |
| 15 | 0.2998              | 0.7002              | 0.0 |
| 16 | 0.33980000000000005 | 0.6602              | 0.0 |
| 17 | 0.37989999999999996 | 0.6201              | 0.0 |
| 18 | 0.4199              | 0.5801              | 0.0 |
| 19 | 0.46                | 0.54                | 0.0 |
| 20 | 0.5                 | 0.5                 | 0.0 |
| 21 | 0.54                | 0.46                | 0.0 |
| 22 | 0.5801              | 0.4199              | 0.0 |
| 23 | 0.6201              | 0.37989999999999996 | 0.0 |
| 24 | 0.6602              | 0.33980000000000005 | 0.0 |
| 25 | 0.7002              | 0.2998              | 0.0 |
| 26 | 0.7402000000000001  | 0.2598              | 0.0 |
| 27 | 0.7798000000000002  | 0.22020000000000003 | 0.0 |
| 28 | 0.8198              | 0.1802              | 0.0 |
| 29 | 0.8599              | 0.1401              | 0.0 |
| 30 | 0.8999              | 0.10010000000000001 | 0.0 |
| 31 | 0.8999              | 0.10010000000000001 | 0.0 |
| 32 | 0.8999              | 0.10010000000000001 | 0.0 |
| 33 | 0.8999              | 0.10010000000000001 | 0.0 |
| 34 | 0.8999              | 0.10010000000000001 | 0.0 |
| 35 | 0.8999              | 0.10010000000000001 | 0.0 |

beta=0.03, budget=0.3, model=mistral7b

- PPL: 8.7855
- PPL Δ%: 4.73
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |
| 13    | 0.25   |
| 14    | 0.25   |
| 15    | 0.25   |
| 16    | 0.255  |
| 17    | 0.285  |
| 18    | 0.315  |
| 19    | 0.345  |
| 20    | 0.375  |
| 21    | 0.405  |
| 22    | 0.435  |
| 23    | 0.465  |
| 24    | 0.495  |
| 25    | 0.525  |
| 26    | 0.555  |
| 27    | 0.585  |
| 28    | 0.615  |
| 29    | 0.645  |
| 30    | 0.675  |
| 31    | 0.705  |
| 32    | 0.735  |
| 33    | 0.765  |
| 34    | 0.795  |
| 35    | 0.825  |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4               |
|-------|---------------------|---------------------|--------------------|
| 0     | 0.0                 | 0.0                 | 1.0                |
| 1     | 0.0                 | 0.0                 | 1.0                |
| 2     | 0.0                 | 0.0                 | 1.0                |
| 3     | 0.0                 | 0.0                 | 1.0                |
| 4     | 0.0                 | 0.0                 | 1.0                |
| 5     | 0.0                 | 0.0                 | 1.0                |
| 6     | 0.0                 | 0.0                 | 1.0                |
| 7     | 0.0                 | 0.0                 | 1.0                |
| 8     | 0.0                 | 0.0                 | 1.0                |
| 9     | 0.0                 | 0.0                 | 1.0                |
| 10    | 0.0                 | 0.0                 | 1.0                |
| 11    | 0.0                 | 0.0                 | 1.0                |
| 12    | 0.0                 | 0.0                 | 1.0                |
| 13    | 0.0                 | 0.0                 | 1.0                |
| 14    | 0.0                 | 0.0                 | 1.0                |
| 15    | 0.0                 | 0.0                 | 1.0                |
| 16    | 0.0                 | 0.0195              | 0.9805             |
| 17    | 0.0                 | 0.1396              | 0.8603999999999999 |
| 18    | 0.0                 | 0.2598              | 0.7402000000000001 |
| 19    | 0.0                 | 0.37989999999999996 | 0.6201             |
| 20    | 0.0                 | 0.5                 | 0.5                |
| 21    | 0.0                 | 0.6196              | 0.3804             |
| 22    | 0.0                 | 0.7397              | 0.2603             |
| 23    | 0.0                 | 0.8599              | 0.1401             |
| 24    | 0.0                 | 0.98                | 0.02               |
| 25    | 0.04979999999999999 | 0.9502              | 0.0                |
| 26    | 0.1099              | 0.8901              | 0.0                |

|    |                     |                     |     |
|----|---------------------|---------------------|-----|
| 27 | 0.16990000000000002 | 0.8301              | 0.0 |
| 28 | 0.23                | 0.77                | 0.0 |
| 29 | 0.29                | 0.71                | 0.0 |
| 30 | 0.3501              | 0.6499              | 0.0 |
| 31 | 0.41019999999999995 | 0.5898              | 0.0 |
| 32 | 0.4702              | 0.5298              | 0.0 |
| 33 | 0.5298              | 0.4702              | 0.0 |
| 34 | 0.5898              | 0.41019999999999995 | 0.0 |
| 35 | 0.6499              | 0.3501              | 0.0 |

beta=0.03, budget=0.5, model=mistral7b

- PPL: 8.703
- PPL Δ%: 3.75
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.275  |
| 11    | 0.305  |
| 12    | 0.335  |
| 13    | 0.365  |
| 14    | 0.395  |
| 15    | 0.425  |
| 16    | 0.455  |
| 17    | 0.485  |
| 18    | 0.515  |
| 19    | 0.545  |
| 20    | 0.575  |
| 21    | 0.605  |
| 22    | 0.635  |
| 23    | 0.665  |
| 24    | 0.695  |
| 25    | 0.725  |
| 26    | 0.755  |
| 27    | 0.785  |
| 28    | 0.815  |
| 29    | 0.845  |
| 30    | 0.875  |
| 31    | 0.905  |
| 32    | 0.935  |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                 | INT8                | INT4                |
|-------|----------------------|---------------------|---------------------|
| 0     | 0.0                  | 0.0                 | 1.0                 |
| 1     | 0.0                  | 0.0                 | 1.0                 |
| 2     | 0.0                  | 0.0                 | 1.0                 |
| 3     | 0.0                  | 0.0                 | 1.0                 |
| 4     | 0.0                  | 0.0                 | 1.0                 |
| 5     | 0.0                  | 0.0                 | 1.0                 |
| 6     | 0.0                  | 0.0                 | 1.0                 |
| 7     | 0.0                  | 0.0                 | 1.0                 |
| 8     | 0.0                  | 0.0                 | 1.0                 |
| 9     | 0.0                  | 0.0                 | 1.0                 |
| 10    | 0.0                  | 0.09959999999999998 | 0.9004              |
| 11    | 0.0                  | 0.2197              | 0.7803              |
| 12    | 0.0                  | 0.33980000000000005 | 0.6602              |
| 13    | 0.0                  | 0.46                | 0.54                |
| 14    | 0.0                  | 0.5796              | 0.42040000000000005 |
| 15    | 0.0                  | 0.6997              | 0.3003              |
| 16    | 0.0                  | 0.8198              | 0.1802              |
| 17    | 0.0                  | 0.9399000000000001  | 0.06010000000000001 |
| 18    | 0.029800000000000004 | 0.97019999999999998 | 0.0                 |
| 19    | 0.0898               | 0.9102              | 0.0                 |
| 20    | 0.1499               | 0.8501              | 0.0                 |
| 21    | 0.21                 | 0.79                | 0.0                 |
| 22    | 0.27                 | 0.73                | 0.0                 |
| 23    | 0.3301               | 0.6699              | 0.0                 |
| 24    | 0.39010000000000006  | 0.6099              | 0.0                 |
| 25    | 0.4502               | 0.5498              | 0.0                 |
| 26    | 0.5098               | 0.4902              | 0.0                 |
| 27    | 0.5698               | 0.43019999999999997 | 0.0                 |
| 28    | 0.6299               | 0.37010000000000004 | 0.0                 |
| 29    | 0.6899000000000001   | 0.3101              | 0.0                 |
| 30    | 0.75                 | 0.25                | 0.0                 |
| 31    | 0.8100999999999999   | 0.18990000000000004 | 0.0                 |
| 32    | 0.8701               | 0.1299              | 0.0                 |
| 33    | 0.8999               | 0.10010000000000001 | 0.0                 |
| 34    | 0.8999               | 0.10010000000000001 | 0.0                 |
| 35    | 0.8999               | 0.10010000000000001 | 0.0                 |

beta=0.03, budget=0.7, model=mistral7b

- PPL: 8.4844
- PPL Δ%: 1.14
- Baseline PPL: 8.3887
- Num layers: 36

- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.265  |
| 4     | 0.295  |
| 5     | 0.325  |
| 6     | 0.355  |
| 7     | 0.385  |
| 8     | 0.415  |
| 9     | 0.445  |
| 10    | 0.475  |
| 11    | 0.505  |
| 12    | 0.535  |
| 13    | 0.565  |
| 14    | 0.595  |
| 15    | 0.625  |
| 16    | 0.655  |
| 17    | 0.685  |
| 18    | 0.715  |
| 19    | 0.745  |
| 20    | 0.775  |
| 21    | 0.805  |
| 22    | 0.835  |
| 23    | 0.865  |
| 24    | 0.895  |
| 25    | 0.925  |
| 26    | 0.95   |
| 27    | 0.95   |
| 28    | 0.95   |
| 29    | 0.95   |
| 30    | 0.95   |
| 31    | 0.95   |
| 32    | 0.95   |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.05960000000000001 | 0.9404              |
| 4     | 0.0                 | 0.1797              | 0.8203              |
| 5     | 0.0                 | 0.2998              | 0.7002              |
| 6     | 0.0                 | 0.4199              | 0.5801              |
| 7     | 0.0                 | 0.5396              | 0.4604              |
| 8     | 0.0                 | 0.6597              | 0.3403              |
| 9     | 0.0                 | 0.7798000000000002  | 0.2202000000000003  |
| 10    | 0.0                 | 0.8999              | 0.10010000000000001 |
| 11    | 0.0098              | 0.9902000000000001  | 0.0                 |
| 12    | 0.0698              | 0.9302              | 0.0                 |
| 13    | 0.1299              | 0.8701              | 0.0                 |
| 14    | 0.18990000000000004 | 0.8100999999999999  | 0.0                 |
| 15    | 0.25                | 0.75                | 0.0                 |
| 16    | 0.3101              | 0.6899000000000001  | 0.0                 |
| 17    | 0.37010000000000004 | 0.6299              | 0.0                 |
| 18    | 0.43019999999999997 | 0.5698              | 0.0                 |
| 19    | 0.4902              | 0.5098              | 0.0                 |
| 20    | 0.5498              | 0.4502              | 0.0                 |
| 21    | 0.6099              | 0.39010000000000006 | 0.0                 |
| 22    | 0.6699              | 0.3301              | 0.0                 |
| 23    | 0.73                | 0.27                | 0.0                 |
| 24    | 0.79                | 0.21                | 0.0                 |
| 25    | 0.8501              | 0.1499              | 0.0                 |
| 26    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 27    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 28    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 29    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 30    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 31    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 32    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 33    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 34    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35    | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.04, budget=0.3, model=mistral7b

- PPL: 8.7831
- PPL Δ%: 4.7
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |

|    |      |
|----|------|
| 10 | 0.25 |
| 11 | 0.25 |
| 12 | 0.25 |
| 13 | 0.25 |
| 14 | 0.25 |
| 15 | 0.25 |
| 16 | 0.25 |
| 17 | 0.28 |
| 18 | 0.32 |
| 19 | 0.36 |
| 20 | 0.4  |
| 21 | 0.44 |
| 22 | 0.48 |
| 23 | 0.52 |
| 24 | 0.56 |
| 25 | 0.6  |
| 26 | 0.64 |
| 27 | 0.68 |
| 28 | 0.72 |
| 29 | 0.76 |
| 30 | 0.8  |
| 31 | 0.84 |
| 32 | 0.88 |
| 33 | 0.92 |
| 34 | 0.95 |
| 35 | 0.95 |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.0                 | 1.0                 |
| 8     | 0.0                 | 0.0                 | 1.0                 |
| 9     | 0.0                 | 0.0                 | 1.0                 |
| 10    | 0.0                 | 0.0                 | 1.0                 |
| 11    | 0.0                 | 0.0                 | 1.0                 |
| 12    | 0.0                 | 0.0                 | 1.0                 |
| 13    | 0.0                 | 0.0                 | 1.0                 |
| 14    | 0.0                 | 0.0                 | 1.0                 |
| 15    | 0.0                 | 0.0                 | 1.0                 |
| 16    | 0.0                 | 0.0                 | 1.0                 |
| 17    | 0.0                 | 0.1196              | 0.8804              |
| 18    | 0.0                 | 0.2798              | 0.7201999999999998  |
| 19    | 0.0                 | 0.4399              | 0.5601              |
| 20    | 0.0                 | 0.5996              | 0.40040000000000003 |
| 21    | 0.0                 | 0.7597999999999999  | 0.2402              |
| 22    | 0.0                 | 0.9199              | 0.0801              |
| 23    | 0.04                | 0.96                | 0.0                 |
| 24    | 0.1201              | 0.8799              | 0.0                 |
| 25    | 0.20020000000000002 | 0.7998              | 0.0                 |
| 26    | 0.2798              | 0.7201999999999998  | 0.0                 |
| 27    | 0.35989999999999994 | 0.6401              | 0.0                 |
| 28    | 0.4399              | 0.5601              | 0.0                 |
| 29    | 0.52                | 0.48                | 0.0                 |
| 30    | 0.6001              | 0.3999              | 0.0                 |
| 31    | 0.6802              | 0.3198              | 0.0                 |
| 32    | 0.7597999999999999  | 0.2402              | 0.0                 |
| 33    | 0.8398              | 0.1602              | 0.0                 |
| 34    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35    | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.04, budget=0.5, model=mistral7b

- PPL: 8.7349
- PPL Δ%: 4.13
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.28   |
| 13    | 0.32   |
| 14    | 0.36   |
| 15    | 0.4    |
| 16    | 0.44   |
| 17    | 0.48   |
| 18    | 0.52   |
| 19    | 0.56   |
| 20    | 0.6    |
| 21    | 0.64   |
| 22    | 0.68   |

|    |      |
|----|------|
| 23 | 0.72 |
| 24 | 0.76 |
| 25 | 0.8  |
| 26 | 0.84 |
| 27 | 0.88 |
| 28 | 0.92 |
| 29 | 0.95 |
| 30 | 0.95 |
| 31 | 0.95 |
| 32 | 0.95 |
| 33 | 0.95 |
| 34 | 0.95 |
| 35 | 0.95 |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.0                 | 1.0                 |
| 8     | 0.0                 | 0.0                 | 1.0                 |
| 9     | 0.0                 | 0.0                 | 1.0                 |
| 10    | 0.0                 | 0.0                 | 1.0                 |
| 11    | 0.0                 | 0.0                 | 1.0                 |
| 12    | 0.0                 | 0.1196              | 0.8804              |
| 13    | 0.0                 | 0.2798              | 0.7201999999999998  |
| 14    | 0.0                 | 0.4399              | 0.5601              |
| 15    | 0.0                 | 0.5996              | 0.40040000000000003 |
| 16    | 0.0                 | 0.7597999999999999  | 0.2402              |
| 17    | 0.0                 | 0.9199              | 0.0801              |
| 18    | 0.04                | 0.96                | 0.0                 |
| 19    | 0.1201              | 0.8799              | 0.0                 |
| 20    | 0.20020000000000002 | 0.7998              | 0.0                 |
| 21    | 0.2798              | 0.7201999999999998  | 0.0                 |
| 22    | 0.35989999999999994 | 0.6401              | 0.0                 |
| 23    | 0.4399              | 0.5601              | 0.0                 |
| 24    | 0.52                | 0.48                | 0.0                 |
| 25    | 0.6001              | 0.3999              | 0.0                 |
| 26    | 0.6802              | 0.3198              | 0.0                 |
| 27    | 0.7597999999999999  | 0.2402              | 0.0                 |
| 28    | 0.8398              | 0.1602              | 0.0                 |
| 29    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 30    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 31    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 32    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 33    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 34    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35    | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.04, budget=0.7, model=mistral7b

- PPL: 8.5685
- PPL Δ%: 2.14
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.28   |
| 8     | 0.32   |
| 9     | 0.36   |
| 10    | 0.4    |
| 11    | 0.44   |
| 12    | 0.48   |
| 13    | 0.52   |
| 14    | 0.56   |
| 15    | 0.6    |
| 16    | 0.64   |
| 17    | 0.68   |
| 18    | 0.72   |
| 19    | 0.76   |
| 20    | 0.8    |
| 21    | 0.84   |
| 22    | 0.88   |
| 23    | 0.92   |
| 24    | 0.95   |
| 25    | 0.95   |
| 26    | 0.95   |
| 27    | 0.95   |
| 28    | 0.95   |
| 29    | 0.95   |
| 30    | 0.95   |
| 31    | 0.95   |
| 32    | 0.95   |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.1196              | 0.8804              |
| 8     | 0.0                 | 0.2798              | 0.7201999999999998  |
| 9     | 0.0                 | 0.4399              | 0.5601              |
| 10    | 0.0                 | 0.5996              | 0.40040000000000003 |
| 11    | 0.0                 | 0.7597999999999999  | 0.2402              |
| 12    | 0.0                 | 0.9199              | 0.0801              |
| 13    | 0.04                | 0.96                | 0.0                 |
| 14    | 0.1201              | 0.8799              | 0.0                 |
| 15    | 0.20020000000000002 | 0.7998              | 0.0                 |
| 16    | 0.2798              | 0.7201999999999998  | 0.0                 |
| 17    | 0.35989999999999994 | 0.6401              | 0.0                 |
| 18    | 0.4399              | 0.5601              | 0.0                 |
| 19    | 0.52                | 0.48                | 0.0                 |
| 20    | 0.6001              | 0.3999              | 0.0                 |
| 21    | 0.6802              | 0.3198              | 0.0                 |
| 22    | 0.7597999999999999  | 0.2402              | 0.0                 |
| 23    | 0.8398              | 0.1602              | 0.0                 |
| 24    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 25    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 26    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 27    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 28    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 29    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 30    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 31    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 32    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 33    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 34    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35    | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.05, budget=0.3, model=mistral7b

- PPL: 8.7787
- PPL Δ%: 4.65
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |
| 13    | 0.25   |
| 14    | 0.25   |
| 15    | 0.25   |
| 16    | 0.25   |
| 17    | 0.275  |
| 18    | 0.325  |
| 19    | 0.375  |
| 20    | 0.425  |
| 21    | 0.475  |
| 22    | 0.525  |
| 23    | 0.575  |
| 24    | 0.625  |
| 25    | 0.675  |
| 26    | 0.725  |
| 27    | 0.775  |
| 28    | 0.825  |
| 29    | 0.875  |
| 30    | 0.925  |
| 31    | 0.95   |
| 32    | 0.95   |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16 | INT8 | INT4 |
|-------|------|------|------|
| 0     | 0.0  | 0.0  | 1.0  |
| 1     | 0.0  | 0.0  | 1.0  |
| 2     | 0.0  | 0.0  | 1.0  |
| 3     | 0.0  | 0.0  | 1.0  |
| 4     | 0.0  | 0.0  | 1.0  |
| 5     | 0.0  | 0.0  | 1.0  |
| 6     | 0.0  | 0.0  | 1.0  |
| 7     | 0.0  | 0.0  | 1.0  |
| 8     | 0.0  | 0.0  | 1.0  |
| 9     | 0.0  | 0.0  | 1.0  |
| 10    | 0.0  | 0.0  | 1.0  |
| 11    | 0.0  | 0.0  | 1.0  |
| 12    | 0.0  | 0.0  | 1.0  |

|    |                     |                     |                     |
|----|---------------------|---------------------|---------------------|
| 13 | 0.0                 | 0.0                 | 1.0                 |
| 14 | 0.0                 | 0.0                 | 1.0                 |
| 15 | 0.0                 | 0.0                 | 1.0                 |
| 16 | 0.0                 | 0.0                 | 1.0                 |
| 17 | 0.0                 | 0.09959999999999998 | 0.9004              |
| 18 | 0.0                 | 0.2998              | 0.7002              |
| 19 | 0.0                 | 0.5                 | 0.5                 |
| 20 | 0.0                 | 0.6997              | 0.3003              |
| 21 | 0.0                 | 0.8999              | 0.10010000000000001 |
| 22 | 0.04979999999999999 | 0.9502              | 0.0                 |
| 23 | 0.1499              | 0.8501              | 0.0                 |
| 24 | 0.25                | 0.75                | 0.0                 |
| 25 | 0.3501              | 0.6499              | 0.0                 |
| 26 | 0.4502              | 0.5498              | 0.0                 |
| 27 | 0.5498              | 0.4502              | 0.0                 |
| 28 | 0.6499              | 0.3501              | 0.0                 |
| 29 | 0.75                | 0.25                | 0.0                 |
| 30 | 0.8501              | 0.1499              | 0.0                 |
| 31 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 32 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 33 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 34 | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35 | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.05, budget=0.5, model=mistral7b

- PPL: 8.7386
- PPL Δ%: 4.17
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |
| 13    | 0.275  |
| 14    | 0.325  |
| 15    | 0.375  |
| 16    | 0.425  |
| 17    | 0.475  |
| 18    | 0.525  |
| 19    | 0.575  |
| 20    | 0.625  |
| 21    | 0.675  |
| 22    | 0.725  |
| 23    | 0.775  |
| 24    | 0.825  |
| 25    | 0.875  |
| 26    | 0.925  |
| 27    | 0.95   |
| 28    | 0.95   |
| 29    | 0.95   |
| 30    | 0.95   |
| 31    | 0.95   |
| 32    | 0.95   |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.0                 | 1.0                 |
| 8     | 0.0                 | 0.0                 | 1.0                 |
| 9     | 0.0                 | 0.0                 | 1.0                 |
| 10    | 0.0                 | 0.0                 | 1.0                 |
| 11    | 0.0                 | 0.0                 | 1.0                 |
| 12    | 0.0                 | 0.0                 | 1.0                 |
| 13    | 0.0                 | 0.09959999999999998 | 0.9004              |
| 14    | 0.0                 | 0.2998              | 0.7002              |
| 15    | 0.0                 | 0.5                 | 0.5                 |
| 16    | 0.0                 | 0.6997              | 0.3003              |
| 17    | 0.0                 | 0.8999              | 0.10010000000000001 |
| 18    | 0.04979999999999999 | 0.9502              | 0.0                 |
| 19    | 0.1499              | 0.8501              | 0.0                 |
| 20    | 0.25                | 0.75                | 0.0                 |
| 21    | 0.3501              | 0.6499              | 0.0                 |
| 22    | 0.4502              | 0.5498              | 0.0                 |
| 23    | 0.5498              | 0.4502              | 0.0                 |
| 24    | 0.6499              | 0.3501              | 0.0                 |
| 25    | 0.75                | 0.25                | 0.0                 |
| 26    | 0.8501              | 0.1499              | 0.0                 |



|    |        |                     |     |
|----|--------|---------------------|-----|
| 27 | 0.8999 | 0.10010000000000001 | 0.0 |
| 28 | 0.8999 | 0.10010000000000001 | 0.0 |
| 29 | 0.8999 | 0.10010000000000001 | 0.0 |
| 30 | 0.8999 | 0.10010000000000001 | 0.0 |
| 31 | 0.8999 | 0.10010000000000001 | 0.0 |
| 32 | 0.8999 | 0.10010000000000001 | 0.0 |
| 33 | 0.8999 | 0.10010000000000001 | 0.0 |
| 34 | 0.8999 | 0.10010000000000001 | 0.0 |
| 35 | 0.8999 | 0.10010000000000001 | 0.0 |

beta=0.05, budget=0.7, model=mistral7b

- PPL: 8.6262
- PPL Δ%: 2.83
- Baseline PPL: 8.3887
- Num layers: 36
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.275  |
| 10    | 0.325  |
| 11    | 0.375  |
| 12    | 0.425  |
| 13    | 0.475  |
| 14    | 0.525  |
| 15    | 0.575  |
| 16    | 0.625  |
| 17    | 0.675  |
| 18    | 0.725  |
| 19    | 0.775  |
| 20    | 0.825  |
| 21    | 0.875  |
| 22    | 0.925  |
| 23    | 0.95   |
| 24    | 0.95   |
| 25    | 0.95   |
| 26    | 0.95   |
| 27    | 0.95   |
| 28    | 0.95   |
| 29    | 0.95   |
| 30    | 0.95   |
| 31    | 0.95   |
| 32    | 0.95   |
| 33    | 0.95   |
| 34    | 0.95   |
| 35    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.0                 | 1.0                 |
| 8     | 0.0                 | 0.0                 | 1.0                 |
| 9     | 0.0                 | 0.09959999999999998 | 0.9004              |
| 10    | 0.0                 | 0.2998              | 0.7002              |
| 11    | 0.0                 | 0.4995              | 0.5005              |
| 12    | 0.0                 | 0.6997              | 0.3003              |
| 13    | 0.0                 | 0.8999              | 0.10010000000000001 |
| 14    | 0.04979999999999999 | 0.9502              | 0.0                 |
| 15    | 0.1499              | 0.8501              | 0.0                 |
| 16    | 0.25                | 0.75                | 0.0                 |
| 17    | 0.3501              | 0.6499              | 0.0                 |
| 18    | 0.4502              | 0.5498              | 0.0                 |
| 19    | 0.5498              | 0.4502              | 0.0                 |
| 20    | 0.6499              | 0.3501              | 0.0                 |
| 21    | 0.75                | 0.25                | 0.0                 |
| 22    | 0.8501              | 0.1499              | 0.0                 |
| 23    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 24    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 25    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 26    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 27    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 28    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 29    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 30    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 31    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 32    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 33    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 34    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 35    | 0.8999              | 0.10010000000000001 | 0.0                 |

exp4\_layer\_budget\_qwen7b.json

beta=0.0, budget=0.3, model=qwen7b

- PPL: 12820.541

- PPL Δ%: 196007.45
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.3    |
| 1     | 0.3    |
| 2     | 0.3    |
| 3     | 0.3    |
| 4     | 0.3    |
| 5     | 0.3    |
| 6     | 0.3    |
| 7     | 0.3    |
| 8     | 0.3    |
| 9     | 0.3    |
| 10    | 0.3    |
| 11    | 0.3    |
| 12    | 0.3    |
| 13    | 0.3    |
| 14    | 0.3    |
| 15    | 0.3    |
| 16    | 0.3    |
| 17    | 0.3    |
| 18    | 0.3    |
| 19    | 0.3    |
| 20    | 0.3    |
| 21    | 0.3    |
| 22    | 0.3    |
| 23    | 0.3    |
| 24    | 0.3    |
| 25    | 0.3    |
| 26    | 0.3    |
| 27    | 0.3    |

Tier Fractions Per Layer:

| Layer | FP16 | INT8   | INT4   |
|-------|------|--------|--------|
| 0     | 0.0  | 0.1997 | 0.8003 |
| 1     | 0.0  | 0.1997 | 0.8003 |
| 2     | 0.0  | 0.1997 | 0.8003 |
| 3     | 0.0  | 0.1997 | 0.8003 |
| 4     | 0.0  | 0.1997 | 0.8003 |
| 5     | 0.0  | 0.1997 | 0.8003 |
| 6     | 0.0  | 0.1997 | 0.8003 |
| 7     | 0.0  | 0.1997 | 0.8003 |
| 8     | 0.0  | 0.1997 | 0.8003 |
| 9     | 0.0  | 0.1997 | 0.8003 |
| 10    | 0.0  | 0.1997 | 0.8003 |
| 11    | 0.0  | 0.1997 | 0.8003 |
| 12    | 0.0  | 0.1997 | 0.8003 |
| 13    | 0.0  | 0.1997 | 0.8003 |
| 14    | 0.0  | 0.1997 | 0.8003 |
| 15    | 0.0  | 0.1997 | 0.8003 |
| 16    | 0.0  | 0.1997 | 0.8003 |
| 17    | 0.0  | 0.1997 | 0.8003 |
| 18    | 0.0  | 0.1997 | 0.8003 |
| 19    | 0.0  | 0.1997 | 0.8003 |
| 20    | 0.0  | 0.1997 | 0.8003 |
| 21    | 0.0  | 0.1997 | 0.8003 |
| 22    | 0.0  | 0.1997 | 0.8003 |
| 23    | 0.0  | 0.1997 | 0.8003 |
| 24    | 0.0  | 0.1997 | 0.8003 |
| 25    | 0.0  | 0.1997 | 0.8003 |
| 26    | 0.0  | 0.1997 | 0.8003 |
| 27    | 0.0  | 0.1997 | 0.8003 |

beta=0.0, budget=0.5, model=qwen7b

- PPL: 6.6659
- PPL Δ%: 1.96
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.5    |
| 1     | 0.5    |
| 2     | 0.5    |
| 3     | 0.5    |
| 4     | 0.5    |
| 5     | 0.5    |
| 6     | 0.5    |
| 7     | 0.5    |
| 8     | 0.5    |
| 9     | 0.5    |
| 10    | 0.5    |
| 11    | 0.5    |
| 12    | 0.5    |
| 13    | 0.5    |
| 14    | 0.5    |
| 15    | 0.5    |
| 16    | 0.5    |
| 17    | 0.5    |
| 18    | 0.5    |
| 19    | 0.5    |
| 20    | 0.5    |
| 21    | 0.5    |
| 22    | 0.5    |

|    |     |
|----|-----|
| 23 | 0.5 |
| 24 | 0.5 |
| 25 | 0.5 |
| 26 | 0.5 |
| 27 | 0.5 |

Tier Fractions Per Layer:

| Layer | FP16 | INT8 | INT4 |
|-------|------|------|------|
| 0     | 0.0  | 1.0  | 0.0  |
| 1     | 0.0  | 1.0  | 0.0  |
| 2     | 0.0  | 1.0  | 0.0  |
| 3     | 0.0  | 1.0  | 0.0  |
| 4     | 0.0  | 1.0  | 0.0  |
| 5     | 0.0  | 1.0  | 0.0  |
| 6     | 0.0  | 1.0  | 0.0  |
| 7     | 0.0  | 1.0  | 0.0  |
| 8     | 0.0  | 1.0  | 0.0  |
| 9     | 0.0  | 1.0  | 0.0  |
| 10    | 0.0  | 1.0  | 0.0  |
| 11    | 0.0  | 1.0  | 0.0  |
| 12    | 0.0  | 1.0  | 0.0  |
| 13    | 0.0  | 1.0  | 0.0  |
| 14    | 0.0  | 1.0  | 0.0  |
| 15    | 0.0  | 1.0  | 0.0  |
| 16    | 0.0  | 1.0  | 0.0  |
| 17    | 0.0  | 1.0  | 0.0  |
| 18    | 0.0  | 1.0  | 0.0  |
| 19    | 0.0  | 1.0  | 0.0  |
| 20    | 0.0  | 1.0  | 0.0  |
| 21    | 0.0  | 1.0  | 0.0  |
| 22    | 0.0  | 1.0  | 0.0  |
| 23    | 0.0  | 1.0  | 0.0  |
| 24    | 0.0  | 1.0  | 0.0  |
| 25    | 0.0  | 1.0  | 0.0  |
| 26    | 0.0  | 1.0  | 0.0  |
| 27    | 0.0  | 1.0  | 0.0  |

beta=0.0, budget=0.7, model=qwen7b

- PPL: 6.5613
- PPL Δ%: 0.36
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.7    |
| 1     | 0.7    |
| 2     | 0.7    |
| 3     | 0.7    |
| 4     | 0.7    |
| 5     | 0.7    |
| 6     | 0.7    |
| 7     | 0.7    |
| 8     | 0.7    |
| 9     | 0.7    |
| 10    | 0.7    |
| 11    | 0.7    |
| 12    | 0.7    |
| 13    | 0.7    |
| 14    | 0.7    |
| 15    | 0.7    |
| 16    | 0.7    |
| 17    | 0.7    |
| 18    | 0.7    |
| 19    | 0.7    |
| 20    | 0.7    |
| 21    | 0.7    |
| 22    | 0.7    |
| 23    | 0.7    |
| 24    | 0.7    |
| 25    | 0.7    |
| 26    | 0.7    |
| 27    | 0.7    |

Tier Fractions Per Layer:

| Layer | FP16   | INT8   | INT4 |
|-------|--------|--------|------|
| 0     | 0.3999 | 0.6001 | 0.0  |
| 1     | 0.3999 | 0.6001 | 0.0  |
| 2     | 0.3999 | 0.6001 | 0.0  |
| 3     | 0.3999 | 0.6001 | 0.0  |
| 4     | 0.3999 | 0.6001 | 0.0  |
| 5     | 0.3999 | 0.6001 | 0.0  |
| 6     | 0.3999 | 0.6001 | 0.0  |
| 7     | 0.3999 | 0.6001 | 0.0  |
| 8     | 0.3999 | 0.6001 | 0.0  |
| 9     | 0.3999 | 0.6001 | 0.0  |
| 10    | 0.3999 | 0.6001 | 0.0  |
| 11    | 0.3999 | 0.6001 | 0.0  |
| 12    | 0.3999 | 0.6001 | 0.0  |
| 13    | 0.3999 | 0.6001 | 0.0  |
| 14    | 0.3999 | 0.6001 | 0.0  |
| 15    | 0.3999 | 0.6001 | 0.0  |
| 16    | 0.3999 | 0.6001 | 0.0  |
| 17    | 0.3999 | 0.6001 | 0.0  |
| 18    | 0.3999 | 0.6001 | 0.0  |
| 19    | 0.3999 | 0.6001 | 0.0  |
| 20    | 0.3999 | 0.6001 | 0.0  |
| 21    | 0.3999 | 0.6001 | 0.0  |

|    |        |        |     |
|----|--------|--------|-----|
| 22 | 0.3999 | 0.6001 | 0.0 |
| 23 | 0.3999 | 0.6001 | 0.0 |
| 24 | 0.3999 | 0.6001 | 0.0 |
| 25 | 0.3999 | 0.6001 | 0.0 |
| 26 | 0.3999 | 0.6001 | 0.0 |
| 27 | 0.3999 | 0.6001 | 0.0 |

beta=0.01, budget=0.3, model=qwen7b

- PPL: 28036.9593
- PPL Δ%: 428763.08
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.255  |
| 10    | 0.265  |
| 11    | 0.275  |
| 12    | 0.285  |
| 13    | 0.295  |
| 14    | 0.305  |
| 15    | 0.315  |
| 16    | 0.325  |
| 17    | 0.335  |
| 18    | 0.345  |
| 19    | 0.355  |
| 20    | 0.365  |
| 21    | 0.375  |
| 22    | 0.385  |
| 23    | 0.395  |
| 24    | 0.405  |
| 25    | 0.415  |
| 26    | 0.425  |
| 27    | 0.435  |

Tier Fractions Per Layer:

| Layer | FP16 | INT8                | INT4                |
|-------|------|---------------------|---------------------|
| 0     | 0.0  | 0.0                 | 1.0                 |
| 1     | 0.0  | 0.0                 | 1.0                 |
| 2     | 0.0  | 0.0                 | 1.0                 |
| 3     | 0.0  | 0.0                 | 1.0                 |
| 4     | 0.0  | 0.0                 | 1.0                 |
| 5     | 0.0  | 0.0                 | 1.0                 |
| 6     | 0.0  | 0.0                 | 1.0                 |
| 7     | 0.0  | 0.0                 | 1.0                 |
| 8     | 0.0  | 0.0                 | 1.0                 |
| 9     | 0.0  | 0.0195              | 0.9805              |
| 10    | 0.0  | 0.05960000000000001 | 0.9404              |
| 11    | 0.0  | 0.09599999999999998 | 0.9004              |
| 12    | 0.0  | 0.1396              | 0.8603999999999999  |
| 13    | 0.0  | 0.1797              | 0.8203              |
| 14    | 0.0  | 0.2197              | 0.7803              |
| 15    | 0.0  | 0.2598              | 0.7402000000000001  |
| 16    | 0.0  | 0.2998              | 0.7002              |
| 17    | 0.0  | 0.33980000000000005 | 0.6602              |
| 18    | 0.0  | 0.37989999999999996 | 0.6201              |
| 19    | 0.0  | 0.4199              | 0.5801              |
| 20    | 0.0  | 0.46                | 0.54                |
| 21    | 0.0  | 0.5                 | 0.5                 |
| 22    | 0.0  | 0.5396              | 0.4604              |
| 23    | 0.0  | 0.5796              | 0.42040000000000005 |
| 24    | 0.0  | 0.6196              | 0.3804              |
| 25    | 0.0  | 0.6597              | 0.3403              |
| 26    | 0.0  | 0.6997              | 0.3003              |
| 27    | 0.0  | 0.7397              | 0.2603              |

beta=0.01, budget=0.5, model=qwen7b

- PPL: 1912.0959
- PPL Δ%: 29148.08
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.365  |
| 1     | 0.375  |
| 2     | 0.385  |
| 3     | 0.395  |
| 4     | 0.405  |
| 5     | 0.415  |
| 6     | 0.425  |
| 7     | 0.435  |
| 8     | 0.445  |
| 9     | 0.455  |
| 10    | 0.465  |
| 11    | 0.475  |
| 12    | 0.485  |
| 13    | 0.495  |
| 14    | 0.505  |

|    |       |
|----|-------|
| 15 | 0.515 |
| 16 | 0.525 |
| 17 | 0.535 |
| 18 | 0.545 |
| 19 | 0.555 |
| 20 | 0.565 |
| 21 | 0.575 |
| 22 | 0.585 |
| 23 | 0.595 |
| 24 | 0.605 |
| 25 | 0.615 |
| 26 | 0.625 |
| 27 | 0.635 |

Tier Fractions Per Layer:

| Layer | FP16                 | INT8               | INT4               |
|-------|----------------------|--------------------|--------------------|
| 0     | 0.0                  | 0.46               | 0.54               |
| 1     | 0.0                  | 0.5                | 0.5                |
| 2     | 0.0                  | 0.5396             | 0.4604             |
| 3     | 0.0                  | 0.5796             | 0.4204000000000005 |
| 4     | 0.0                  | 0.6196             | 0.3804             |
| 5     | 0.0                  | 0.6597             | 0.3403             |
| 6     | 0.0                  | 0.6997             | 0.3003             |
| 7     | 0.0                  | 0.7397             | 0.2603             |
| 8     | 0.0                  | 0.7798000000000002 | 0.2202000000000003 |
| 9     | 0.0                  | 0.8198             | 0.1802             |
| 10    | 0.0                  | 0.8599             | 0.1401             |
| 11    | 0.0                  | 0.8999             | 0.1001000000000001 |
| 12    | 0.0                  | 0.9399000000000001 | 0.0601000000000001 |
| 13    | 0.0                  | 0.98               | 0.02               |
| 14    | 0.0098               | 0.9902000000000001 | 0.0                |
| 15    | 0.029800000000000004 | 0.9701999999999998 | 0.0                |
| 16    | 0.04979999999999999  | 0.9502             | 0.0                |
| 17    | 0.0698               | 0.9302             | 0.0                |
| 18    | 0.0898               | 0.9102             | 0.0                |
| 19    | 0.1099               | 0.8901             | 0.0                |
| 20    | 0.1299               | 0.8701             | 0.0                |
| 21    | 0.1499               | 0.8501             | 0.0                |
| 22    | 0.16990000000000002  | 0.8301             | 0.0                |
| 23    | 0.18990000000000004  | 0.8100999999999999 | 0.0                |
| 24    | 0.21                 | 0.79               | 0.0                |
| 25    | 0.23                 | 0.77               | 0.0                |
| 26    | 0.25                 | 0.75               | 0.0                |
| 27    | 0.27                 | 0.73               | 0.0                |

beta=0.01, budget=0.7, model=qwen7b

- PPL: 6.5296
- PPL Δ%: -0.12
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.565  |
| 1     | 0.575  |
| 2     | 0.585  |
| 3     | 0.595  |
| 4     | 0.605  |
| 5     | 0.615  |
| 6     | 0.625  |
| 7     | 0.635  |
| 8     | 0.645  |
| 9     | 0.655  |
| 10    | 0.665  |
| 11    | 0.675  |
| 12    | 0.685  |
| 13    | 0.695  |
| 14    | 0.705  |
| 15    | 0.715  |
| 16    | 0.725  |
| 17    | 0.735  |
| 18    | 0.745  |
| 19    | 0.755  |
| 20    | 0.765  |
| 21    | 0.775  |
| 22    | 0.785  |
| 23    | 0.795  |
| 24    | 0.805  |
| 25    | 0.815  |
| 26    | 0.825  |
| 27    | 0.835  |

Tier Fractions Per Layer:

| Layer | FP16                | INT8               | INT4 |
|-------|---------------------|--------------------|------|
| 0     | 0.1299              | 0.8701             | 0.0  |
| 1     | 0.1499              | 0.8501             | 0.0  |
| 2     | 0.16990000000000002 | 0.8301             | 0.0  |
| 3     | 0.18990000000000004 | 0.8100999999999999 | 0.0  |
| 4     | 0.21                | 0.79               | 0.0  |
| 5     | 0.23                | 0.77               | 0.0  |
| 6     | 0.25                | 0.75               | 0.0  |
| 7     | 0.27                | 0.73               | 0.0  |
| 8     | 0.29                | 0.71               | 0.0  |
| 9     | 0.3101              | 0.6899000000000001 | 0.0  |
| 10    | 0.3301              | 0.6699             | 0.0  |
| 11    | 0.3501              | 0.6499             | 0.0  |
| 12    | 0.37010000000000004 | 0.6299             | 0.0  |
| 13    | 0.39010000000000006 | 0.6099             | 0.0  |

|    |                    |                    |     |
|----|--------------------|--------------------|-----|
| 14 | 0.4101999999999995 | 0.5898             | 0.0 |
| 15 | 0.4301999999999997 | 0.5698             | 0.0 |
| 16 | 0.4502             | 0.5498             | 0.0 |
| 17 | 0.4702             | 0.5298             | 0.0 |
| 18 | 0.4902             | 0.5098             | 0.0 |
| 19 | 0.5098             | 0.4902             | 0.0 |
| 20 | 0.5298             | 0.4702             | 0.0 |
| 21 | 0.5498             | 0.4502             | 0.0 |
| 22 | 0.5698             | 0.4301999999999997 | 0.0 |
| 23 | 0.5898             | 0.4101999999999995 | 0.0 |
| 24 | 0.6099             | 0.3901000000000006 | 0.0 |
| 25 | 0.6299             | 0.3701000000000004 | 0.0 |
| 26 | 0.6499             | 0.3501             | 0.0 |
| 27 | 0.6699             | 0.3301             | 0.0 |

beta=0.02, budget=0.3, model=qwen7b

- PPL: 28316.446
- PPL Δ%: 433038.2
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.27   |
| 13    | 0.29   |
| 14    | 0.31   |
| 15    | 0.33   |
| 16    | 0.35   |
| 17    | 0.37   |
| 18    | 0.39   |
| 19    | 0.41   |
| 20    | 0.43   |
| 21    | 0.45   |
| 22    | 0.47   |
| 23    | 0.49   |
| 24    | 0.51   |
| 25    | 0.53   |
| 26    | 0.55   |
| 27    | 0.57   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8               | INT4               |
|-------|---------------------|--------------------|--------------------|
| 0     | 0.0                 | 0.0                | 1.0                |
| 1     | 0.0                 | 0.0                | 1.0                |
| 2     | 0.0                 | 0.0                | 1.0                |
| 3     | 0.0                 | 0.0                | 1.0                |
| 4     | 0.0                 | 0.0                | 1.0                |
| 5     | 0.0                 | 0.0                | 1.0                |
| 6     | 0.0                 | 0.0                | 1.0                |
| 7     | 0.0                 | 0.0                | 1.0                |
| 8     | 0.0                 | 0.0                | 1.0                |
| 9     | 0.0                 | 0.0                | 1.0                |
| 10    | 0.0                 | 0.0                | 1.0                |
| 11    | 0.0                 | 0.0                | 1.0                |
| 12    | 0.0                 | 0.0796             | 0.9204             |
| 13    | 0.0                 | 0.1597             | 0.8403             |
| 14    | 0.0                 | 0.2397             | 0.7603             |
| 15    | 0.0                 | 0.3198             | 0.6802             |
| 16    | 0.0                 | 0.3999             | 0.6001             |
| 17    | 0.0                 | 0.48               | 0.52               |
| 18    | 0.0                 | 0.5596             | 0.4404000000000007 |
| 19    | 0.0                 | 0.6396             | 0.3604             |
| 20    | 0.0                 | 0.7197             | 0.2803             |
| 21    | 0.0                 | 0.7998             | 0.2002000000000002 |
| 22    | 0.0                 | 0.8799             | 0.1201             |
| 23    | 0.0                 | 0.96               | 0.04               |
| 24    | 0.02                | 0.98               | 0.0                |
| 25    | 0.06010000000000001 | 0.9399000000000001 | 0.0                |
| 26    | 0.10010000000000001 | 0.8999             | 0.0                |
| 27    | 0.1401              | 0.8599             | 0.0                |

beta=0.02, budget=0.5, model=qwen7b

- PPL: 26038.9746
- PPL Δ%: 398201.21
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.27   |
| 3     | 0.29   |
| 4     | 0.31   |
| 5     | 0.33   |
| 6     | 0.35   |

|    |      |
|----|------|
| 7  | 0.37 |
| 8  | 0.39 |
| 9  | 0.41 |
| 10 | 0.43 |
| 11 | 0.45 |
| 12 | 0.47 |
| 13 | 0.49 |
| 14 | 0.51 |
| 15 | 0.53 |
| 16 | 0.55 |
| 17 | 0.57 |
| 18 | 0.59 |
| 19 | 0.61 |
| 20 | 0.63 |
| 21 | 0.65 |
| 22 | 0.67 |
| 23 | 0.69 |
| 24 | 0.71 |
| 25 | 0.73 |
| 26 | 0.75 |
| 27 | 0.77 |

Tier Fractions Per Layer:

| Layer | FP16                 | INT8                | INT4                |
|-------|----------------------|---------------------|---------------------|
| 0     | 0.0                  | 0.0                 | 1.0                 |
| 1     | 0.0                  | 0.0                 | 1.0                 |
| 2     | 0.0                  | 0.0796              | 0.9204              |
| 3     | 0.0                  | 0.1597              | 0.8403              |
| 4     | 0.0                  | 0.2397              | 0.7603              |
| 5     | 0.0                  | 0.3198              | 0.6802              |
| 6     | 0.0                  | 0.3999              | 0.6001              |
| 7     | 0.0                  | 0.48                | 0.52                |
| 8     | 0.0                  | 0.5596              | 0.44040000000000007 |
| 9     | 0.0                  | 0.6396              | 0.3604              |
| 10    | 0.0                  | 0.7197              | 0.2803              |
| 11    | 0.0                  | 0.7998              | 0.20020000000000002 |
| 12    | 0.0                  | 0.8799              | 0.1201              |
| 13    | 0.0                  | 0.96                | 0.04                |
| 14    | 0.02                 | 0.98                | 0.0                 |
| 15    | 0.060100000000000001 | 0.93990000000000001 | 0.0                 |
| 16    | 0.100100000000000001 | 0.8999              | 0.0                 |
| 17    | 0.1401               | 0.8599              | 0.0                 |
| 18    | 0.1802               | 0.8198              | 0.0                 |
| 19    | 0.220200000000000003 | 0.77980000000000002 | 0.0                 |
| 20    | 0.2598               | 0.74020000000000001 | 0.0                 |
| 21    | 0.2998               | 0.7002              | 0.0                 |
| 22    | 0.339800000000000005 | 0.6602              | 0.0                 |
| 23    | 0.379899999999999996 | 0.6201              | 0.0                 |
| 24    | 0.4199               | 0.5801              | 0.0                 |
| 25    | 0.46                 | 0.54                | 0.0                 |
| 26    | 0.5                  | 0.5                 | 0.0                 |
| 27    | 0.54                 | 0.46                | 0.0                 |

beta=0.02, budget=0.7, model=qwen7b

- PPL: 130.7586
- PPL Δ%: 1900.13
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.43   |
| 1     | 0.45   |
| 2     | 0.47   |
| 3     | 0.49   |
| 4     | 0.51   |
| 5     | 0.53   |
| 6     | 0.55   |
| 7     | 0.57   |
| 8     | 0.59   |
| 9     | 0.61   |
| 10    | 0.63   |
| 11    | 0.65   |
| 12    | 0.67   |
| 13    | 0.69   |
| 14    | 0.71   |
| 15    | 0.73   |
| 16    | 0.75   |
| 17    | 0.77   |
| 18    | 0.79   |
| 19    | 0.81   |
| 20    | 0.83   |
| 21    | 0.85   |
| 22    | 0.87   |
| 23    | 0.89   |
| 24    | 0.91   |
| 25    | 0.93   |
| 26    | 0.95   |
| 27    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                 | INT8                | INT4                |
|-------|----------------------|---------------------|---------------------|
| 0     | 0.0                  | 0.7197              | 0.2803              |
| 1     | 0.0                  | 0.7998              | 0.20020000000000002 |
| 2     | 0.0                  | 0.8799              | 0.1201              |
| 3     | 0.0                  | 0.96                | 0.04                |
| 4     | 0.02                 | 0.98                | 0.0                 |
| 5     | 0.060100000000000001 | 0.93990000000000001 | 0.0                 |

|    |                     |                     |     |
|----|---------------------|---------------------|-----|
| 6  | 0.10010000000000001 | 0.8999              | 0.0 |
| 7  | 0.1401              | 0.8599              | 0.0 |
| 8  | 0.1802              | 0.8198              | 0.0 |
| 9  | 0.22020000000000003 | 0.7798000000000002  | 0.0 |
| 10 | 0.2598              | 0.7402000000000001  | 0.0 |
| 11 | 0.2998              | 0.7002              | 0.0 |
| 12 | 0.33980000000000005 | 0.6602              | 0.0 |
| 13 | 0.37989999999999996 | 0.6201              | 0.0 |
| 14 | 0.4199              | 0.5801              | 0.0 |
| 15 | 0.46                | 0.54                | 0.0 |
| 16 | 0.5                 | 0.5                 | 0.0 |
| 17 | 0.54                | 0.46                | 0.0 |
| 18 | 0.5801              | 0.4199              | 0.0 |
| 19 | 0.6201              | 0.37989999999999996 | 0.0 |
| 20 | 0.6602              | 0.33980000000000005 | 0.0 |
| 21 | 0.7002              | 0.2998              | 0.0 |
| 22 | 0.74020000000000001 | 0.2598              | 0.0 |
| 23 | 0.77980000000000002 | 0.22020000000000003 | 0.0 |
| 24 | 0.8198              | 0.1802              | 0.0 |
| 25 | 0.8599              | 0.1401              | 0.0 |
| 26 | 0.8999              | 0.10010000000000001 | 0.0 |
| 27 | 0.8999              | 0.10010000000000001 | 0.0 |

beta=0.03, budget=0.3, model=qwen7b

- PPL: 28252.0098
- PPL Δ%: 432052.56
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.255  |
| 13    | 0.285  |
| 14    | 0.315  |
| 15    | 0.345  |
| 16    | 0.375  |
| 17    | 0.405  |
| 18    | 0.435  |
| 19    | 0.465  |
| 20    | 0.495  |
| 21    | 0.525  |
| 22    | 0.555  |
| 23    | 0.585  |
| 24    | 0.615  |
| 25    | 0.645  |
| 26    | 0.675  |
| 27    | 0.705  |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4               |
|-------|---------------------|---------------------|--------------------|
| 0     | 0.0                 | 0.0                 | 1.0                |
| 1     | 0.0                 | 0.0                 | 1.0                |
| 2     | 0.0                 | 0.0                 | 1.0                |
| 3     | 0.0                 | 0.0                 | 1.0                |
| 4     | 0.0                 | 0.0                 | 1.0                |
| 5     | 0.0                 | 0.0                 | 1.0                |
| 6     | 0.0                 | 0.0                 | 1.0                |
| 7     | 0.0                 | 0.0                 | 1.0                |
| 8     | 0.0                 | 0.0                 | 1.0                |
| 9     | 0.0                 | 0.0                 | 1.0                |
| 10    | 0.0                 | 0.0                 | 1.0                |
| 11    | 0.0                 | 0.0                 | 1.0                |
| 12    | 0.0                 | 0.0195              | 0.9805             |
| 13    | 0.0                 | 0.1396              | 0.8603999999999999 |
| 14    | 0.0                 | 0.2598              | 0.7402000000000001 |
| 15    | 0.0                 | 0.37989999999999996 | 0.6201             |
| 16    | 0.0                 | 0.5                 | 0.5                |
| 17    | 0.0                 | 0.6196              | 0.3804             |
| 18    | 0.0                 | 0.7397              | 0.2603             |
| 19    | 0.0                 | 0.8599              | 0.1401             |
| 20    | 0.0                 | 0.98                | 0.02               |
| 21    | 0.04979999999999999 | 0.9502              | 0.0                |
| 22    | 0.1099              | 0.8901              | 0.0                |
| 23    | 0.16990000000000002 | 0.8301              | 0.0                |
| 24    | 0.23                | 0.77                | 0.0                |
| 25    | 0.29                | 0.71                | 0.0                |
| 26    | 0.3501              | 0.6499              | 0.0                |
| 27    | 0.41019999999999995 | 0.5898              | 0.0                |

beta=0.03, budget=0.5, model=qwen7b

- PPL: 28865.4798
- PPL Δ%: 441436.41
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|



| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.275  |
| 7     | 0.305  |
| 8     | 0.335  |
| 9     | 0.365  |
| 10    | 0.395  |
| 11    | 0.425  |
| 12    | 0.455  |
| 13    | 0.485  |
| 14    | 0.515  |
| 15    | 0.545  |
| 16    | 0.575  |
| 17    | 0.605  |
| 18    | 0.635  |
| 19    | 0.665  |
| 20    | 0.695  |
| 21    | 0.725  |
| 22    | 0.755  |
| 23    | 0.785  |
| 24    | 0.815  |
| 25    | 0.845  |
| 26    | 0.875  |
| 27    | 0.905  |

Tier Fractions Per Layer:

| Layer | FP16                 | INT8                | INT4                |
|-------|----------------------|---------------------|---------------------|
| 0     | 0.0                  | 0.0                 | 1.0                 |
| 1     | 0.0                  | 0.0                 | 1.0                 |
| 2     | 0.0                  | 0.0                 | 1.0                 |
| 3     | 0.0                  | 0.0                 | 1.0                 |
| 4     | 0.0                  | 0.0                 | 1.0                 |
| 5     | 0.0                  | 0.0                 | 1.0                 |
| 6     | 0.0                  | 0.09959999999999998 | 0.9004              |
| 7     | 0.0                  | 0.2197              | 0.7803              |
| 8     | 0.0                  | 0.33980000000000005 | 0.6602              |
| 9     | 0.0                  | 0.46                | 0.54                |
| 10    | 0.0                  | 0.5796              | 0.42040000000000005 |
| 11    | 0.0                  | 0.6997              | 0.3003              |
| 12    | 0.0                  | 0.8198              | 0.1802              |
| 13    | 0.0                  | 0.9399000000000001  | 0.06010000000000001 |
| 14    | 0.029800000000000004 | 0.9701999999999998  | 0.0                 |
| 15    | 0.0898               | 0.9102              | 0.0                 |
| 16    | 0.1499               | 0.8501              | 0.0                 |
| 17    | 0.21                 | 0.79                | 0.0                 |
| 18    | 0.27                 | 0.73                | 0.0                 |
| 19    | 0.3301               | 0.6699              | 0.0                 |
| 20    | 0.39010000000000006  | 0.6099              | 0.0                 |
| 21    | 0.4502               | 0.5498              | 0.0                 |
| 22    | 0.5098               | 0.4902              | 0.0                 |
| 23    | 0.5698               | 0.43019999999999997 | 0.0                 |
| 24    | 0.6299               | 0.37010000000000004 | 0.0                 |
| 25    | 0.6899000000000001   | 0.3101              | 0.0                 |
| 26    | 0.75                 | 0.25                | 0.0                 |
| 27    | 0.8100999999999999   | 0.18990000000000004 | 0.0                 |

beta=0.03, budget=0.7, model=qwen7b

- PPL: 13096.9987
- PPL Δ%: 200236.24
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.295  |
| 1     | 0.325  |
| 2     | 0.355  |
| 3     | 0.385  |
| 4     | 0.415  |
| 5     | 0.445  |
| 6     | 0.475  |
| 7     | 0.505  |
| 8     | 0.535  |
| 9     | 0.565  |
| 10    | 0.595  |
| 11    | 0.625  |
| 12    | 0.655  |
| 13    | 0.685  |
| 14    | 0.715  |
| 15    | 0.745  |
| 16    | 0.775  |
| 17    | 0.805  |
| 18    | 0.835  |
| 19    | 0.865  |
| 20    | 0.895  |
| 21    | 0.925  |
| 22    | 0.95   |
| 23    | 0.95   |
| 24    | 0.95   |
| 25    | 0.95   |
| 26    | 0.95   |
| 27    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.1797              | 0.8203              |
| 1     | 0.0                 | 0.2998              | 0.7002              |
| 2     | 0.0                 | 0.4199              | 0.5801              |
| 3     | 0.0                 | 0.5396              | 0.4604              |
| 4     | 0.0                 | 0.6597              | 0.3403              |
| 5     | 0.0                 | 0.7798000000000002  | 0.2202000000000003  |
| 6     | 0.0                 | 0.8999              | 0.10010000000000001 |
| 7     | 0.0098              | 0.9902000000000001  | 0.0                 |
| 8     | 0.0698              | 0.9302              | 0.0                 |
| 9     | 0.1299              | 0.8701              | 0.0                 |
| 10    | 0.18990000000000004 | 0.8100999999999999  | 0.0                 |
| 11    | 0.25                | 0.75                | 0.0                 |
| 12    | 0.3101              | 0.6899000000000001  | 0.0                 |
| 13    | 0.37010000000000004 | 0.6299              | 0.0                 |
| 14    | 0.43019999999999997 | 0.5698              | 0.0                 |
| 15    | 0.4902              | 0.5098              | 0.0                 |
| 16    | 0.5498              | 0.4502              | 0.0                 |
| 17    | 0.6099              | 0.39010000000000006 | 0.0                 |
| 18    | 0.6699              | 0.3301              | 0.0                 |
| 19    | 0.73                | 0.27                | 0.0                 |
| 20    | 0.79                | 0.21                | 0.0                 |
| 21    | 0.8501              | 0.1499              | 0.0                 |
| 22    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 23    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 24    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 25    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 26    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 27    | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.04, budget=0.3, model=qwen7b

- PPL: 29224.1048
- PPL Δ%: 446922.07
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |
| 13    | 0.28   |
| 14    | 0.32   |
| 15    | 0.36   |
| 16    | 0.4    |
| 17    | 0.44   |
| 18    | 0.48   |
| 19    | 0.52   |
| 20    | 0.56   |
| 21    | 0.6    |
| 22    | 0.64   |
| 23    | 0.68   |
| 24    | 0.72   |
| 25    | 0.76   |
| 26    | 0.8    |
| 27    | 0.84   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8               | INT4                |
|-------|---------------------|--------------------|---------------------|
| 0     | 0.0                 | 0.0                | 1.0                 |
| 1     | 0.0                 | 0.0                | 1.0                 |
| 2     | 0.0                 | 0.0                | 1.0                 |
| 3     | 0.0                 | 0.0                | 1.0                 |
| 4     | 0.0                 | 0.0                | 1.0                 |
| 5     | 0.0                 | 0.0                | 1.0                 |
| 6     | 0.0                 | 0.0                | 1.0                 |
| 7     | 0.0                 | 0.0                | 1.0                 |
| 8     | 0.0                 | 0.0                | 1.0                 |
| 9     | 0.0                 | 0.0                | 1.0                 |
| 10    | 0.0                 | 0.0                | 1.0                 |
| 11    | 0.0                 | 0.0                | 1.0                 |
| 12    | 0.0                 | 0.0                | 1.0                 |
| 13    | 0.0                 | 0.1196             | 0.8804              |
| 14    | 0.0                 | 0.2798             | 0.7201999999999998  |
| 15    | 0.0                 | 0.4399             | 0.5601              |
| 16    | 0.0                 | 0.5996             | 0.40040000000000003 |
| 17    | 0.0                 | 0.7597999999999999 | 0.2402              |
| 18    | 0.0                 | 0.9199             | 0.0801              |
| 19    | 0.04                | 0.96               | 0.0                 |
| 20    | 0.1201              | 0.8799             | 0.0                 |
| 21    | 0.20020000000000002 | 0.7998             | 0.0                 |
| 22    | 0.2798              | 0.7201999999999998 | 0.0                 |
| 23    | 0.35989999999999994 | 0.6401             | 0.0                 |
| 24    | 0.4399              | 0.5601             | 0.0                 |
| 25    | 0.52                | 0.48               | 0.0                 |
| 26    | 0.6001              | 0.3999             | 0.0                 |
| 27    | 0.6802              | 0.3198             | 0.0                 |

beta=0.04, budget=0.5, model=qwen7b

- PPL: 28961.3366
- PPL Δ%: 442902.67
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.28   |
| 9     | 0.32   |
| 10    | 0.36   |
| 11    | 0.4    |
| 12    | 0.44   |
| 13    | 0.48   |
| 14    | 0.52   |
| 15    | 0.56   |
| 16    | 0.6    |
| 17    | 0.64   |
| 18    | 0.68   |
| 19    | 0.72   |
| 20    | 0.76   |
| 21    | 0.8    |
| 22    | 0.84   |
| 23    | 0.88   |
| 24    | 0.92   |
| 25    | 0.95   |
| 26    | 0.95   |
| 27    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.0                 | 1.0                 |
| 8     | 0.0                 | 0.1196              | 0.8804              |
| 9     | 0.0                 | 0.2798              | 0.7201999999999998  |
| 10    | 0.0                 | 0.4399              | 0.5601              |
| 11    | 0.0                 | 0.5996              | 0.40040000000000003 |
| 12    | 0.0                 | 0.7597999999999999  | 0.2402              |
| 13    | 0.0                 | 0.9199              | 0.0801              |
| 14    | 0.04                | 0.96                | 0.0                 |
| 15    | 0.1201              | 0.8799              | 0.0                 |
| 16    | 0.20020000000000002 | 0.7998              | 0.0                 |
| 17    | 0.2798              | 0.7201999999999998  | 0.0                 |
| 18    | 0.35989999999999994 | 0.6401              | 0.0                 |
| 19    | 0.4399              | 0.5601              | 0.0                 |
| 20    | 0.52                | 0.48                | 0.0                 |
| 21    | 0.6001              | 0.3999              | 0.0                 |
| 22    | 0.6802              | 0.3198              | 0.0                 |
| 23    | 0.7597999999999999  | 0.2402              | 0.0                 |
| 24    | 0.8398              | 0.1602              | 0.0                 |
| 25    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 26    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 27    | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.04, budget=0.7, model=qwen7b

- PPL: 27857.321
- PPL Δ%: 426015.27
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.28   |
| 4     | 0.32   |
| 5     | 0.36   |
| 6     | 0.4    |
| 7     | 0.44   |
| 8     | 0.48   |
| 9     | 0.52   |
| 10    | 0.56   |
| 11    | 0.6    |
| 12    | 0.64   |
| 13    | 0.68   |
| 14    | 0.72   |
| 15    | 0.76   |
| 16    | 0.8    |
| 17    | 0.84   |
| 18    | 0.88   |
| 19    | 0.92   |
| 20    | 0.95   |

|    |      |
|----|------|
| 21 | 0.95 |
| 22 | 0.95 |
| 23 | 0.95 |
| 24 | 0.95 |
| 25 | 0.95 |
| 26 | 0.95 |
| 27 | 0.95 |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.1196              | 0.8804              |
| 4     | 0.0                 | 0.2798              | 0.7201999999999998  |
| 5     | 0.0                 | 0.4399              | 0.5601              |
| 6     | 0.0                 | 0.5996              | 0.40040000000000003 |
| 7     | 0.0                 | 0.7597999999999999  | 0.2402              |
| 8     | 0.0                 | 0.9199              | 0.0801              |
| 9     | 0.04                | 0.96                | 0.0                 |
| 10    | 0.1201              | 0.8799              | 0.0                 |
| 11    | 0.20020000000000002 | 0.7998              | 0.0                 |
| 12    | 0.2798              | 0.7201999999999998  | 0.0                 |
| 13    | 0.35989999999999994 | 0.6401              | 0.0                 |
| 14    | 0.4399              | 0.5601              | 0.0                 |
| 15    | 0.52                | 0.48                | 0.0                 |
| 16    | 0.6001              | 0.3999              | 0.0                 |
| 17    | 0.6802              | 0.3198              | 0.0                 |
| 18    | 0.7597999999999999  | 0.2402              | 0.0                 |
| 19    | 0.8398              | 0.1602              | 0.0                 |
| 20    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 21    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 22    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 23    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 24    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 25    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 26    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 27    | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.05, budget=0.3, model=qwen7b

- PPL: 29567.5846
- PPL Δ%: 452176.05
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.25   |
| 10    | 0.25   |
| 11    | 0.25   |
| 12    | 0.25   |
| 13    | 0.275  |
| 14    | 0.325  |
| 15    | 0.375  |
| 16    | 0.425  |
| 17    | 0.475  |
| 18    | 0.525  |
| 19    | 0.575  |
| 20    | 0.625  |
| 21    | 0.675  |
| 22    | 0.725  |
| 23    | 0.775  |
| 24    | 0.825  |
| 25    | 0.875  |
| 26    | 0.925  |
| 27    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.0                 | 1.0                 |
| 8     | 0.0                 | 0.0                 | 1.0                 |
| 9     | 0.0                 | 0.0                 | 1.0                 |
| 10    | 0.0                 | 0.0                 | 1.0                 |
| 11    | 0.0                 | 0.0                 | 1.0                 |
| 12    | 0.0                 | 0.0                 | 1.0                 |
| 13    | 0.0                 | 0.09559999999999998 | 0.9004              |
| 14    | 0.0                 | 0.2998              | 0.7002              |
| 15    | 0.0                 | 0.5                 | 0.5                 |
| 16    | 0.0                 | 0.6997              | 0.3003              |
| 17    | 0.0                 | 0.8999              | 0.10010000000000001 |
| 18    | 0.04979999999999999 | 0.9502              | 0.0                 |
| 19    | 0.1499              | 0.8501              | 0.0                 |

|    |        |                     |     |
|----|--------|---------------------|-----|
| 20 | 0.25   | 0.75                | 0.0 |
| 21 | 0.3501 | 0.6499              | 0.0 |
| 22 | 0.4502 | 0.5498              | 0.0 |
| 23 | 0.5498 | 0.4502              | 0.0 |
| 24 | 0.6499 | 0.3501              | 0.0 |
| 25 | 0.75   | 0.25                | 0.0 |
| 26 | 0.8501 | 0.1499              | 0.0 |
| 27 | 0.8999 | 0.10010000000000001 | 0.0 |

beta=0.05, budget=0.5, model=qwen7b

- PPL: 29092.0775
- PPL Δ%: 444902.53
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.25   |
| 6     | 0.25   |
| 7     | 0.25   |
| 8     | 0.25   |
| 9     | 0.275  |
| 10    | 0.325  |
| 11    | 0.375  |
| 12    | 0.425  |
| 13    | 0.475  |
| 14    | 0.525  |
| 15    | 0.575  |
| 16    | 0.625  |
| 17    | 0.675  |
| 18    | 0.725  |
| 19    | 0.775  |
| 20    | 0.825  |
| 21    | 0.875  |
| 22    | 0.925  |
| 23    | 0.95   |
| 24    | 0.95   |
| 25    | 0.95   |
| 26    | 0.95   |
| 27    | 0.95   |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.0                 | 1.0                 |
| 6     | 0.0                 | 0.0                 | 1.0                 |
| 7     | 0.0                 | 0.0                 | 1.0                 |
| 8     | 0.0                 | 0.0                 | 1.0                 |
| 9     | 0.0                 | 0.09559999999999998 | 0.9004              |
| 10    | 0.0                 | 0.2998              | 0.7002              |
| 11    | 0.0                 | 0.5                 | 0.5                 |
| 12    | 0.0                 | 0.6997              | 0.3003              |
| 13    | 0.0                 | 0.8999              | 0.10010000000000001 |
| 14    | 0.04979999999999999 | 0.9502              | 0.0                 |
| 15    | 0.1499              | 0.8501              | 0.0                 |
| 16    | 0.25                | 0.75                | 0.0                 |
| 17    | 0.3501              | 0.6499              | 0.0                 |
| 18    | 0.4502              | 0.5498              | 0.0                 |
| 19    | 0.5498              | 0.4502              | 0.0                 |
| 20    | 0.6499              | 0.3501              | 0.0                 |
| 21    | 0.75                | 0.25                | 0.0                 |
| 22    | 0.8501              | 0.1499              | 0.0                 |
| 23    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 24    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 25    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 26    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 27    | 0.8999              | 0.10010000000000001 | 0.0                 |

beta=0.05, budget=0.7, model=qwen7b

- PPL: 29115.2552
- PPL Δ%: 445257.06
- Baseline PPL: 6.5375
- Num layers: 28
- Seq len: 2048
- Num samples: 3

Layer Budgets:

| Layer | Budget |
|-------|--------|
| 0     | 0.25   |
| 1     | 0.25   |
| 2     | 0.25   |
| 3     | 0.25   |
| 4     | 0.25   |
| 5     | 0.275  |
| 6     | 0.325  |
| 7     | 0.375  |
| 8     | 0.425  |
| 9     | 0.475  |
| 10    | 0.525  |
| 11    | 0.575  |
| 12    | 0.625  |

|    |       |
|----|-------|
| 13 | 0.675 |
| 14 | 0.725 |
| 15 | 0.775 |
| 16 | 0.825 |
| 17 | 0.875 |
| 18 | 0.925 |
| 19 | 0.95  |
| 20 | 0.95  |
| 21 | 0.95  |
| 22 | 0.95  |
| 23 | 0.95  |
| 24 | 0.95  |
| 25 | 0.95  |
| 26 | 0.95  |
| 27 | 0.95  |

Tier Fractions Per Layer:

| Layer | FP16                | INT8                | INT4                |
|-------|---------------------|---------------------|---------------------|
| 0     | 0.0                 | 0.0                 | 1.0                 |
| 1     | 0.0                 | 0.0                 | 1.0                 |
| 2     | 0.0                 | 0.0                 | 1.0                 |
| 3     | 0.0                 | 0.0                 | 1.0                 |
| 4     | 0.0                 | 0.0                 | 1.0                 |
| 5     | 0.0                 | 0.09559999999999998 | 0.9004              |
| 6     | 0.0                 | 0.2998              | 0.7002              |
| 7     | 0.0                 | 0.4995              | 0.5005              |
| 8     | 0.0                 | 0.6997              | 0.3003              |
| 9     | 0.0                 | 0.8999              | 0.10010000000000001 |
| 10    | 0.04979999999999999 | 0.9502              | 0.0                 |
| 11    | 0.1499              | 0.8501              | 0.0                 |
| 12    | 0.25                | 0.75                | 0.0                 |
| 13    | 0.3501              | 0.6499              | 0.0                 |
| 14    | 0.4502              | 0.5498              | 0.0                 |
| 15    | 0.5498              | 0.4502              | 0.0                 |
| 16    | 0.6499              | 0.3501              | 0.0                 |
| 17    | 0.75                | 0.25                | 0.0                 |
| 18    | 0.8501              | 0.1499              | 0.0                 |
| 19    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 20    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 21    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 22    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 23    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 24    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 25    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 26    | 0.8999              | 0.10010000000000001 | 0.0                 |
| 27    | 0.8999              | 0.10010000000000001 | 0.0                 |